

HEALTHCONNECT CLINIC — DATA ANALYTICS INTEGRATION, ADVANCED ANALYSIS & VALIDATION

Abstract This study evaluates HealthConnect Clinic's appointment attendance patterns using advanced segmentation and KPI validation. Week 6 analysis identified booking lead time and previous no-show history as key risk indicators, with a ****70.52% no-show rate**** among appointments booked 31–60 days in advance for patients with previous no-shows. The findings refine Week 5 insights, strengthen decision support, and provide analytical inputs for AI modelling and subsequent testing.

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Improving Patient
Appointment
Attendance and
Healthcare
Support Using
Data and AI

Executive Summary

This Week 6 report presents the **HealthConnect Clinic** progression from initial appointment attendance analysis to advanced analytical validation, segmentation, integration, and decision support. The work builds directly on the findings and dashboard developed in Week 5 and is aligned with the overall project objective of determining how HealthConnect can use data and AI to reduce missed appointments and improve the patient support experience.

Analysis of the 5,000 appointment records identified **4,737 eligible appointments**, comprising 2,423 no-shows and 2,314 attended appointments. The validated overall **No-Show Rate was 51.15%**, while the **Attendance Rate was 48.85%**. Week 6 testing confirmed the core Week 5 KPIs and strengthened their interpretation through deeper segmentation.

The most important finding was the interaction between **booking lead time and previous no-show history**. Appointments booked **31–60 days in advance** had a 63.95% no-show rate overall, compared with 25.93% for same-day bookings. Patients with at least one previous no-show also demonstrated substantially higher observed risk, with a **57.83% no-show rate compared with 46.30% among patients with no previous no-shows**, a difference of 11.53 percentage points. When these characteristics were examined jointly, appointments booked 31–60 days in advance for patients with previous no-shows produced the highest identified high-risk segment, with a **70.52% no-show rate**.

Further segmentation showed that this high-risk group remained vulnerable across other operational characteristics. Within the segment, no-show rates generally increased with distance from the clinic, reaching 77.94% among appointments 20–30 km away, although small groups require caution. Follow-up appointments had the highest observed no-show rate among appointment types at 72.76%, but appointment type showed a smaller difference than booking lead time and previous no-show history.

Reminder analysis provided a more nuanced result. Overall, appointments with reminders had a **49.86% no-show rate**, compared with **54.63% without reminders**, a 4.77-percentage-point difference. Within the high-risk segment, the corresponding rates were 69.21% with a reminder and 73.98% without one. However, the relationship varied across segments and the dataset does

not establish causality. Therefore, reminder status is treated as an **observational association rather than evidence that reminders directly reduce no-shows**.

Based on the deeper analysis, the Week 6 KPI framework was refined rather than replaced. The core No-Show Rate, Attendance Rate, and Repeat No-Show Rate were retained. Reminder analysis was reframed as **No-Show Rate by Reminder Status**, while the **70.52% High-Risk Segment No-Show Rate** was introduced as a decision-support indicator. Cancellation Rate remains a contextual metric rather than a primary attendance KPI.

The Week 6 dashboard was consequently upgraded from broad performance monitoring to **targeted decision support**. It now highlights the highest-risk appointment segments and their relationships with booking lead time, previous no-show history, reminder status, distance, and appointment type. The analysis also provides Data Science with candidate modelling variables and, importantly, identifies **booking lead time × previous no-show history** as an interaction requiring further modelling investigation. Potential feature-leakage issues, particularly the timing of reminder and waiting-time variables, were also documented.

Overall, Week 6 establishes that HealthConnect should move beyond monitoring its overall 51.15% no-show rate toward **targeted, evidence-based attendance support for identifiable high-risk appointment segments**. The findings remain observational and should not be interpreted as causal effects. Week 7 should therefore focus on testing the stability of these findings, validating model features and timing, assessing subgroup performance, testing dashboard behavior, and evaluating whether the identified risk segments can support practical and measurable interventions.

List of Abbreviations

Abbreviation	Definition
AI	Artificial Intelligence
BI	Business Intelligence
DAX	Data Analysis Expressions
EDA	Exploratory Data Analysis
KPI	Key Performance Indicator
KPIs	Key Performance Indicators
PDF	Portable Document Format
SMS	Short Message Service
SQL	Structured Query Language

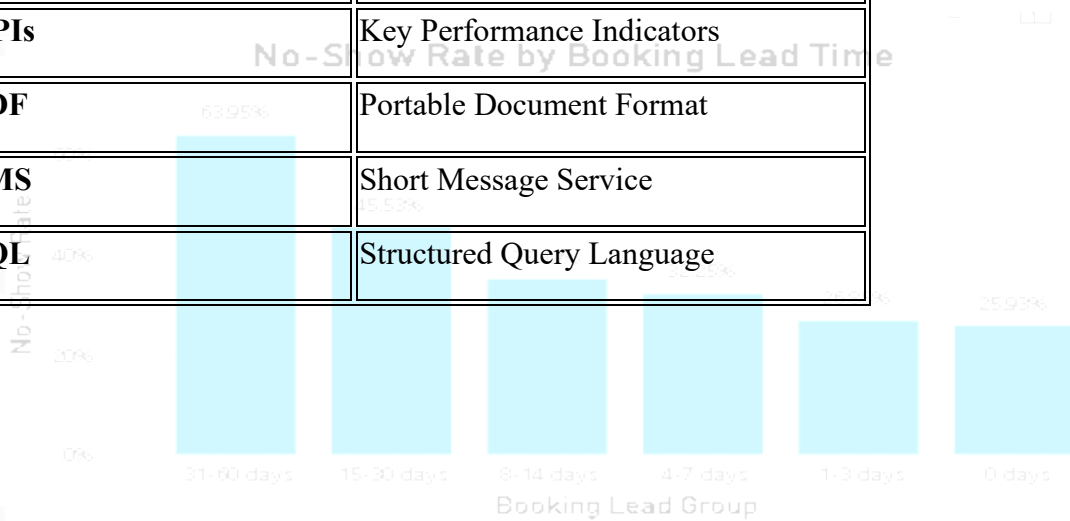


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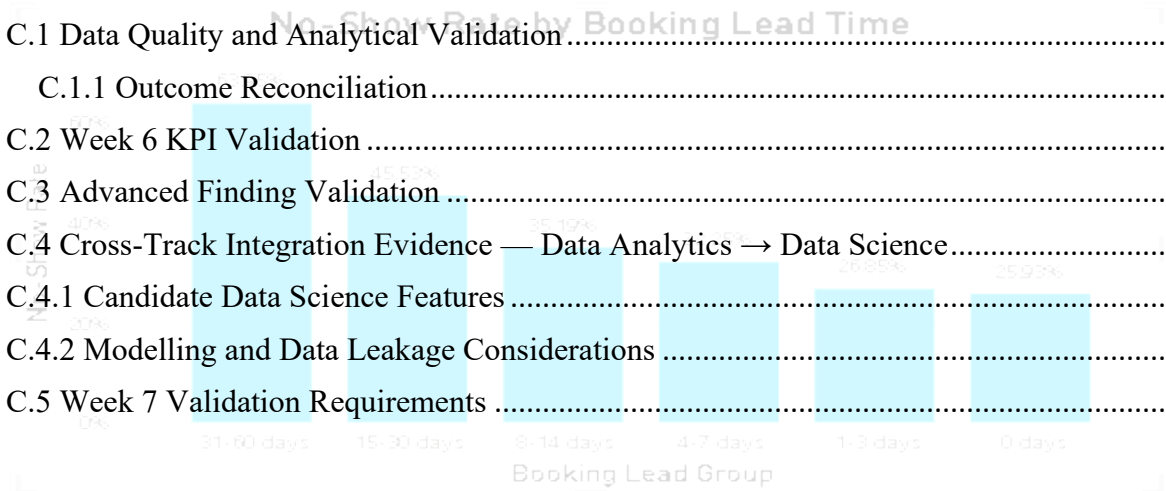
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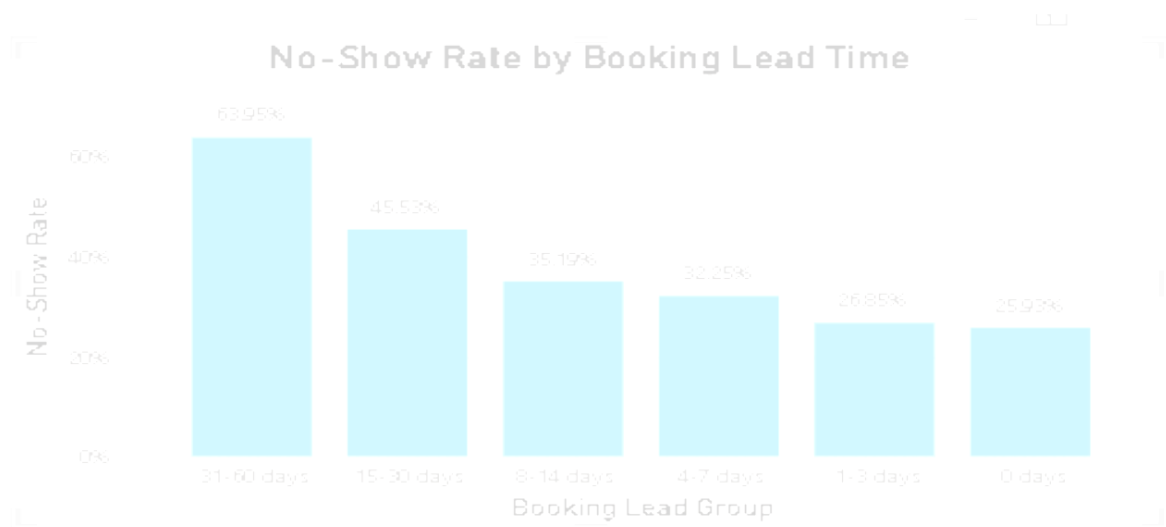


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Chapter 1

Week 5 → Week 6 Transition

1.1 Introduction

Week 5 established the initial analytical foundation for the HealthConnect Clinic project by preparing and validating the appointment dataset, conducting exploratory analysis, developing key performance indicators (KPIs), identifying important attendance patterns, and producing an initial analytical dashboard.

The Week 5 analysis established an overall **No-Show Rate of 51.15%** among 4,737 eligible appointments, with **2,423 no-shows** and **2,314 attended appointments**. Booking lead time and previous no-show history emerged as the most important observed indicators of attendance risk, while reminder status, distance to the clinic, appointment type, day, and time provided additional segmentation context.

Week 6 was designed to build directly on these findings rather than repeat the Week 5 analysis. The objective was to investigate the most important patterns more deeply, determine whether they remained meaningful when variables were examined jointly, validate and refine the KPI framework, improve the analytical dashboard, and translate the strongest findings into practical decision-support information.

This progression is consistent with the Week 6 requirement to advance the Week 5 work through integration, advanced development, validation, and evidence-based refinement.

1.2 What Was Established in Week 5

The Week 5 analysis established several important findings:

- The overall no-show rate was **51.15%** among eligible appointments.
- The attendance rate was **48.85%**.

- Appointments booked **31–60 days in advance** had the highest observed no-show rate among the booking-lead groups at **63.95%**.
- Patients with at least one previous no-show had an observed no-show rate of **57.83%**, compared with **46.30%** among patients with no previous no-shows.
- Appointments with a recorded reminder had an observed no-show rate of **49.86%**, compared with **54.63%** where no reminder was recorded.
- No-show rates generally increased across greater distance-to-clinic groups.
- Follow-up appointments had the highest observed no-show rate among the major appointment types at **54.17%**.
- Day-of-week and time-of-day differences were comparatively modest.

These findings provided the basis for selecting the highest-priority areas for deeper Week 6 investigation.

1.3 Why Further Analysis Was Required

The Week 5 findings identified individual patterns, but individual-variable comparisons did not fully establish whether the strongest indicators remained important when considered together.

For example, booking lead time and previous no-show history were both associated with higher observed no-show rates. Week 6 therefore investigated whether their combination identified a more specific high-risk appointment segment.

Similarly, the overall reminder comparison showed a difference in observed no-show rates, but this did not establish whether the relationship was consistent across different booking-lead groups or patient-history groups. Week 6 therefore examined reminder status within relevant segments rather than treating the overall difference as evidence of effectiveness.

Additional segmentation by distance and appointment type was also performed within the identified high-risk segment to determine whether these variables provided useful secondary prioritization information.

1.4 Week 6 Analytical Objectives

The Week 6 Data Analytics work therefore focused on five main objectives:

- ✓ **Deepen the most important Week 5 findings** through additional segmentation and interaction analysis.
- ✓ **Validate the Week 5 KPIs** and determine whether any required refinement.
- ✓ **Identify the highest-impact appointment segments** for decision support.
- ✓ **Improve the analytical dashboard** so that it supports prioritization rather than only descriptive monitoring.
- ✓ **Provide validated analytical inputs for cross-track integration**, particularly for Data Science modelling decisions.

1.5 Week 5 → Week 6 Analytical Progression

Table 1.1 — Week 5 → Week 6 Transition

Week 5	Week 6 Development
Identified booking lead time as an important pattern	Tested booking lead time jointly with previous no-show history and reminder status
Identified previous no-show history as an important indicator	Validated its importance across booking-lead segments
Observed a reminder/no-reminder difference	Tested the association across multiple segments
Examined distance generally	Investigated distance within the highest-risk segment
Compared appointment types	Evaluated appointment type as a secondary differentiator within the high-risk segment
Developed initial KPIs	Validated and refined the KPI framework
Produced an initial dashboard	Upgraded the dashboard toward decision support
Produced broad business findings	Prioritized findings according to magnitude, consistency, and actionability
Identified potential risk indicators	Defined candidate analytical inputs for Data Science

The analytical progression can be summarized as follows:

1.6 Scope of the Week 6 Analysis

The deeper analysis focused primarily on:

- **Booking lead time**
- **Previous no-show history**
- **Reminder status**
- **Distance to clinic**
- **Appointment type**
- **Interactions between these variables**

The analysis continued to use **eligible appointments**, defined as appointments with an outcome of either **Attended** or **No-Show**, for primary attendance and no-show calculations. Cancelled appointments were excluded from these denominators.

This maintained methodological consistency with Week 5 while allowing the Week 6 analysis to focus on attendance behavior rather than cancellation behavior.

1.7 Week 6 Analytical Direction

The central analytical question evolved from identifying broad patterns to identifying actionable combinations of characteristics:

Which appointment and patient characteristics, when considered together, identify the highest observed attendance-risk segments that HealthConnect Clinic should prioritize for further support and testing?

This change in analytical direction was important because the objective of Week 6 was not simply to produce more charts. It was to determine whether the strongest Week 5 findings could be combined into more useful decision-support indicators.

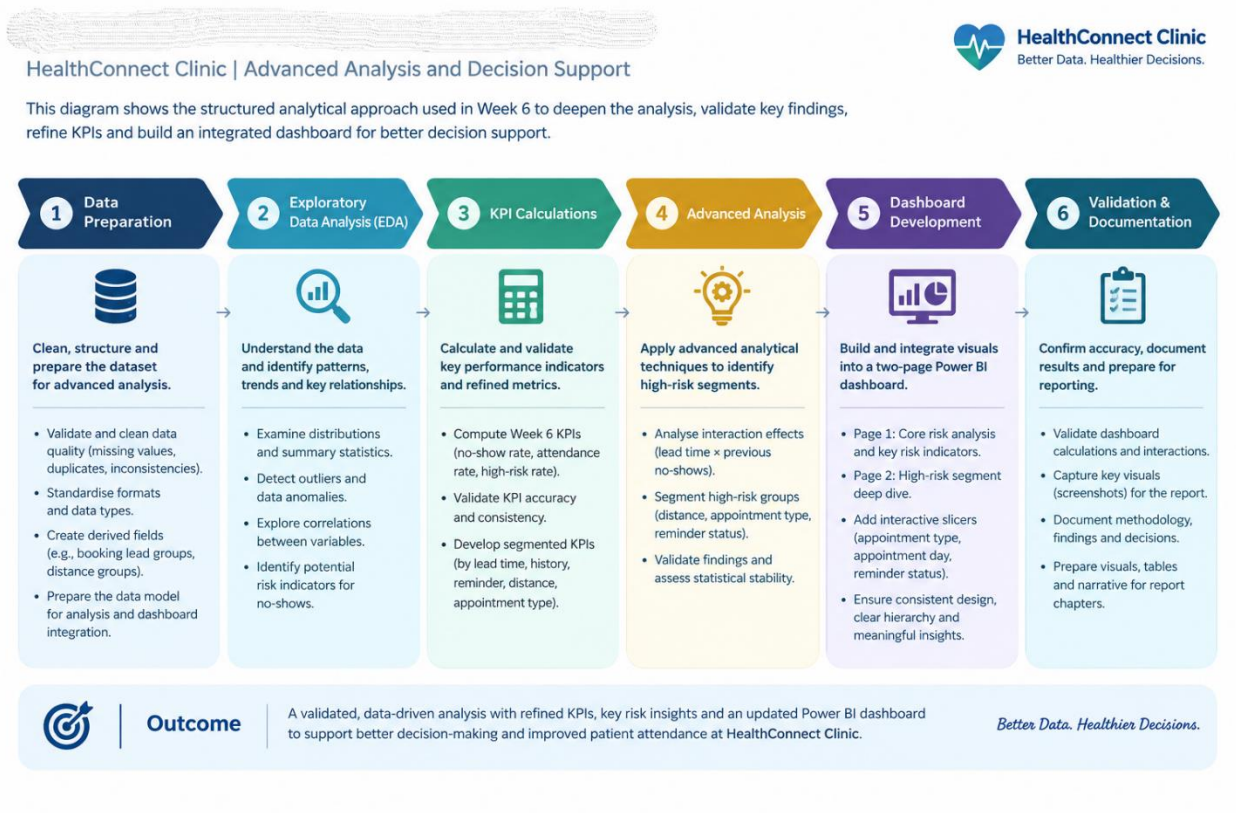
The subsequent analysis identified a high-risk segment consisting of appointments booked **31–60 days in advance for patients with at least one previous no-show**. This segment had an observed no-show rate of **70.52%**, substantially above the overall eligible-appointment no-show rate of **51.15%**.

This finding became the fundamental focus of the Week 6 decision-support analysis.

Chapter 2

Advanced Analytical Approach

Figure 2.1 — Week 6 Analytical Approach



2.1 Purpose of the Advanced Analysis

The purpose of the Week 6 analysis was to determine whether the most important Week 5 findings remained meaningful when relevant variables were examined jointly and across more specific appointment segments.

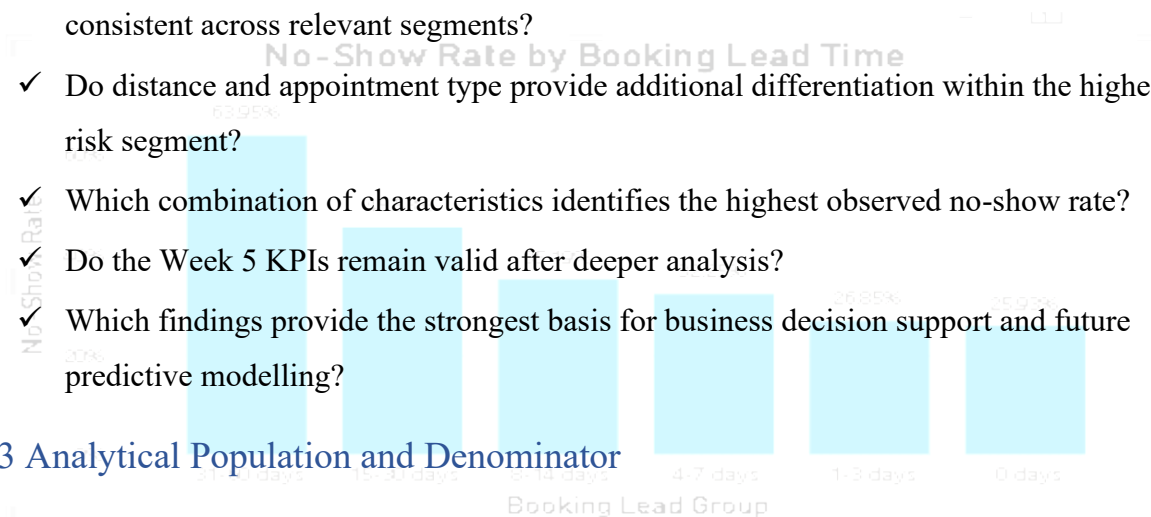
Rather than repeating the full exploratory analysis conducted in Week 5, the analysis focused on the strongest observed indicators of missed appointments and investigated their interactions, consistency, and practical decision-support value.

The analysis therefore moved from identifying individual patterns to examining whether combinations of appointment and patient characteristics could identify higher-risk segments.

2.2 Analytical Questions

The Week 6 analysis was guided by the following questions:

- ✓ Does the relationship between booking lead time and no-show behavior remain meaningful when previous no-show history is considered?
- ✓ Does previous no-show history remain an important differentiator across different booking-lead groups?
- ✓ Does the observed association between reminder status and no-show behavior remain consistent across relevant segments?
- ✓ Do distance and appointment type provide additional differentiation within the highest-risk segment?
- ✓ Which combination of characteristics identifies the highest observed no-show rate?
- ✓ Do the Week 5 KPIs remain valid after deeper analysis?
- ✓ Which findings provide the strongest basis for business decision support and future predictive modelling?



2.3 Analytical Population and Denominator

The primary analysis continued to use **eligible appointments**, defined as appointments with an outcome of either:

- **Attended**
- **No-Show**

This produced an analytical population of **4,737 eligible appointments**.

Cancelled appointments were excluded from the primary no-show and attendance calculations because a cancellation does not represent the same attendance outcome as a missed appointment.

The primary denominator therefore remained:

Eligible Appointments = 4,737

This ensured consistency between the Week 5 and Week 6 analyses.

2.4 Selection of Variables for Deeper Investigation

The variables selected for deeper investigation were based primarily on the strength and business relevance of the Week 5 findings.

Primary analytical variables

Booking lead time

Week 5 identified a substantial increase in observed no-show rates as booking lead time increased, with the **31–60-day group reaching 63.95%**.

Previous no-show history

Patients with at least one previous no-show had an observed no-show rate of **57.83%**, compared with **46.30%** among patients with no previous no-shows.

Reminder status

Appointments with a recorded reminder had an observed no-show rate of **49.86%**, compared with **54.63%** where no reminder was recorded.

Secondary analytical variables

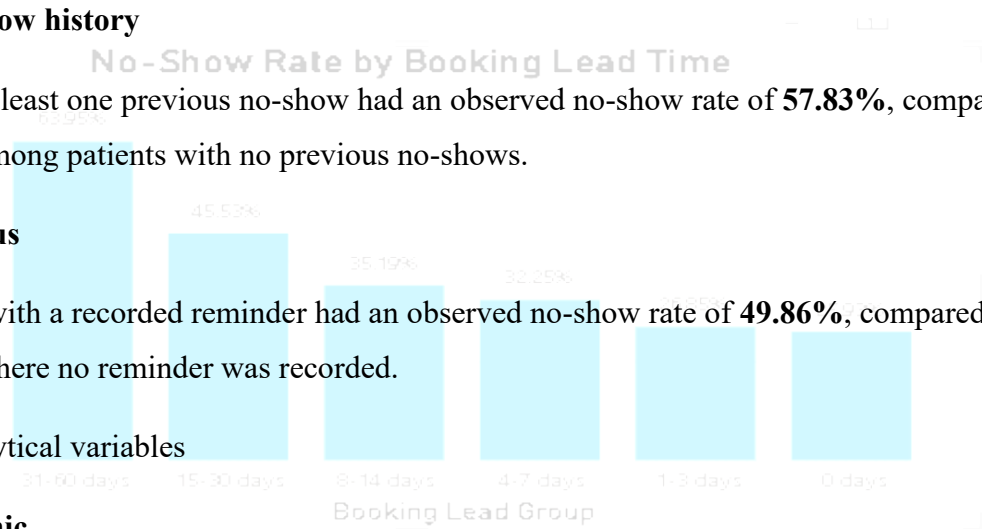
Distance to clinic

Distance was investigated as an additional segmentation factor, particularly within the highest-risk segment.

Appointment type

Appointment type was examined to determine whether it provided meaningful additional differentiation after the strongest risk characteristics had already been identified.

These variables were selected because they offered the greatest potential to explain differences in observed attendance behavior and support practical prioritization.



2.5 Segmentation Strategy

The Week 6 analysis used progressive segmentation.

The first stage examined the strongest individual indicators.

The second stage examined two-variable relationships, including:

- Booking lead time × previous no-show history
- Booking lead time × reminder status
- Previous no-show history × reminder status

The third stage examined the interaction of:

- Booking lead time
- Previous no-show history
- Reminder status

The final stage examined additional characteristics within the identified high-risk segment:

- Distance to clinic
- Appointment type

This progressive approach allowed the analysis to move from broad patterns toward increasingly specific decision-support segments without immediately creating excessively granular combinations.

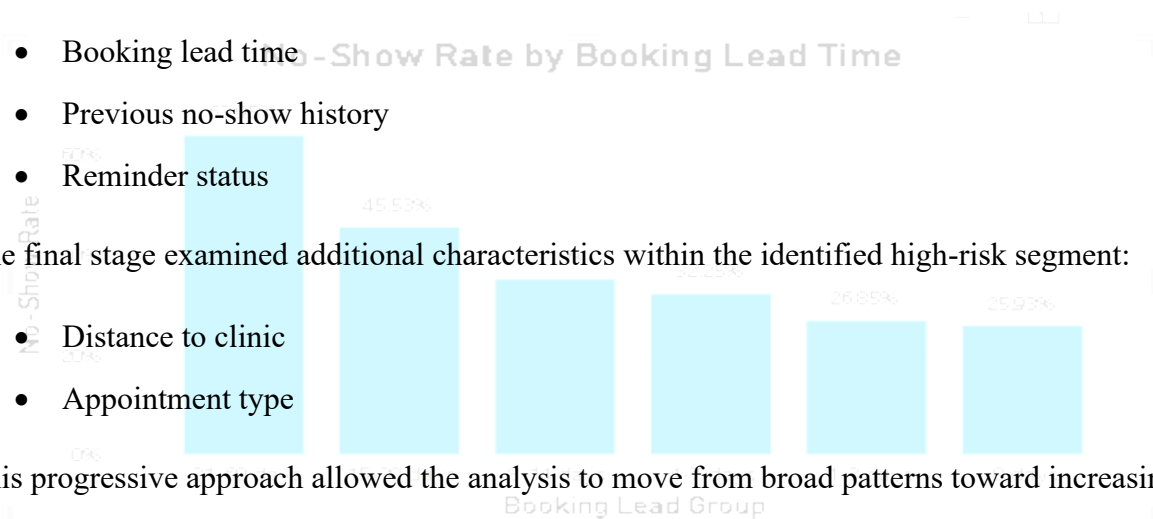
2.6 High-Risk Segment Definition

The deeper analysis identified a particularly important segment consisting of:

Appointments booked 31–60 days in advance for patients with at least one previous no-show.

This segment had an observed no-show rate of:

70.52%



This became the principal **High-Risk Segment No-Show Rate** used in Week 6 decision-support analysis.

The segment should be interpreted as an analytical targeting indicator rather than a causal classification. Its purpose is to identify appointments with a substantially higher observed no-show rate for further investigation and potential attendance-support prioritization.

2.7 Reminder Analysis Approach

Reminder status was treated as an **observational variable**.

The analysis compared observed no-show rates between appointments where a reminder was recorded and those where no reminder was recorded.

However, the analysis did not interpret the difference as evidence that reminders caused lower no-show rates.

This distinction was necessary because the available dataset does not establish:

- the exact timing of the reminder;
- the reminder content;
- whether the reminder was opened or acknowledged;
- whether reminders were targeted to particular patient groups; or
- whether other operational factors influenced who received a reminder.

The Week 6 analysis therefore evaluated **association and consistency**, rather than intervention effectiveness.

2.8 KPI Validation Approach

The Week 5 KPI framework was reviewed against the deeper analytical results.

Validation considered:

- whether the KPI definitions remained appropriate;
- whether the denominators were consistent;
- whether the results reconciled to the underlying appointment outcomes;

- whether deeper segmentation supported or contradicted the original findings;
- whether any KPI required a change in interpretation; and
- whether an additional decision-support indicator was warranted.

This process resulted in the retention of the core Week 5 KPIs while adding a more targeted high-risk segment indicator.

2.9 Prioritization Framework

Not every observed difference was treated as equally important.

Week 6 findings were prioritized using three criteria:

Magnitude

How large is the observed difference or no-show rate?

Consistency

Does the pattern remain visible across relevant segments rather than appearing in only one small subgroup?

Actionability

Can the finding reasonably inform a business decision, attendance-support strategy, dashboard indicator, or future modelling activity?

This framework helped prevent small or unstable subgroup differences from being treated as major business findings.

2.10 Sample-Size Considerations

Observed rates were interpreted together with the number of eligible appointments in each segment.

This was particularly important for smaller groups. For example, the **>30 km** distance category within the high-risk segment contained only **16 eligible appointments**.

Although its observed no-show rate was **75.00%**, the small sample means that this result should not be treated as strong standalone evidence.

Consequently, Week 6 prioritized patterns supported by larger groups and repeated across multiple relevant segments.

2.11 Analytical Interpretation Principles

The following principles were applied throughout the analysis:

- ✓ **Association was not interpreted as causation.**
- ✓ **Cancellations were excluded from primary attendance/no-show denominators.**
- ✓ **Small subgroups were interpreted cautiously.**
- ✓ **Observed rates were considered alongside sample size.**
- ✓ **Reminder status was treated as an association measure rather than an effectiveness measure.**
- ✓ **Waiting time was not used as a pre-appointment no-show predictor because its measurement timing was not established.**
- ✓ **Appointment outcome was treated as the outcome variable and not as a predictor.**
- ✓ **Potential data leakage was considered when identifying variables for future predictive modelling.**
- ✓ **Over-segmentation was avoided where it could produce unstable or operationally impractical results.**

2.12 Relationship to Future Data Science Modelling

The advanced analysis also provided a stronger analytical basis for potential predictive modelling.

Candidate variables identified for further Data Science consideration included:

- Previous no-show history
- Previous appointment history
- Booking lead time

- Distance to clinic
- Appointment type
- Reminder status, subject to timing validation
- Appointment day and time
- Relevant interaction between booking lead time and previous no-show history

The analysis particularly supports investigation of the interaction between **booking lead time** and **previous no-show history**, because their combination identified a substantially higher-risk observed segment.

However, variables should only be used in predictive modelling after their timing and potential leakage have been validated.

For example, appointment_outcome must not be used as a predictor, while waiting_time_minutes should not be used as a pre-appointment predictor unless its measurement timing is confirmed to occur before the prediction point.

2.13 Summary of the Analytical Approach

The Week 6 methodology can therefore be summarized as:

Week 5 findings → Variable prioritization → Segment analysis → Interaction analysis → High-risk segment identification → KPI validation → Decision-support prioritization → Data Science analytical inputs → Week 7 testing requirements

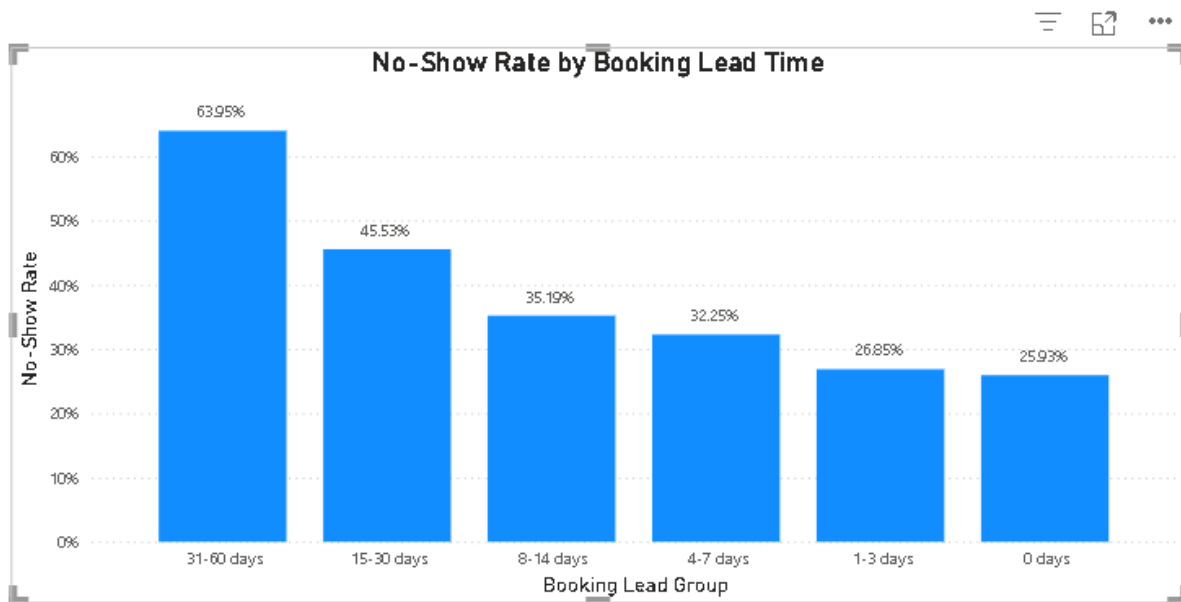
This approach ensured that Week 6 extended the existing HealthConnect analytical work rather than duplicating the Week 5 exploratory analysis.

The resulting analysis provided a stronger evidence base for identifying high-risk appointment segments, refining the dashboard, supporting business decisions, and preparing the analytical outputs for the testing and validation stage in Week 7.

Chapter 3

Advanced Analysis of Booking Lead Time

Figure 3.1 — No-Show Rate by Booking Lead Time



3.1 Purpose of the Analysis

Week 5 identified booking lead time as one of the strongest observed indicators of appointment non-attendance. The no-show rate increased substantially as the interval between booking and appointment increased, reaching **63.95%** for appointments booked 31–60 days in advance compared with **25.93%** for same-day bookings.

Week 6 extended this finding by examining booking lead time jointly with previous no-show history and reminder status. The purpose was to determine whether the booking-lead pattern remained meaningful after considering other relevant characteristics and whether it could help identify specific high-risk appointment segments.

3.2 Booking Lead Time and Previous No-Show History

The first deeper analysis examined booking lead time against previous no-show history.

Table 3.1 —booking lead time against previous no-show history.

Booking Lead Time	No Previous No-Shows	1+ Previous No-Shows
0 days	18.75%	36.36%
1–3 days	17.74%	39.13%
4–7 days	27.32%	39.52%
8–14 days	33.06%	38.79%
15–30 days	41.09%	51.27%
31–60 days	59.07%	70.52%

No observations were available in the 61+ day category.

3.2.1 Key Observation

The booking-lead pattern remains evident after segmentation by previous no-show history.

For patients with no previous no-shows, the observed no-show rate increased from **18.75%** for same-day appointments to **59.07%** for appointments booked 31–60 days in advance.

For patients with one or more previous no-shows, the observed rate increased from **36.36%** for same-day appointments to **70.52%** for appointments booked 31–60 days in advance.

The highest observed rate was therefore found at the intersection of the two strongest Week 5 indicators:

31–60-day booking lead time + 1 or more previous no-shows = 70.52% no-show rate.

Within the 31–60-day booking group, patients with previous no-shows had an observed no-show rate of **70.52%**, compared with **59.07%** for those without previous no-shows. This represents an observed difference of **11.45 percentage points**.

3.3 Interpretation

The Week 6 analysis strengthens the Week 5 conclusion that booking lead time is an important attendance-risk indicator.

More importantly, the deeper analysis demonstrates that booking lead time does not operate in isolation. Previous no-show history further differentiates the observed attendance behavior within the longer booking-lead groups.

The findings therefore suggest that:

- Long booking lead times are associated with higher observed no-show rates.
- Previous no-show history is associated with higher observed no-show rates.
- The combination of the two characteristics identifies a substantially higher-risk segment.
- The highest observed rate occurs among patients with previous no-shows whose appointments were booked 31–60 days in advance.

These results represent **observed associations within the HealthConnect dataset** and should not be interpreted as evidence that longer booking lead time or previous no-show history directly causes missed appointments.

3.4 Booking Lead Time and Reminder Status

The second analysis examined whether the observed relationship between reminder status and no-show behavior varied across booking-lead groups.

Table 3.2— Booking Lead Time and Reminder Status

Booking Lead Time	Reminder Sent	No Reminder	Difference
0 days	20.69%	39.13%	+18.44 pp
1–3 days	26.92%	26.67%	–0.26 pp
4–7 days	32.17%	32.47%	+0.29 pp
8–14 days	34.43%	37.33%	+2.90 pp
15–30 days	43.91%	49.72%	+5.81 pp
31–60 days	62.58%	67.63%	+5.05 pp

The analysis shows that the reminder association is **not uniform across booking-lead groups**.

The largest observed differences occur in:

- **15–30 days:** 5.81 percentage points lower with a recorded reminder.

- **31–60 days:** 5.05 percentage points lower with a recorded reminder.

By contrast, the differences in the 1–3-day and 4–7-day groups are close to zero.

The same-day difference of 18.44 percentage points should also be interpreted cautiously because the no-reminder group contains only **23 eligible appointments**.

3.5 Implications for Decision Support

The deeper booking-lead analysis changes the practical interpretation of the Week 5 findings.

The Week 5 analysis showed that booking lead time was associated with no-show behavior. Week 6 demonstrates that the strongest signal becomes more useful when combined with previous no-show history.

This suggests that HealthConnect could consider prioritizing attendance-support attention toward:

- ✓ Appointments booked substantially in advance.
- ✓ Patients with a history of previous no-shows.
- ✓ Particularly, appointments combining both characteristics.

The analysis also indicates that reminder status may provide additional segmentation information, but it should not be treated as a standalone intervention-effectiveness measure.

3.6 Data Science Implication

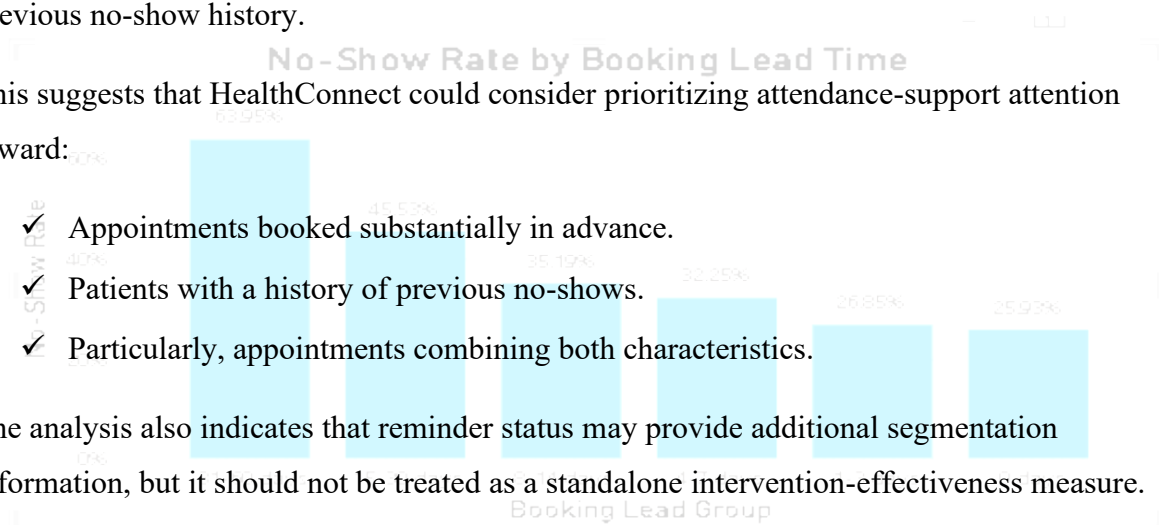
The interaction between booking lead time and previous no-show history provides a potentially important analytical input for the Data Science track.

A future predictive model could investigate whether the combination of:

Booking Lead Time × Previous No-Show History

provides additional predictive value beyond either variable individually.

This should be tested empirically rather than assumed. Feature timing and potential leakage should also be assessed before these variables are incorporated into a production-oriented prediction workflow.



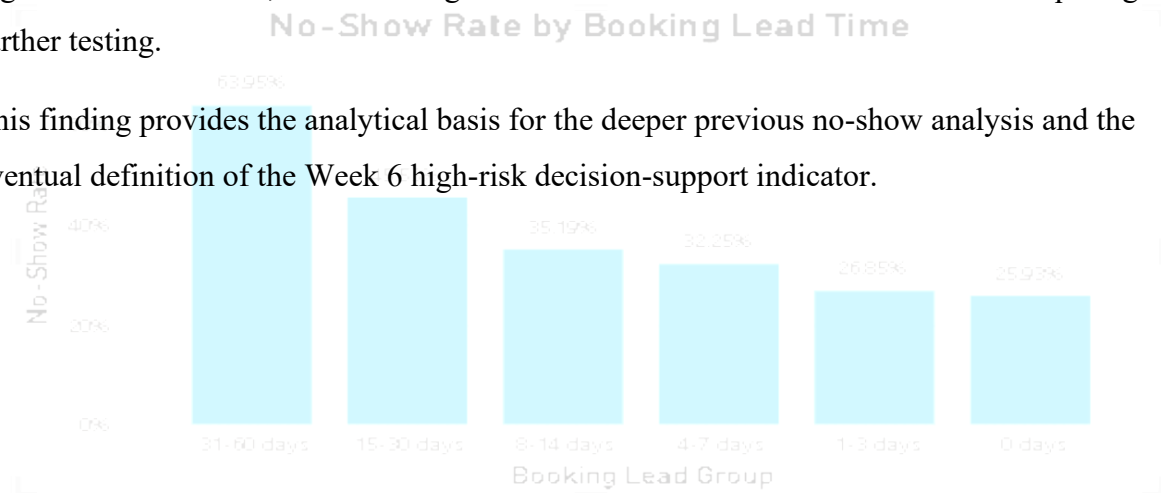
3.7 Chapter Conclusion

The Week 6 analysis validates and strengthens the Week 5 booking-lead finding.

The most important result is that the relationship between booking lead time and no-show behavior remains visible after accounting for previous no-show history. The combination of **31–60-day booking lead time and one or more previous no-shows** identifies an observed no-show rate of **70.52%**, making it the most important high-risk segment identified during the Week 6 analysis.

Reminder status shows a more modest and variable association across booking-lead groups. The evidence therefore supports using booking lead time and previous no-show history as primary segmentation variables, while treating reminder status as a contextual association requiring further testing.

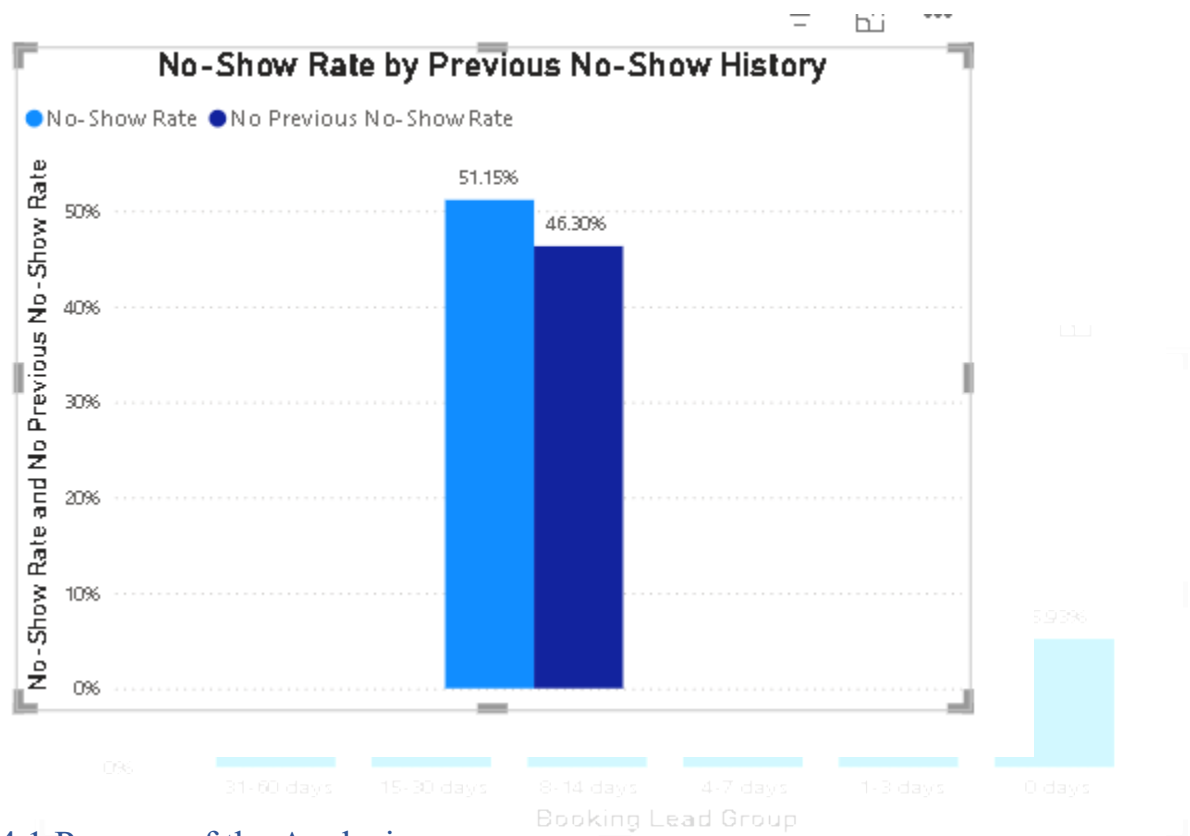
This finding provides the analytical basis for the deeper previous no-show analysis and the eventual definition of the Week 6 high-risk decision-support indicator.



Chapter 4

Advanced Analysis of Previous No-Show History

Figure 4.1 — No-Show Rate by Previous No-Show History



4.1 Purpose of the Analysis

Week 5 identified previous no-show history as one of the strongest observed indicators of future non-attendance. Patients with at least one previous no-show had an observed no-show rate of **57.83%**, compared with **46.30%** among patients with no previous no-shows.

Week 6 examined this relationship in greater depth to determine whether the difference remained meaningful across booking lead-time groups and reminder-status groups.

The purpose was also to determine whether previous no-show history contributed additional decision-support value when combined with other important appointment characteristics.

4.2 Previous No-Show History as a Standalone Indicator

The Week 5 comparison established:

Patient History	Eligible Appointments	No-Show Rate
No previous no-shows	2,745	46.30%
1+ previous no-shows	1,992	57.83%

The difference is:

$$57.83\% - 46.30\% = 11.53 \text{ percentage points}$$

This makes previous no-show history one of the strongest individual differentiators identified in the HealthConnect analysis.

The Week 6 analysis retained this finding and investigated whether it remained evident when booking lead time and reminder status were considered.

4.3 Previous No-Show History × Booking Lead Time

The interaction between previous no-show history and booking lead time produced the following results:

Table 4.1— Previous No-Show History × Booking Lead Time

Booking Lead Time	No Previous No-Shows	1+ Previous No-Shows
0 days	18.75%	36.36%
1–3 days	17.74%	39.13%
4–7 days	27.32%	39.52%
8–14 days	33.06%	38.79%
15–30 days	41.09%	51.27%
31–60 days	59.07%	70.52%

The difference associated with previous no-show history is visible across every booking-lead group shown in the analysis.

The largest difference occurs in the **31–60-day group**:

- No previous no-shows: **59.07%**
- 1+ previous no-shows: **70.52%**
- Difference: **11.45 percentage points**

This demonstrates that previous no-show history continues to differentiate attendance behavior even within the highest booking-lead group.

4.4 Previous No-Show History × Reminder Status

The next analysis examined whether the observed reminder association differed according to previous no-show history.

Table 4.2— Previous No-Show History × Reminder Status

Previous No-Show History	Reminder Status	Eligible Appointments	No-Shows	No-Show Rate
No previous no-shows	Reminder sent	2,013	912	45.31%
No previous no-shows	No reminder	732	359	49.04%
1+ previous no-shows	Reminder sent	1,439	809	56.22%
1+ previous no-shows	No reminder	553	343	62.03%

4.4.1 Patients with No Previous No-Shows

Among patients with no previous no-shows:

- Reminder sent: **45.31%**
- No reminder: **49.04%**
- Observed difference: **3.74 percentage points**

4.4.2 Patients with Previous No-Shows

Among patients with at least one previous no-show:

- Reminder sent: **56.22%**

- No reminder: **62.03%**
- Observed difference: **5.81 percentage points**

The reminder/no-reminder difference is therefore present in both groups, but it is larger among patients who have previously missed appointments.

However, the result remains observational. The data does not establish that receiving a reminder caused the lower observed no-show rate.

4.5 Relative Importance of Previous No-Show History

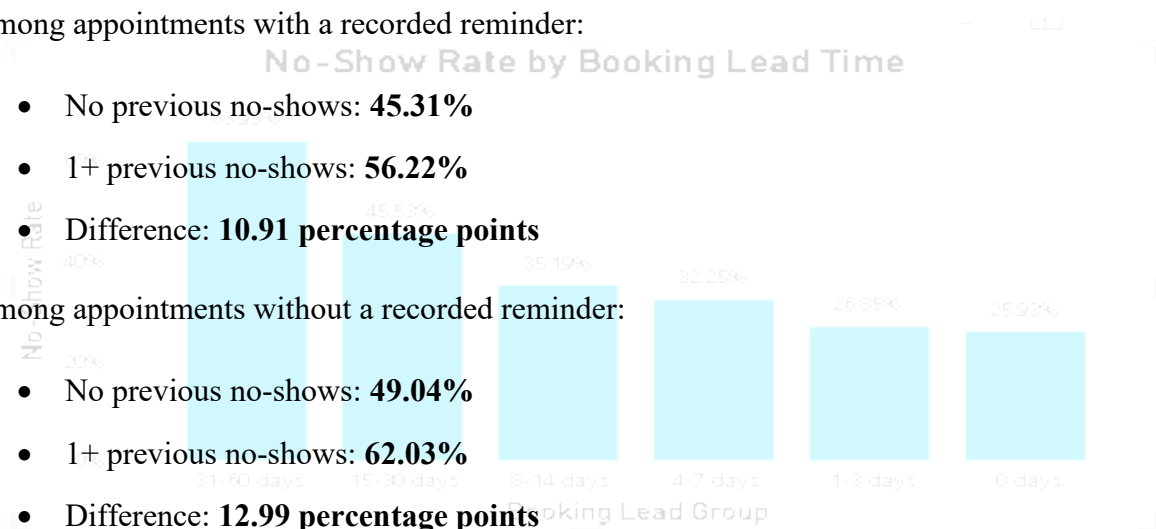
The results provide an important distinction between the two variables.

Among appointments with a recorded reminder:

- No previous no-shows: **45.31%**
- 1+ previous no-shows: **56.22%**
- Difference: **10.91 percentage points**

Among appointments without a recorded reminder:

- No previous no-shows: **49.04%**
- 1+ previous no-shows: **62.03%**
- Difference: **12.99 percentage points**



This indicates that previous no-show history remains a substantial differentiator regardless of reminder status.

Therefore, the Week 6 evidence suggests that **previous no-show history is a stronger and more consistent attendance-risk indicator than reminder status alone.**

4.6 Combined Interpretation

The deeper analysis provides three important observations.

First: Previous no-show history remains important

The 11.53-percentage-point overall difference identified in Week 5 remains meaningful when the data is segmented by other important characteristics.

Second: Booking lead time strengthens the risk signal

The highest observed rate occurs when previous no-show history and long booking lead time overlap.

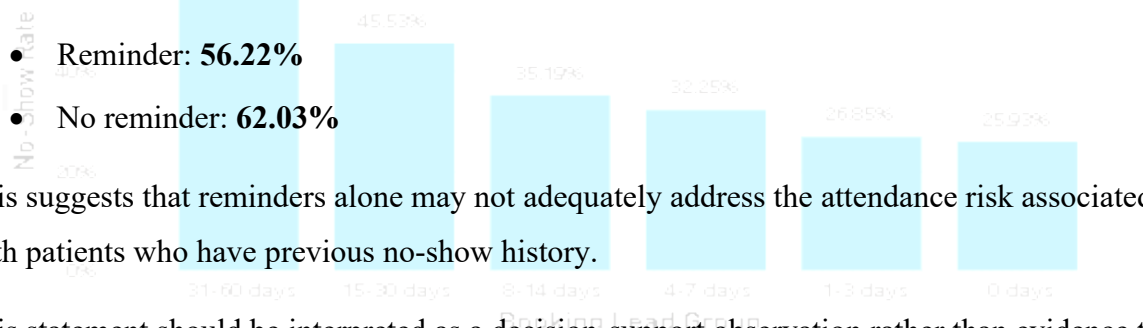
The combination of:

1+ previous no-shows + 31–60-day booking lead

produces a **70.52% observed no-show rate**.

Third: Reminder status does not eliminate the risk

Within the high-risk patient-history group, the observed no-show rate remains high whether a reminder was recorded or not:



This suggests that reminders alone may not adequately address the attendance risk associated with patients who have previous no-show history.

This statement should be interpreted as a decision-support observation rather than evidence that reminders are ineffective or that alternative interventions will necessarily reduce no-shows.

4.7 Implications for HealthConnect

Previous no-show history should therefore be considered a priority variable when HealthConnect evaluates appointment attendance risk.

A practical decision-support approach could prioritize appointments involving patients with previous no-shows, particularly where other high-risk characteristics are also present.

The strongest priority emerges when previous no-show history is combined with a **31–60-day booking lead time**.

This supports a move away from treating all appointments uniformly toward a more targeted attendance-support approach.

Potential operational responses could include:

- earlier confirmation for selected high-risk appointments;
- closer monitoring of appointments booked substantially in advance;
- targeted attendance-support communication;
- additional confirmation closer to the appointment date; and
- further investigation through predictive modelling.

These should be treated as **candidate actions for testing**, rather than proven interventions based on the observational analysis.

4.8 Data Science Implication

Previous no-show history should be considered a candidate feature for future predictive modelling because it demonstrates a substantial and repeated association with appointment non-attendance.

The Week 6 analysis also supports investigating interactions between:

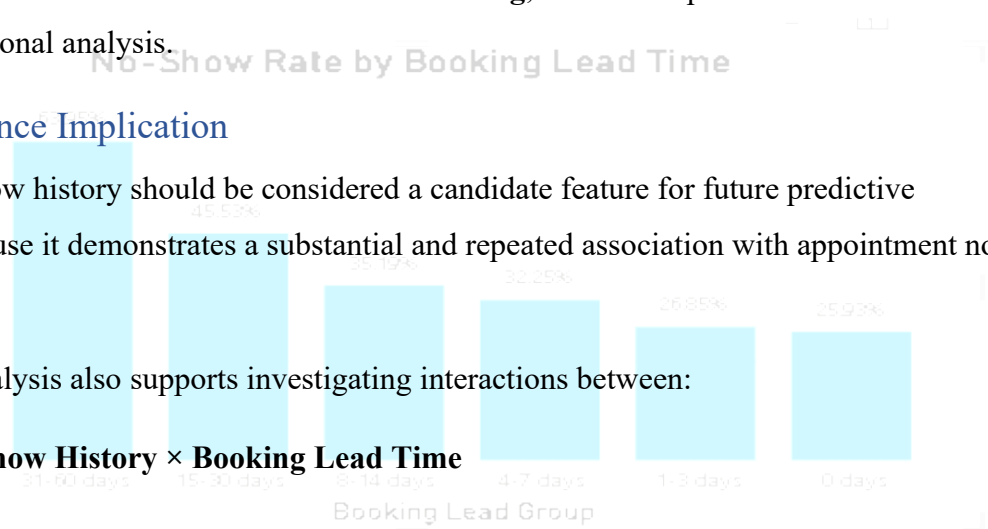
Previous No-Show History × Booking Lead Time

and potentially:

Previous No-Show History × Reminder Status

The value of these interactions should be determined through model testing rather than assumed from the descriptive analysis.

Feature timing should also be confirmed to ensure that all variables are available at the intended prediction point.



4.9 Analytical Limitation

The analysis identifies associations within historical appointment records. It does not establish why patients previously missed appointments or whether their previous behavior will necessarily continue in future appointments.

Similarly, the observed reminder differences cannot be interpreted as causal effects because the dataset does not provide sufficient information about reminder timing, content, delivery, patient response, or the operational rules determining which appointments received reminders.

These limitations should remain explicit in the final HealthConnect recommendations.

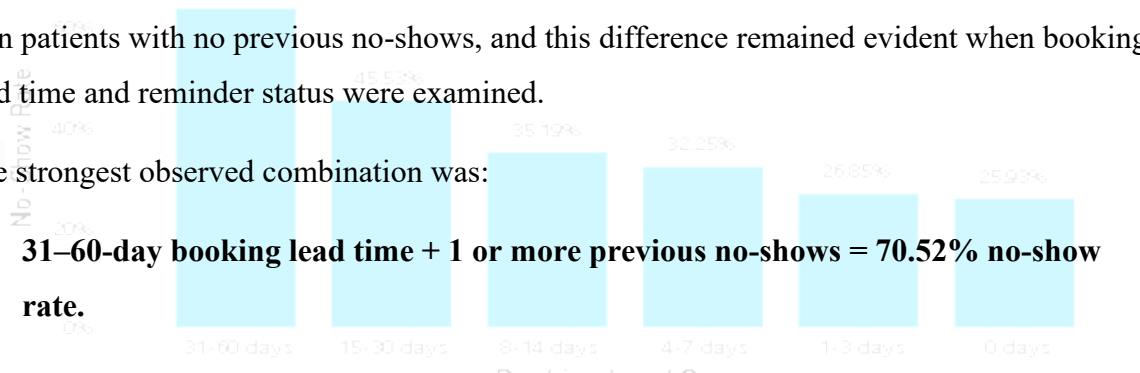
4.10 Chapter Conclusion

The Week 6 analysis validates previous no-show history as a major attendance-risk indicator.

Patients with one or more previous no-shows had a substantially higher observed no-show rate than patients with no previous no-shows, and this difference remained evident when booking lead time and reminder status were examined.

The strongest observed combination was:

31–60-day booking lead time + 1 or more previous no-shows = 70.52% no-show rate.



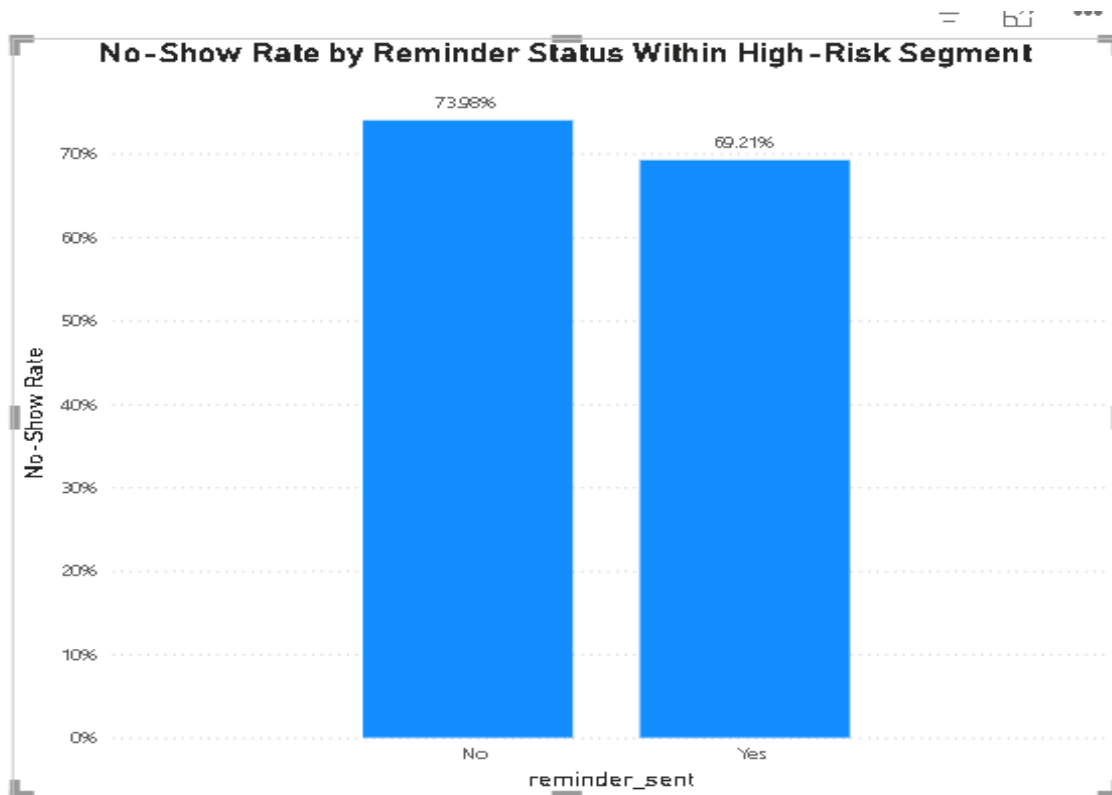
Reminder status showed a modestly lower observed no-show rate within both patient-history groups, but previous no-show history remained the stronger differentiator.

The evidence therefore supports using previous no-show history as a primary decision-support variable and combining it with booking lead time to identify appointments requiring closer analytical and operational attention.

Chapter 5

Advanced Reminder Analysis

Figure 5.1 — No-Show Rate by Reminder Status



5.1 Purpose of the Analysis

Week 5 identified a lower observed no-show rate among appointments for which a reminder was recorded. Among eligible appointments, the no-show rate was **49.86% when a reminder was sent**, compared with **54.63% when no reminder was recorded**, representing a difference of **4.77 percentage points**.

However, this overall comparison did not establish whether the relationship remained consistent across different appointment characteristics.

Week 6 therefore examined reminder status across booking lead-time groups, previous no-show history, and the identified high-risk segment. The objective was to determine whether reminder status provides consistent additional decision-support value and whether the Week 5 conclusion required refinement.

5.2 Week 5 Reminder Finding

The Week 5 results were:

Reminder Status	Eligible Appointments	No-Show Rate
Reminder sent	3,452	49.86%
No reminder	1,285	54.63%

The observed difference was:

54.63% – 49.86% = 4.77 percentage points

This indicated an association between reminder status and lower observed no-show rates.

Because the dataset does not establish the timing, content, delivery success, or patient response to reminders, this relationship was treated as **observational rather than causal**.

5.3 Reminder Status × Booking Lead Time

The Week 6 segmentation produced the following results:

Table 5.1 —Booking Lead Time and Reminder status Differences

Booking Lead Time	Reminder Sent	No Reminder	Difference
0 days	20.69%	39.13%	+18.44 pp
1–3 days	26.92%	26.67%	–0.26 pp
4–7 days	32.17%	32.47%	+0.29 pp
8–14 days	34.43%	37.33%	+2.90 pp
15–30 days	43.91%	49.72%	+5.81 pp
31–60 days	62.58%	67.63%	+5.05 pp

Positive values indicate a lower observed no-show rate among appointments where a reminder was recorded.

5.3.1 Interpretation

The relationship between reminder status and no-show rate is **not uniform across booking lead-time groups**.

The largest observed difference occurs among same-day appointments, at **18.44 percentage points**. However, this comparison should be interpreted cautiously because the no-reminder group contains only **23 eligible appointments**.

The differences in the **1–3-day** and **4–7-day** groups are close to zero, indicating little observed difference between reminder-status groups in those segments.

More consistent differences appear in the longer booking windows:

- **15–30 days:** 5.81 percentage points
- **31–60 days:** 5.05 percentage points

This provides stronger evidence that reminder status is associated with lower observed no-show rates in some longer-lead appointment groups, but not consistently across every booking window.

5.4 Reminder Status × Previous No-Show History

Table 5.2 —Table Reminder status and No-Show Rate

Previous No-Show History	Reminder Status	Eligible Appointments	No-Show Rate
No previous no-shows	Reminder sent	2,013	45.31%
No previous no-shows	No reminder	732	49.04%
1+ previous no-shows	Reminder sent	1,439	56.22%
1+ previous no-shows	No reminder	553	62.03%

Reminder status was also examined according to patient attendance history. Among patients with no previous no-shows, the observed difference was:

$$49.04\% - 45.31\% = 3.74 \text{ percentage points}$$

Among patients with one or more previous no-shows:

62.03% – 56.22% = 5.81 percentage points

Therefore, the reminder/no-reminder difference appears in both patient-history groups and is somewhat larger among patients with previous no-shows.

However, previous no-show history remains a stronger differentiator than reminder status.

For example, among appointments where a reminder was recorded:

- No previous no-shows: **45.31%**
- 1+ previous no-shows: **56.22%**
- Difference: **10.91 percentage points**

This reinforces the Week 6 finding that previous attendance behavior should remain a core risk indicator.

5.5 Reminder Status Within the High-Risk Segment

The most important Week 6 segment is defined as:

31–60-day booking lead time + 1 or more previous no-shows

This segment has an overall observed no-show rate of:

70.52%

Reminder status was then examined within this segment.

Reminder Status	Eligible Appointments	No-Shows	No-Show Rate
Reminder sent	708	—	69.21%
No reminder	269	—	73.98%

The observed difference is:

73.98% – 69.21% = 4.77 percentage points

This is important because the reminder association remains visible within the highest-risk segment.

However, the absolute no-show rates remain very high in both groups. Therefore, the evidence does not support treating reminder status as a sufficient solution for this segment.

The more appropriate interpretation is:

Reminder status is associated with a somewhat lower observed no-show rate within the high-risk segment, but the no-show rate remains high regardless of reminder status.

5.6 Three-Way Interpretation

The Week 6 analysis demonstrates that three variables can provide complementary information:

Booking lead time + previous no-show history + reminder status

The strongest observed subgroup is:

31–60 days + 1+ previous no-shows + no reminder = 73.98% no-show rate (n=269).

The corresponding subgroup where a reminder was recorded has:

69.21% no-show rate (n=708).

The difference is therefore **4.77 percentage points**.

This finding should not be interpreted as evidence that the absence of a reminder causes the higher no-show rate. It instead identifies a subgroup where attendance risk is particularly high and where reminder status provides additional descriptive segmentation.

5.7 Business Interpretation

The Week 6 evidence suggests that HealthConnect should avoid treating reminders as a standalone attendance-management strategy.

Instead, reminder status is more appropriately viewed as one component within a broader attendance-support framework.

The strongest decision-support signal comes from the combination of:

- ✓ **Previous no-show history**
- ✓ **Long booking lead time**

✓ Additional appointment characteristics such as distance

Reminder status can then be considered as an additional operational variable when prioritising appointments for attendance support.

This approach is more consistent with the evidence than simply assuming that sending reminders will reduce no-shows.

5.8 Implications for Attendance-Support Planning

Based on the observed relationships, HealthConnect could consider testing targeted attendance-support processes for appointments that combine multiple risk characteristics.

Potential actions include:

- earlier confirmation for appointments booked substantially in advance;
- additional confirmation closer to the appointment date;
- targeted follow-up for patients with previous no-show history;
- enhanced attendance-support communication for selected high-risk appointments; and
- monitoring whether these interventions produce measurable improvements during future testing.

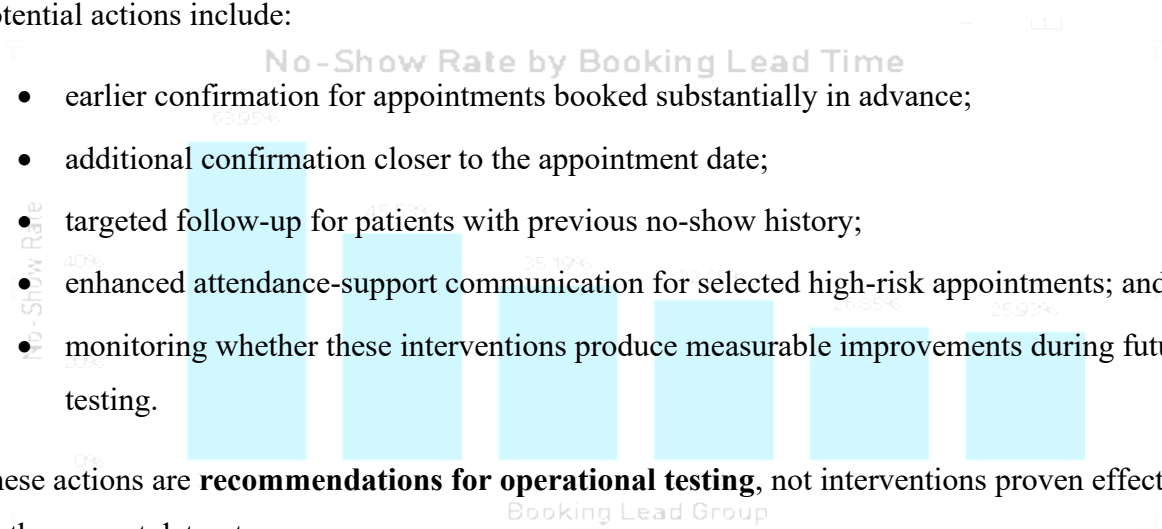
These actions are **recommendations for operational testing**, not interventions proven effective by the current dataset.

A controlled or longitudinal evaluation would be required to determine whether a particular reminder or support intervention actually reduces no-show rates.

5.9 Data Science Implications

The Week 6 reminder analysis provides useful information for the Data Science track.

Reminder status can be considered a candidate modelling feature, but its usefulness depends on the intended prediction point.



If the model is intended to predict no-show risk **at the time of booking**, the timing of the reminder variable must be investigated. A variable that becomes known only after booking or after other operational decisions could introduce information leakage.

The analysis also supports testing interactions involving:

- reminder_sent
- booking lead time
- previous no-show history
- the combined high-risk segment

The Data Science team should determine whether these variables improve predictive performance through formal model testing rather than relying on descriptive differences alone.

5.10 Analytical Limitations

Several limitations affect interpretation of the reminder analysis.

Reminder timing is unknown

The dataset records whether a reminder was sent but does not establish precisely when it was sent relative to the appointment.

Reminder content is unknown

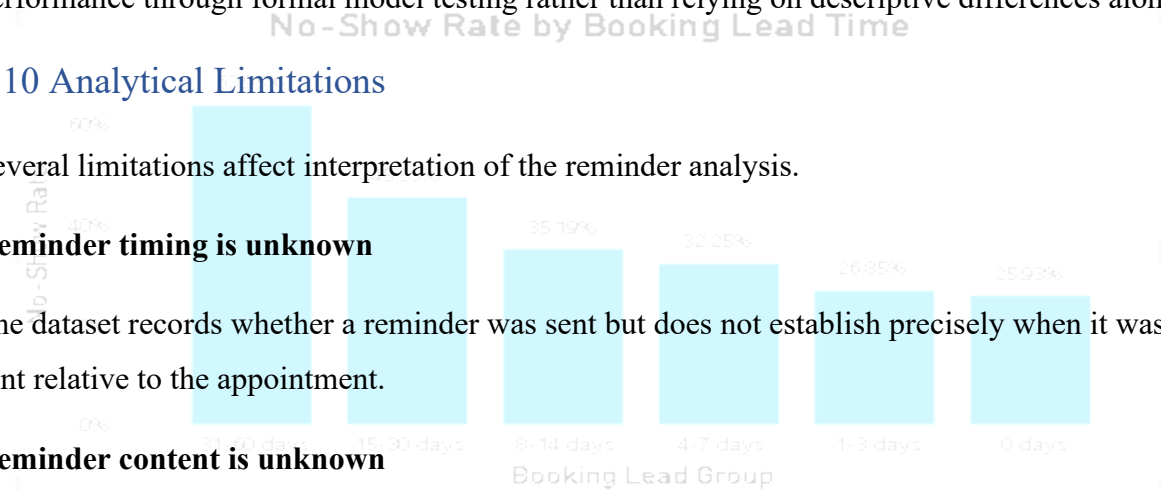
The analysis cannot determine whether SMS, Email, or WhatsApp messages differed in content, timing, frequency, or effectiveness of delivery.

Patient response is unknown

The data does not indicate whether patients received, opened, read, acknowledged, or acted on a reminder.

Operational assignment may not be random

Appointments receiving reminders may differ systematically from appointments that did not receive them. Therefore, observed differences cannot be interpreted as causal effects.



Small subgroup sizes

Some booking-lead/reminder combinations contain relatively few observations. The same-day no-reminder group, for example, contains only **23 eligible appointments**, making its 39.13% rate less stable than results from larger groups.

5.11 Refinement of the Week 5 Conclusion

The Week 5 conclusion that reminder status was associated with a lower observed no-show rate remains valid, but Week 6 provides a more precise interpretation.

The finding should therefore be refined from a simple reminder comparison to:

Reminder status is associated with lower observed no-show rates overall, but the association varies substantially across booking lead-time and patient-history groups. The difference is more consistent among appointments booked 15–60 days in advance and remains visible within the high-risk segment. However, no-show rates remain high even when reminders are recorded, and the observational design does not establish a causal effect.

This refinement prevents the dashboard and final report from presenting reminders as a proven intervention.

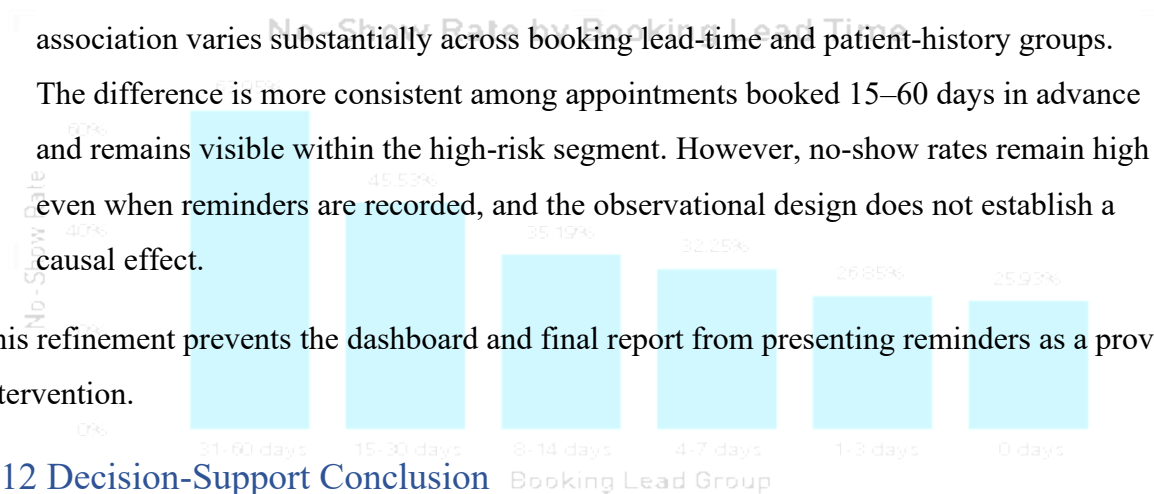
5.12 Decision-Support Conclusion

The Week 6 reminder analysis demonstrates that reminder status has **moderate decision-support value**, but it should not be treated as the primary attendance-risk indicator.

The strongest analytical signals remain:

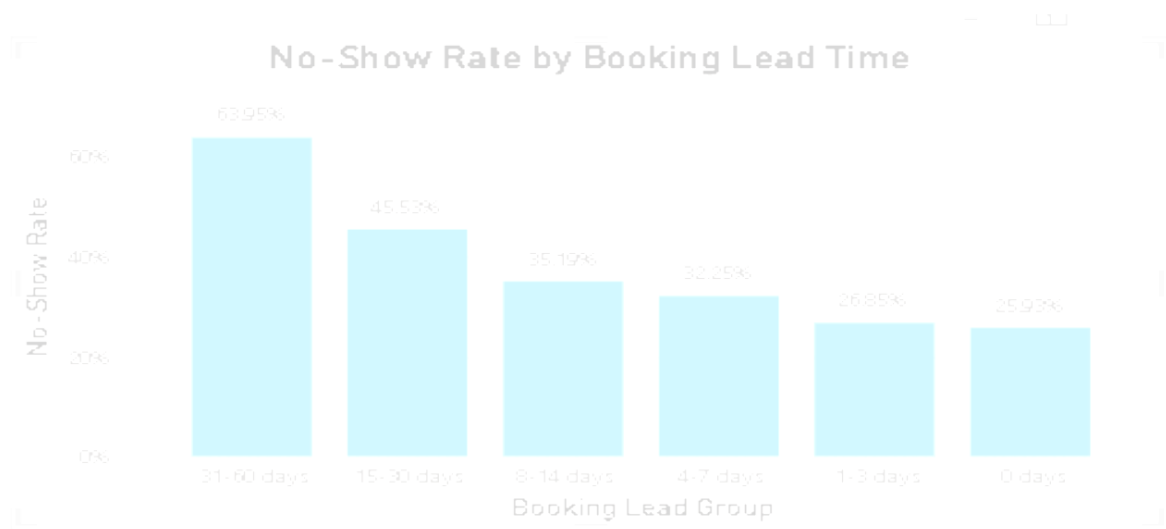
- previous no-show history;
- booking lead time; and
- their interaction.

Reminder status provides an additional operational dimension that can help describe and prioritize high-risk appointments, but it does not explain the full attendance-risk pattern.



The evidence therefore supports a **targeted, multi-factor attendance-support approach** rather than a reminder-only strategy.

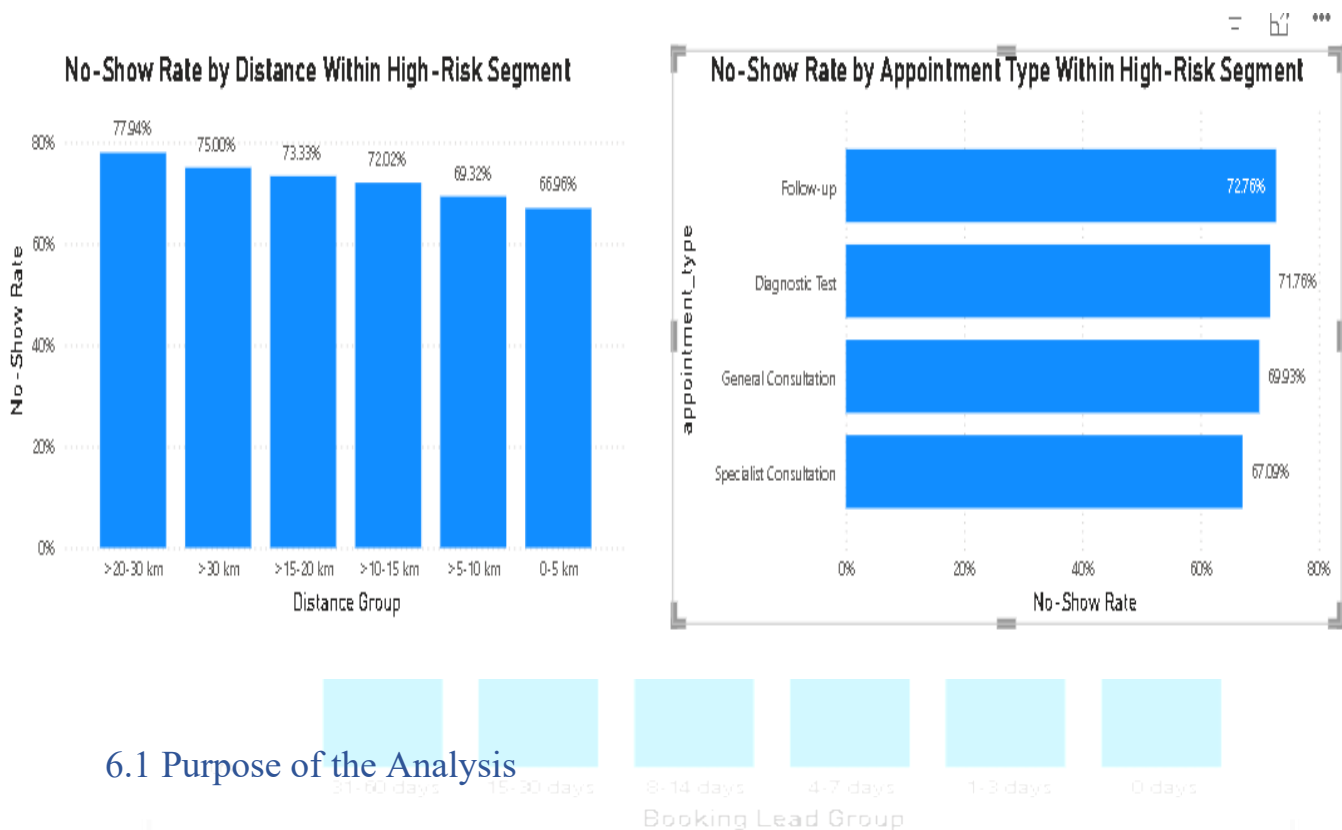
This conclusion will be incorporated into the refined KPI framework and the upgraded Week 6 dashboard.



Chapter 6

Advanced Operational Segmentation

Figure 6.1 —Distance and appointment-type visuals for the high-risk segment



6.1 Purpose of the Analysis

Week 6 moved beyond the identification of the primary high-risk segment to determine whether additional operational characteristics could further differentiate attendance risk.

Two variables were examined:

- **Distance to clinic**
- **Appointment type**

The analysis was conducted specifically within the previously identified high-risk segment:

Appointments booked 31–60 days in advance for patients with at least one previous no-show.

This segment had an overall observed no-show rate of **70.52%**.

The purpose of the additional segmentation was to determine whether distance and appointment type provide useful secondary information for decision support without displacing the stronger indicators already identified.

6.2 High-Risk Segment Definition

The high-risk segment used for this analysis combines two characteristics:

- ✓ Booking lead time of **31–60 days**
- ✓ **One or more previous no-shows**

The resulting no-show rate is:

70.52%

This segment was selected because the interaction between booking lead time and previous no-show history produced the strongest and most consistent high-risk pattern identified during Week 6.

Distance and appointment type were then examined within this segment.

6.3 High-Risk Segment × Distance to Clinic

The relationship between distance and attendance was examined using distance bands.

Table 6.1 — High-Risk Segment × Distance to Clinic

Distance to Clinic	Eligible Appointments	No-Shows	No-Show Rate
0–5 km	224	150	66.96%
>5–10 km	339	235	69.32%
>10–15 km	218	157	72.02%
>15–20 km	90	66	73.33%
>20–30 km	68	53	77.94%
>30 km	16	12	75.00%

6.3.1 Observed Pattern

The results show a generally increasing pattern in observed no-show rates as distance from the clinic increases.

Within the high-risk segment:

- **0–5 km:** 66.96%
- **>5–10 km:** 69.32%
- **>10–15 km:** 72.02%
- **>15–20 km:** 73.33%
- **>20–30 km:** 77.94%
- **>30 km:** 75.00%

The difference between the closest group and the **>20–30 km** group is approximately **11 percentage points**.

This provides additional evidence that distance may have secondary segmentation value within an already high-risk appointment group.

However, the **>30 km** category contains only **16 eligible appointments**. Its 75.00% rate should therefore not be interpreted as a stable estimate or as evidence that the **>30 km** group has greater risk than the **>20–30 km** group.

6.4 Interpretation of Distance

The distance analysis strengthens the Week 6 understanding of overlapping risk characteristics.

The strongest observed risk pattern is not simply associated with distance alone. Rather, higher distance appears to provide additional differentiation **within a segment that has already been identified as high risk because of booking lead time and previous no-show history**.

This distinction is important.

Distance should therefore be treated as a **secondary prioritization variable**, rather than as the primary basis for identifying attendance risk.

The results support further investigation of whether patients facing greater travel requirements may benefit from additional attendance-support processes.

However, the current analysis cannot establish that distance causes missed appointments.

Other factors may be associated with both distance and attendance behavior, and the dataset does not contain sufficient information to determine the underlying mechanism.

6.5 High-Risk Segment × Appointment Type

Appointment type was also examined within the same high-risk segment.

Table 6.2— High-Risk Segment × Appointment Type

Appointment Type	Eligible Appointments	No-Shows	No-Show Rate
Follow-up	279	203	72.76%
Diagnostic Test	131	94	71.76%
General Consultation	409	286	69.93%
Specialist Consultation	158	106	67.09%

6.5.1 Observed Pattern

Follow-up appointments have the highest observed no-show rate within the high-risk segment at **72.76%**, followed by Diagnostic Tests at **71.76%**.

General Consultations have an observed rate of **69.93%**, while Specialist Consultations have the lowest rate at **67.09%**.

The difference between the highest and lowest appointment-type groups is:

$$72.76\% - 67.09\% = 5.67 \text{ percentage points}$$

This is substantially smaller than the differences observed for the strongest Week 6 variables.

6.6 Interpretation of Appointment Type

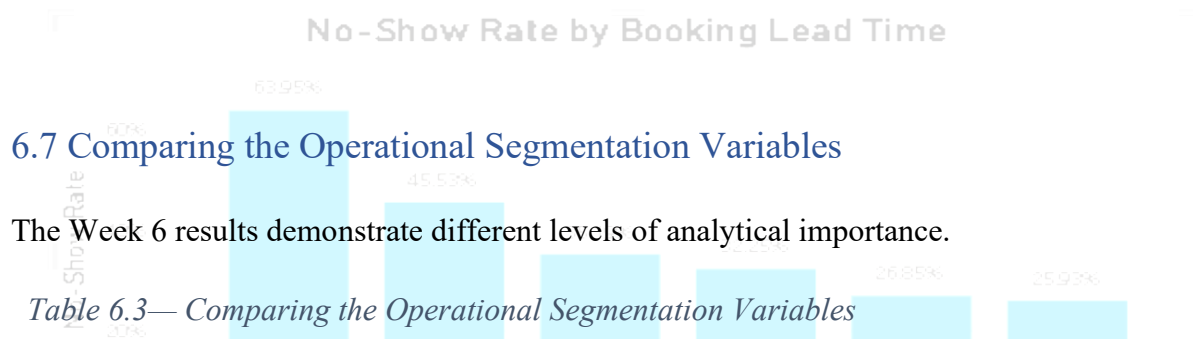
Appointment type provides some additional differentiation, but the observed variation is relatively modest compared with booking lead time and previous no-show history.

This means appointment type should **not be used as the primary rule for identifying high-risk appointments**.

Instead, it can be used as a secondary segmentation variable after the primary high-risk characteristics have been identified.

For example, within the high-risk segment, Follow-up appointments could receive additional operational attention because they have the highest observed rate. However, the difference between appointment types is not sufficiently large to justify treating appointment type as an independent primary risk indicator.

The Diagnostic Test category also requires consideration of sample size because it contains **131 eligible appointments**, which is smaller than the General Consultation and Follow-up groups.



6.7 Comparing the Operational Segmentation Variables

The Week 6 results demonstrate different levels of analytical importance.

Table 6.3— Comparing the Operational Segmentation Variables

Variable	Key Observation	Decision-Support Value
Booking lead time	31–60 days = 63.95% overall; 70.52% when combined with previous no-shows	Very High
Previous no-show history	57.83% vs 46.30%; +11.53 pp	Very High
Distance within high-risk segment	66.96% at 0–5 km vs 77.94% at >20–30 km	High / Secondary
Reminder status	49.86% vs 54.63% overall; association varies by segment	Moderate
Appointment type within high-risk segment	72.76% Follow-up vs 67.09% Specialist	Moderate / Secondary

This comparison reinforces the Week 6 prioritization framework.

The analysis should not treat every statistically or descriptively different segment as equally important. The variables must be considered according to their **magnitude, consistency, sample size, and operational usefulness**.

6.8 Combined Operational Interpretation

The deeper segmentation indicates that attendance risk is better understood as a combination of characteristics rather than as the result of one isolated variable.

The strongest observed combination remains:

31–60-day booking lead time + 1+ previous no-shows = 70.52% no-show rate.

Within this segment, distance provides further differentiation, with observed no-show rates generally increasing across distance bands.

Appointment type provides additional but smaller variation.

This creates a potential prioritization hierarchy:

Primary risk indicators

- Previous no-show history
- Booking lead time

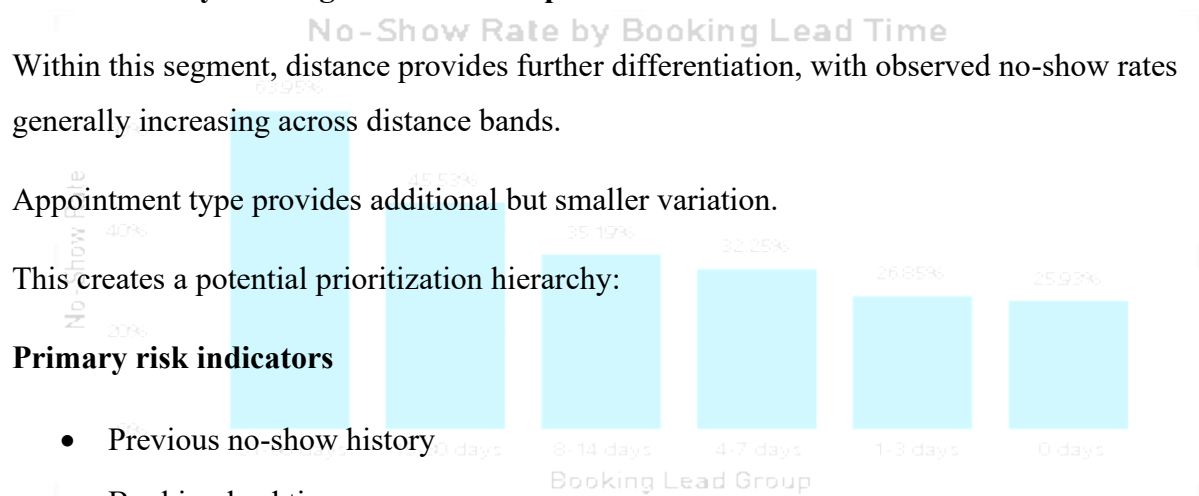
Secondary prioritization indicators

- Distance to clinic
- Appointment type

Additional operational variable

- Reminder status

This hierarchy is more appropriate for decision support than treating every observed association as an equivalent predictor.



6.9 Implications for HealthConnect

The segmentation findings provide HealthConnect with a more targeted basis for attendance-support planning.

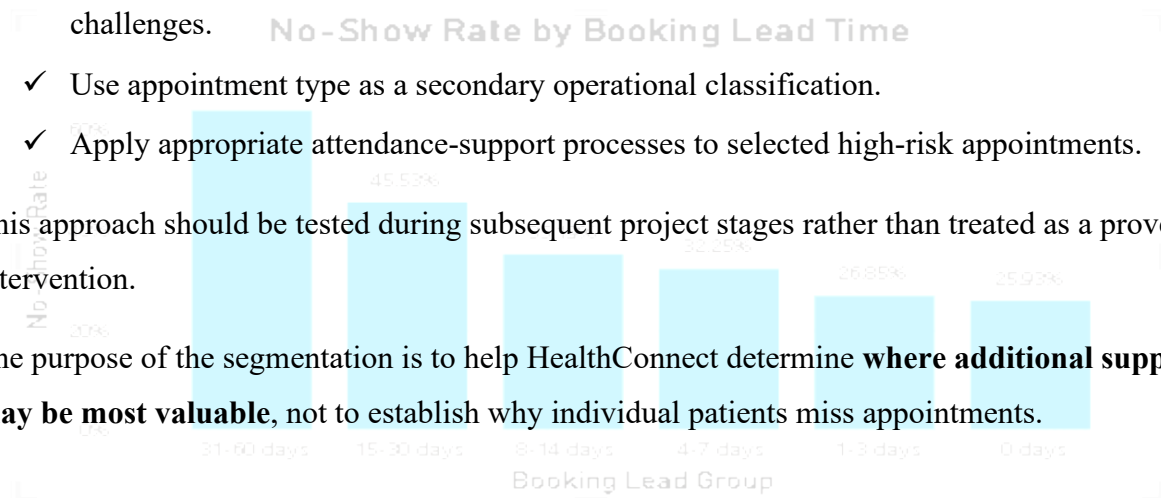
A potential operational prioritization approach could identify appointments that meet the primary high-risk criteria and then use distance and appointment type to further prioritize support.

For example:

- ✓ Identify appointments booked **31–60 days in advance**.
- ✓ Identify patients with **one or more previous no-shows**.
- ✓ Use distance to identify patients who may face greater travel-related attendance challenges.
- ✓ Use appointment type as a secondary operational classification.
- ✓ Apply appropriate attendance-support processes to selected high-risk appointments.

This approach should be tested during subsequent project stages rather than treated as a proven intervention.

The purpose of the segmentation is to help HealthConnect determine **where additional support may be most valuable**, not to establish why individual patients miss appointments.



6.10 Data Science Implications

The segmentation results provide additional candidate variables for the Data Science track.

Potential modelling variables include:

- booking_lead_days
- previous no-show history
- distance_to_clinic_km
- appointment_type
- reminder_sent

- appointment day and time

The analysis particularly supports investigating interactions between:

Booking Lead Time × Previous No-Show History

and potentially:

High-Risk Segment × Distance

The results also suggest that distance may contribute incremental information after the primary risk segment has been identified.

However, this should be established through formal model evaluation rather than assumed from descriptive analysis.

The Data Science team should also consider sample-size limitations when evaluating granular distance categories, particularly the **>30 km group with only 16 eligible appointments**.

6.11 Analytical Limitations

Several limitations apply to the operational segmentation analysis.

6.11.1 Observational relationships

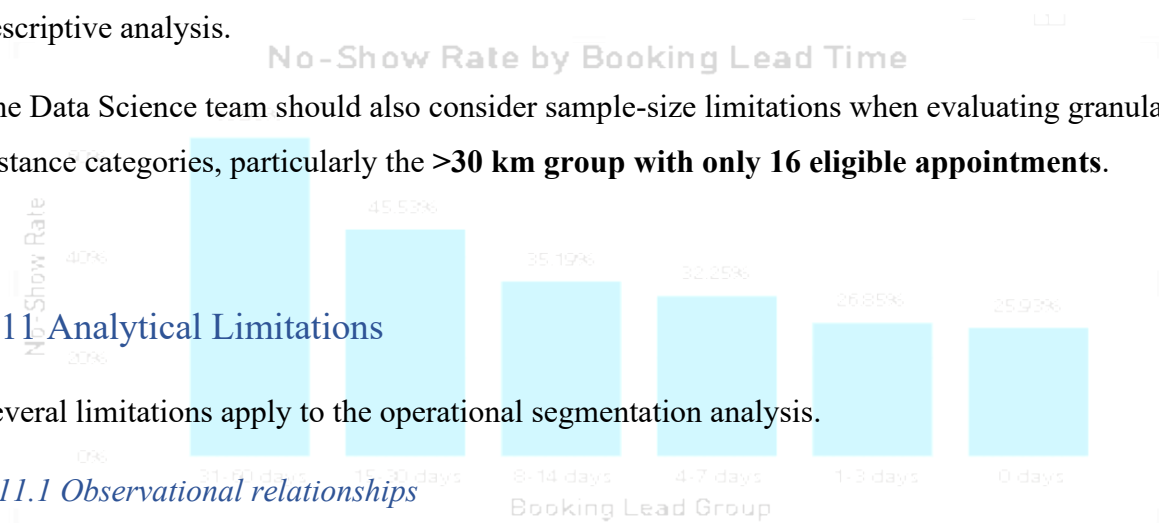
The analysis identifies patterns in historical appointment records but does not establish causal relationships.

6.11.2 Small subgroups

The >30 km group contains only 16 eligible appointments within the high-risk segment. Its 75.00% rate should therefore be interpreted cautiously.

6.11.3 Potential unobserved factors

Distance may represent or correlate with other factors such as transportation availability, travel time, geographic accessibility, or patient circumstances. These factors are not directly captured in the available data.



6.11.4 Appointment-type differences

The observed differences between appointment types are relatively modest. Appointment type should therefore remain a secondary segmentation variable.

6.11.5 Risk of over-segmentation

Creating increasingly granular combinations of variables can produce small cells and unstable rates.

For this reason, HealthConnect should prioritize segments that demonstrate sufficient sample size, meaningful differences, and clear operational relevance.

6.12 Decision-Support Conclusion

The Week 6 operational segmentation analysis strengthens the evidence for a **multi-factor attendance-risk framework**.

Booking lead time and previous no-show history remain the primary indicators of observed attendance risk.

Within the resulting high-risk segment, distance provides additional segmentation value, with observed no-show rates generally increasing as distance increases. Appointment type provides smaller additional differences and should therefore remain a secondary classification variable.

The evidence supports the following analytical hierarchy:

Primary: Booking lead time + previous no-show history

Secondary: Distance to clinic + appointment type

Additional operational context: Reminder status

This hierarchy provides a more disciplined basis for HealthConnect's decision-support approach and helps prevent weaker associations from being treated as primary risk indicators.

The findings also provide additional candidate variables and interaction structures for future Data Science modelling and subsequent Week 7 validation.

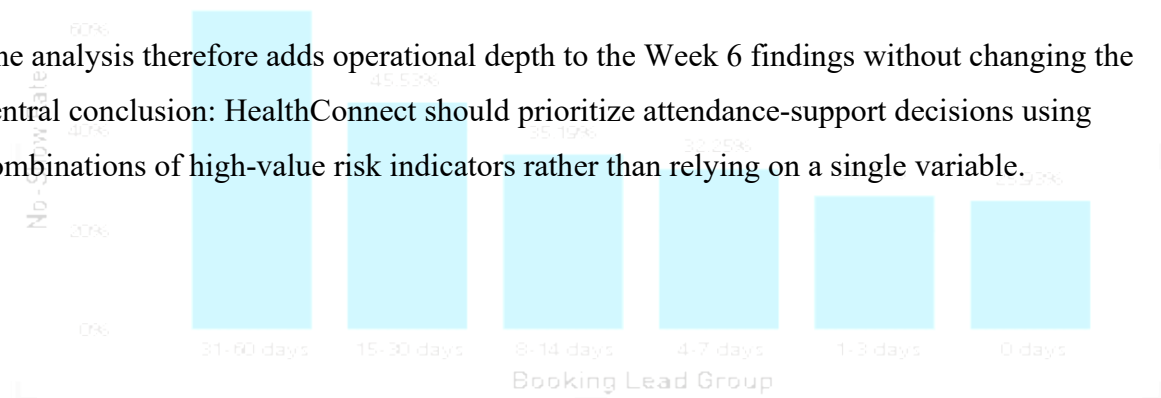
6.13 Chapter Summary

The advanced segmentation confirms that the strongest attendance-risk pattern remains concentrated around **long booking lead times and previous no-show history**.

Within this high-risk segment:

- no-show rates generally increase with distance;
- the **>20–30 km** group reaches **77.94%**, although the >30 km group requires caution because of its small sample;
- Follow-up appointments have the highest appointment-type rate at **72.76%**;
- Specialist Consultations have the lowest at **67.09%**; and
- the relatively modest appointment-type differences indicate that appointment type is a secondary rather than primary risk differentiator.

The analysis therefore adds operational depth to the Week 6 findings without changing the central conclusion: HealthConnect should prioritize attendance-support decisions using combinations of high-value risk indicators rather than relying on a single variable.



Chapter 7

KPI Validation and Refinement

Figure 7.1 — Validated Week 6 KPI Framework



7.1 Purpose of KPI Validation

Week 5 established a KPI framework for monitoring appointment attendance and missed appointments at HealthConnect Clinic.

Week 6 required these KPIs to be validated and refined using the deeper analysis rather than simply reproduced. The objective was to determine:

- whether the Week 5 KPI definitions remain analytically appropriate;
- whether the calculations remain consistent after deeper segmentation;
- whether any KPI requires reframing;
- whether a new decision-support indicator should be introduced; and
- whether the KPI framework remains aligned with the HealthConnect business objective.

This approach is consistent with the Week 6 requirement to **validate and refine important KPIs** as part of the transition from initial analysis to advanced decision-support analysis.

7.2 KPI Validation Principles

The Week 6 validation applied the following principles.

7.2.1 Denominator consistency

Primary attendance KPIs use **eligible appointments**, defined as appointments with an outcome of either:

- Attended
- No-Show

Therefore:

Eligible Appointments = 4,737

Cancelled appointments are excluded from the primary No-Show Rate and Attendance Rate because a cancellation does not represent the same attendance outcome as a missed appointment.

7.2.2 Outcome integrity

The primary outcome categories remain:

- Attended
- No-Show
- Cancelled

This distinction prevents cancellations from being incorrectly classified as missed appointments.

7.2.3 Observational interpretation

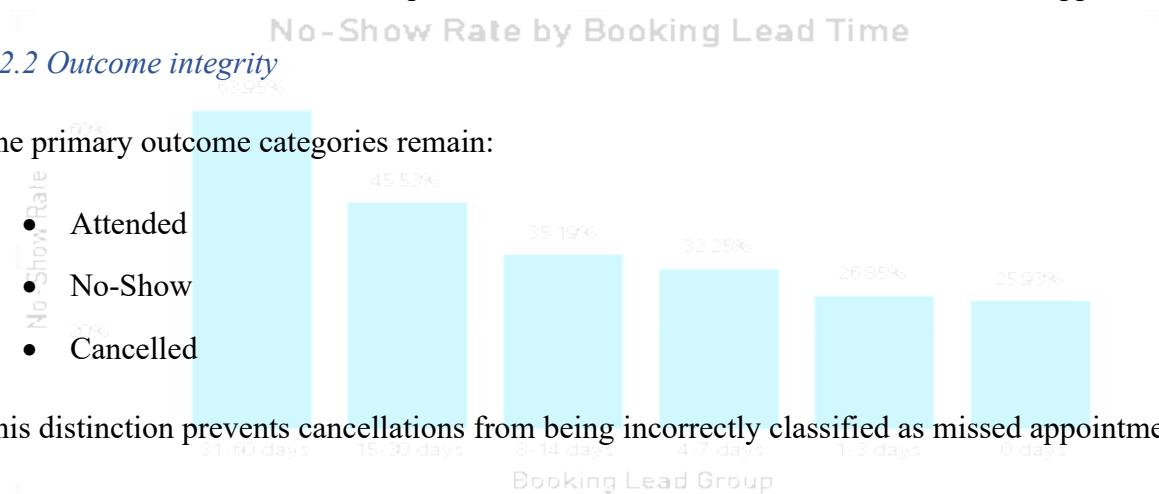
Variables such as reminder status are interpreted as observed associations rather than causal effects.

7.2.4 Decision-support relevance

A KPI should not be retained simply because it can be calculated. It should provide useful information for monitoring, prioritization, or decision-making.

7.2.5 Segment stability

Highly granular indicators require consideration of sample size and should not be elevated to primary KPIs when the underlying subgroup is too small or unstable.



7.3 KPI 1 — No-Show Rate

Definition

The proportion of eligible appointments that resulted in a No-Show outcome.

Validated result

2,423 No-Shows / 4,737 Eligible Appointments = 51.15%

Measure	Result
Eligible Appointments	4,737
No-Shows	2,423
No-Show Rate	51.15%

Validation outcome

Retained unchanged.

The Week 6 analysis does not invalidate the overall No-Show Rate. Instead, the deeper analysis explains where the overall rate is concentrated.

The 51.15% figure therefore remains the primary headline attendance-risk KPI.

Interpretation

More than half of eligible appointments in the dataset resulted in a No-Show outcome.

However, the overall rate alone does not identify which appointment groups require the greatest attention. Week 6 segmentation therefore complements this KPI with more targeted indicators.

7.4 KPI 2 — Attendance Rate

Definition

The proportion of eligible appointments resulting in an Attended outcome.

Validated result

2,314 Attended / 4,737 Eligible Appointments = 48.85%

Measure	Result
Eligible Appointments	4,737
Attended Appointments	2,314
Attendance Rate	48.85%

Validation outcome

Retained unchanged.

The Attendance Rate remains the complementary measure to the No-Show Rate.

Because the primary eligible population consists only of Attended and No-Show appointments:

$$51.15\% + 48.85\% = 100.00\%$$

This provides an internal reconciliation check for the two primary attendance KPIs.

7.5 KPI 3 — No-Show Rate by Reminder Status

Week 5 definition

Week 5 examined the difference between appointments where a reminder was recorded and those where no reminder was recorded.

The validated results are:

Reminder Status	Eligible Appointments	No-Show Rate
Reminder sent	3,452	49.86%
No reminder	1,285	54.63%

Observed difference:

4.77 percentage points

Week 6 validation

The deeper analysis showed that this relationship is not uniform across booking lead-time groups.

For example:

- 15–30 days: **5.81 pp** difference
- 31–60 days: **5.05 pp** difference
- 1–3 days: approximately **0 pp**
- 4–7 days: approximately **0 pp**

The same-day comparison produced a larger difference, but the no-reminder subgroup contained only 23 eligible appointments and therefore requires caution.

Refinement

The original concept is **retained but reframed**.

The KPI should be presented as:

No-Show Rate by Reminder Status

rather than language such as:

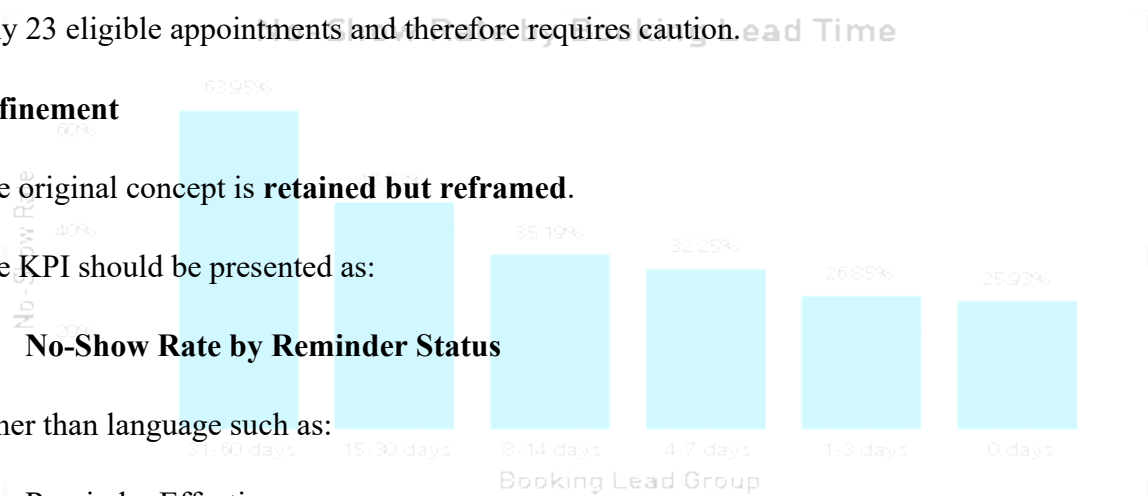
Reminder Effectiveness

The revised terminology is more analytically appropriate because the dataset supports an association between reminder status and observed no-show rates, but does not establish a causal intervention effect.

Validation outcome

Retained with revised interpretation.

The KPI is classified as an **association/operational indicator**, not a causal performance measure.



7.6 KPI 4 — Repeat No-Show Rate

Definition

The observed no-show rate among eligible appointments for patients with at least one previous no-show.

Validated results

Previous No-Show History	Eligible Appointments	No-Show Rate
1+ previous no-shows	1,992	57.83%
No previous no-shows	2,745	46.30%

Observed difference:

57.83% – 46.30% = 11.53 percentage points

Week 6 validation

The relationship remained evident after segmentation by booking lead time.

For example:

Table 7.1 —Booking Lead Time (Days) and No-Show History

Booking Lead Time	No Previous No-Shows	1+ Previous No-Shows
0 days	18.75%	36.36%
1–3 days	17.74%	39.13%
4–7 days	27.32%	39.52%
8–14 days	33.06%	38.79%
15–30 days	41.09%	51.27%
31–60 days	59.07%	70.52%

The finding therefore remains meaningful when examined jointly with another major risk indicator.

Validation outcome

Retained as a priority behavioral KPI.

Previous no-show history remains one of the strongest observed differentiators of attendance risk.

It should therefore continue to receive prominent attention in the HealthConnect analytical framework.

7.7 KPI 5 — High-Risk Segment No-Show Rate

Week 5 used broader appointment segmentation to compare no-show rates across appointment characteristics.

Week 6 provides sufficient evidence to refine this approach into a more targeted decision-support indicator.

New Week 6 indicator

High-Risk Segment No-Show Rate

Definition:

The observed no-show rate among eligible appointments booked **31–60 days in advance** for patients with **at least one previous no-show**.

Validated result

70.52%

This represents the strongest identified high-risk segment in the Week 6 analysis.

The result is particularly important because it combines two independently strong indicators:

- long booking lead time; and
- previous no-show history.

Validation outcome

Introduced as a Week 6 decision-support indicator.

This indicator should complement rather than replace the overall No-Show Rate.

The overall KPI answers:

How large is the clinic's overall no-show problem?

The high-risk segment indicator answers:

Where is the highest observed concentration of attendance risk?

7.8 High-Risk Segment and Reminder Status

The high-risk segment was further examined according to reminder status.

High-Risk Segment	Eligible Appointments	No-Show Rate
Reminder sent	708	69.21%
No reminder	269	73.98%

The observed difference is:

4.77 percentage points

This provides additional context for the High-Risk Segment KPI.

The rate remains very high in both groups, indicating that the high-risk segment cannot be reduced to reminder status alone.

The **73.98%** subgroup is useful as a supporting decision-support statistic, but it should not replace the primary 70.52% High-Risk Segment KPI because the latter provides a cleaner and more stable definition of the broader high-risk group.

7.9 Cancellation Rate — Contextual Metric

The dataset contains:

- 263 Cancelled appointments
- 5,000 total appointments

Therefore:

Cancellation Rate = 5.26%

Validation outcome

Retained as a contextual metric rather than a primary attendance KPI.

Cancellations remain operationally relevant to HealthConnect, but they represent a different outcome from No-Show and Attended appointments.

The metric should therefore be monitored separately and should not be combined with the primary No-Show denominator.

7.10 Final Week 6 KPI Framework

Following the validation and refinement process, the HealthConnect KPI framework becomes:

Table 7.2—Final Week Six KPI Framework

KPI / Indicator	Result	Week 6 Decision	Role
No-Show Rate	51.15%	Retain	Core KPI
Attendance Rate	48.85%	Retain	Core KPI
Repeat No-Show Rate	57.83%	Retain	Behavioral KPI
No-Show Rate by Reminder Status	49.86% vs 54.63%	Reframe	Association / operational indicator
High-Risk Segment No-Show Rate	70.52%	Introduce	Decision-support indicator

KPI / Indicator	Result	Week 6 Decision	Role
Cancellation Rate	5.26%	Retain	Contextual metric

This represents an important improvement from Week 5.

The KPI framework now distinguishes between:

- overall performance;
- attendance behavior;
- operational associations;
- high-risk decision-support information; and
- contextual appointment outcomes.

7.11 KPI Reconciliation

The validated figures provide several internal consistency checks.

Appointment reconciliation

5,000 total appointments

consist of:

- 2,423 No-Shows
- 2,314 Attended
- 263 Cancelled

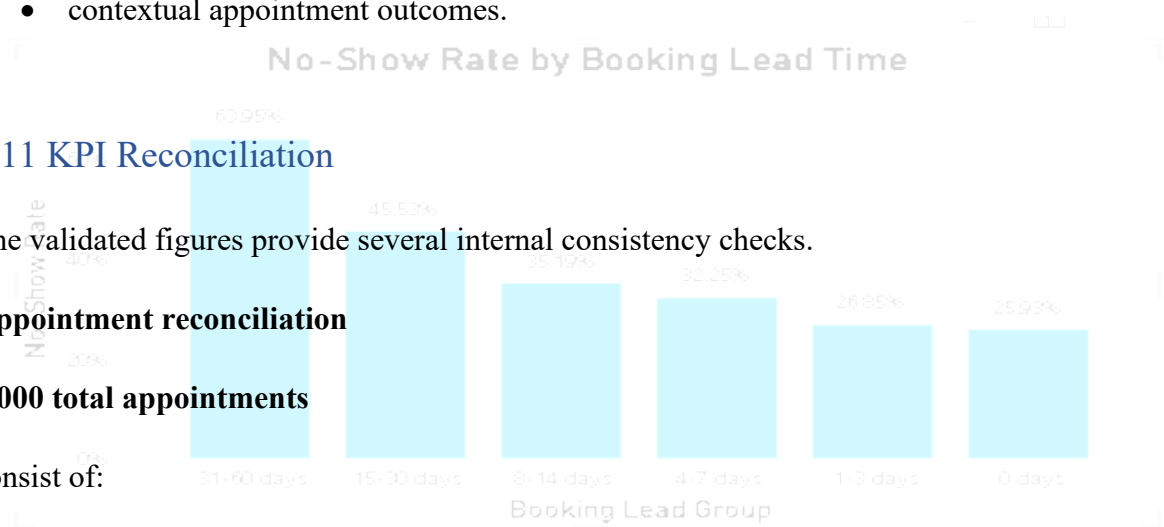
Therefore:

$$2,423 + 2,314 + 263 = 5,000$$

Eligible appointment reconciliation

$$2,423 \text{ No-Shows} + 2,314 \text{ Attended} = 4,737 \text{ Eligible Appointments}$$

Attendance outcome reconciliation



51.15% No-Show Rate + 48.85% Attendance Rate = 100.00%

Cancellation reconciliation

263 / 5,000 = 5.26%

These checks support the reliability and internal consistency of the KPI calculations.

7.12 What Changed from Week 5 to Week 6

The KPI framework was not replaced during Week 6.

Instead, it was strengthened through validation and refinement.

Table 7.3 —Week 5 to 6 KPI Framework

Week 5	Week 6 Improvement
Overall No-Show Rate	Validated and retained
Attendance Rate	Validated and retained
Reminder comparison	Reframed as an observational association
Repeat No-Show Rate	Validated across additional segments
Appointment segmentation	Refined into a defined high-risk segment
Broad findings	Converted into targeted decision-support indicators
Dashboard KPIs	Expanded to support prioritization

This demonstrates the required progression from Week 5 analysis toward Week 6 advanced decision support rather than simply repeating the previous week's calculations.

7.13 Implications for the Week 6 Dashboard

The KPI validation directly informs the upgraded dashboard.

The dashboard should prominently display:

Core KPIs

- Total Appointments: **5,000**
- Eligible Appointments: **4,737**
- No-Show Rate: **51.15%**
- Attendance Rate: **48.85%**

Decision-Support KPI

- High-Risk Segment No-Show Rate: **70.52%**

Supporting Metrics

- Repeat No-Show Rate: **57.83%**
- No-Show Rate by Reminder Status: **49.86% vs 54.63%**
- Cancellation Rate: **5.26%**

The new dashboard should therefore shift emphasis from simply describing attendance patterns to helping HealthConnect identify **where additional attention may be required**.

7.14 Implications for Decision-Making

The validated KPI framework supports a more targeted decision-making approach.

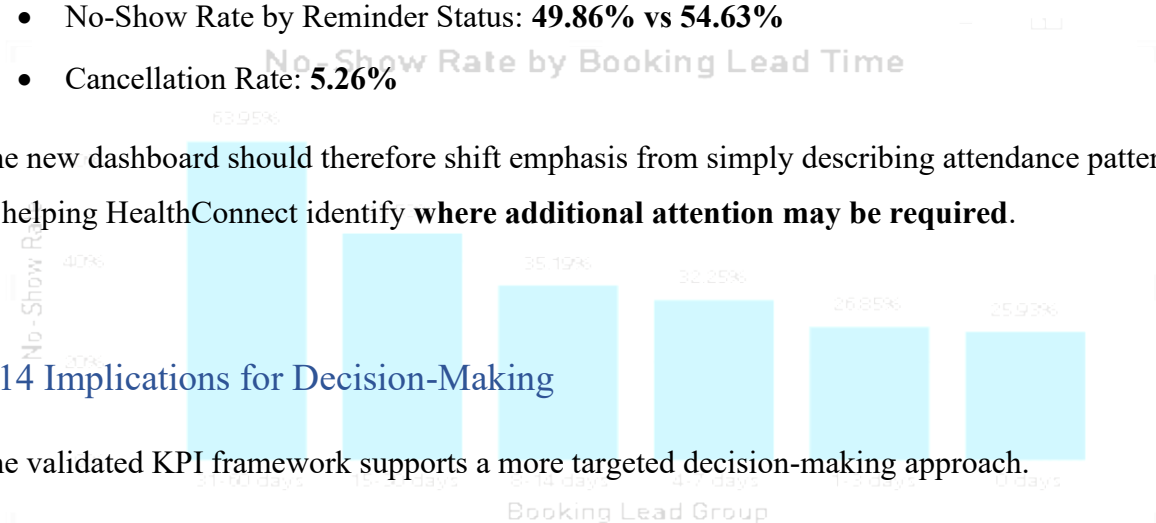
The evidence suggests that HealthConnect should monitor overall attendance performance while simultaneously identifying appointment groups with substantially elevated observed risk.

The most important decision-support signal is:

31–60-day booking lead time + 1+ previous no-shows = 70.52% observed no-show rate.

This indicator can be used to identify a group for further attendance-support consideration and subsequent testing.

It should not be interpreted as a causal risk score or as proof that every appointment in this segment will result in a no-show.



7.15 Implications for Data Science

The KPI validation also provides a clearer analytical foundation for the Data Science track.

The validated findings support consideration of:

- previous no-show history;
- booking lead time;
- distance to clinic;
- appointment type;
- reminder status; and
- interactions between booking lead time and previous no-show history.

The strongest interaction identified analytically is:

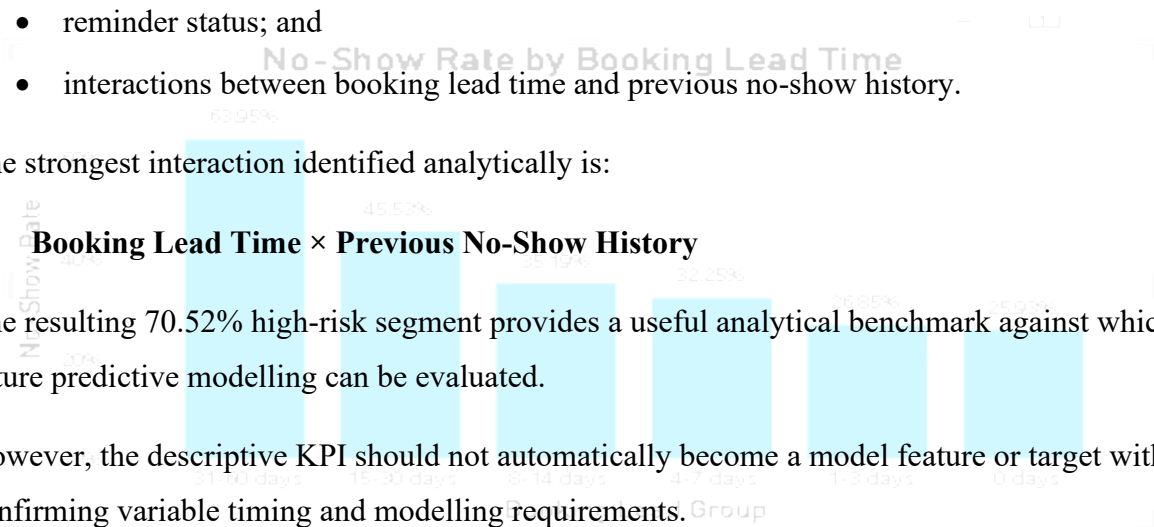
Booking Lead Time × Previous No-Show History

The resulting 70.52% high-risk segment provides a useful analytical benchmark against which future predictive modelling can be evaluated.

However, the descriptive KPI should not automatically become a model feature or target without confirming variable timing and modelling requirements.

In particular:

- appointment_outcome must remain the outcome variable rather than a predictor;
- reminder timing should be confirmed before using reminder_sent as a prediction feature; and
- waiting_time_minutes should not be treated as a pre-appointment no-show predictor unless its timing is confirmed.



7.16 KPI Limitations

The validated KPI framework remains subject to several limitations.

Observational data

The KPIs describe historical patterns and associations. They do not establish causality.

Segment dependence

Some KPI results vary substantially across appointment and patient segments. The overall rate should therefore not be assumed to apply equally to every appointment.

Small subgroups

Some detailed segments contain relatively few observations. These should be treated as supporting evidence rather than primary KPIs.

Reminder interpretation

Reminder status indicates whether a reminder was recorded but does not establish whether the reminder was received, read, or effective.

Predictive interpretation

The 70.52% high-risk segment is a descriptive decision-support indicator. It should not be presented as a validated predictive probability for individual patients.

7.17 Chapter Conclusion

The Week 6 KPI validation confirms that the core Week 5 attendance measures remain reliable and relevant.

The **51.15% No-Show Rate** and **48.85% Attendance Rate** remain the primary performance KPIs, while the **57.83% Repeat No-Show Rate** remains an important behavioral indicator.

The reminder metric remains useful but has been deliberately reframed as **No-Show Rate by Reminder Status** because the observed relationship varies across segments and does not establish causality.

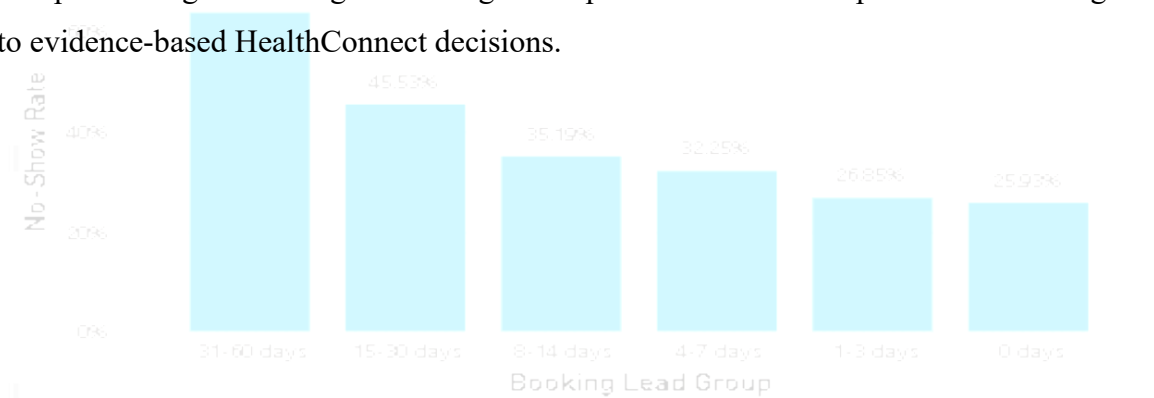
The most important Week 6 refinement is the introduction of the:

High-Risk Segment No-Show Rate — 70.52%

defined as appointments booked 31–60 days in advance for patients with at least one previous no-show.

This refinement transforms the KPI framework from a primarily descriptive monitoring system into a more targeted **decision-support framework**.

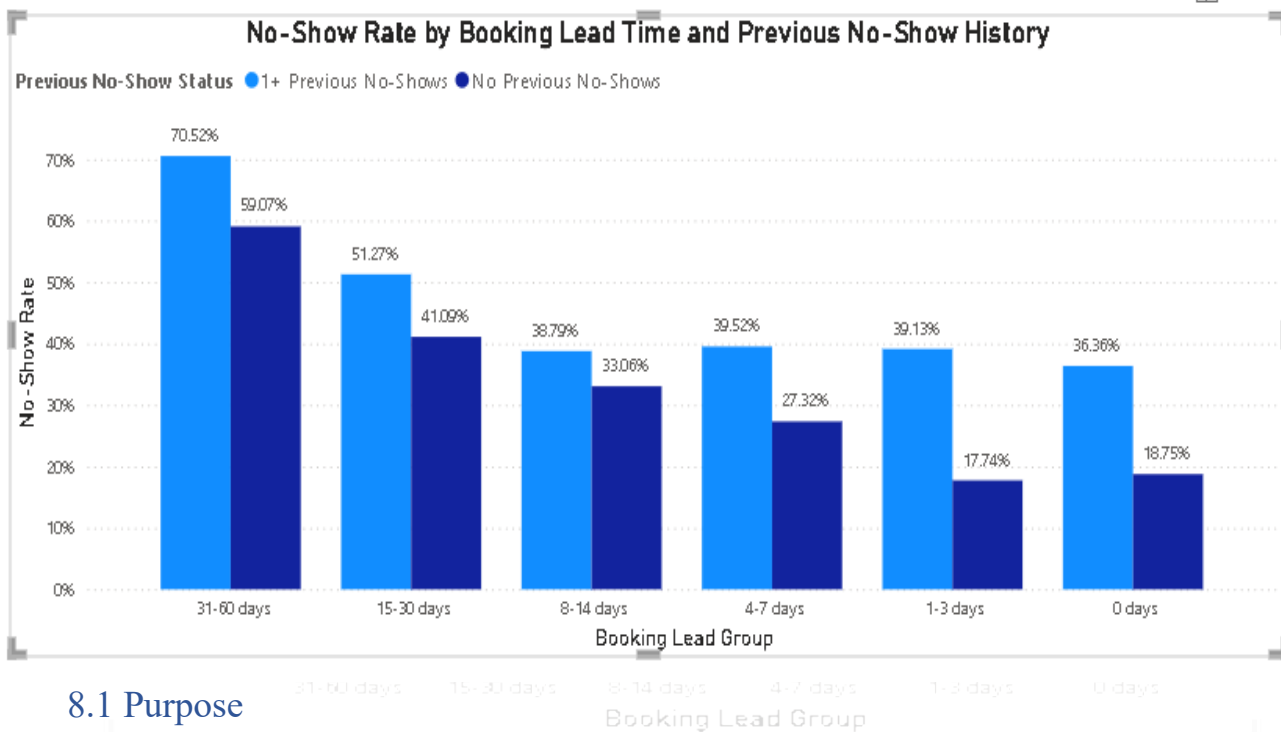
The validated KPI structure now provides a stronger foundation for the next stage of the Week 6 work: prioritizing the findings with the greatest potential business impact and translating them into evidence-based HealthConnect decisions.



Chapter 8

Validated Findings and Decision Support

Figure 8.1—Booking Lead Time × Previous No-Show History



8.1 Purpose

The purpose of this chapter is to translate the advanced analytical results from Week 6 into validated, prioritized business findings that can support HealthConnect Clinic’s decision-making.

Week 5 established that HealthConnect has a high overall no-show rate and identified booking lead time, previous no-show history, reminder status, appointment type, distance, and other appointment characteristics as relevant areas for investigation. Week 6 extended this analysis by examining these factors jointly and across operational segments.

The objective was therefore not simply to identify which variables have high no-show rates, but to determine:

- ✓ which findings remain meaningful after deeper segmentation;

- ✓ which characteristics identify the highest observed-risk appointments;
- ✓ which findings are sufficiently consistent to support decision-making;
- ✓ which findings have practical operational relevance; and
- ✓ which analytical findings should be provided as inputs for future Data Science modelling.

The analysis indicates that **booking lead time and previous no-show history are the strongest observed indicators of attendance risk when considered jointly**. Their combination identifies a substantially higher-risk appointment segment than the overall clinic average.

8.2 From Analytical Findings to Decision Support

The Week 6 analysis moves the project from descriptive reporting toward decision support.

The Week 5 dashboard primarily answered:

Where are no-show rates higher?

The Week 6 analysis addresses a more operational question:

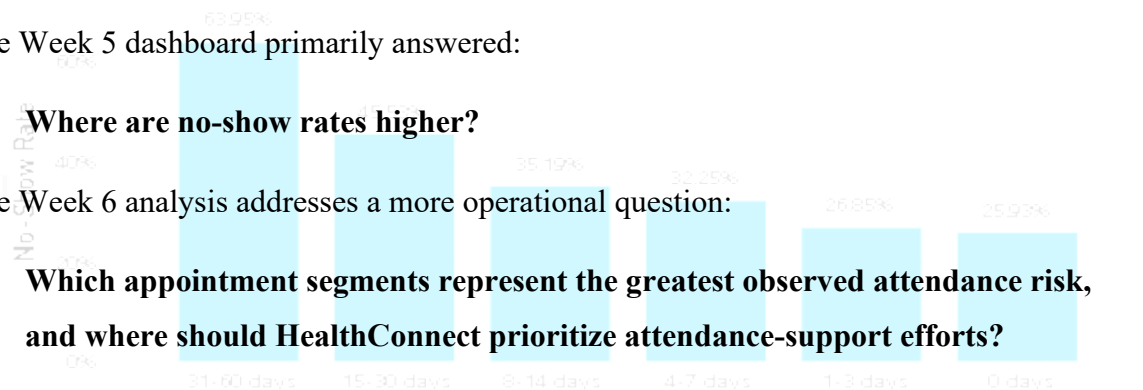
Which appointment segments represent the greatest observed attendance risk, and where should HealthConnect prioritize attendance-support efforts?

This distinction is important because a high no-show rate by itself does not necessarily identify an actionable target. Decision support requires the finding to be sufficiently large, reasonably consistent across related analyses, and operationally meaningful.

Accordingly, the Week 6 findings were evaluated using three criteria:

- ✓ **Magnitude** — How large is the observed difference or risk level?
- ✓ **Consistency** — Does the pattern remain visible when the data is segmented or examined through another relevant variable?
- ✓ **Actionability** — Can HealthConnect reasonably use the finding to inform a decision, prioritization rule, workflow, or future modelling activity?

These criteria were used to rank the findings below.



8.3 Finding Prioritization Framework

The following framework was applied to the major Week 6 findings.

Table 8.1—Week Six Findings Prioritization Matrix

Finding	Evidence	Magnitude	Consistency	Actionability	Priority
31–60-day booking + 1+ previous no-shows	70.52% no-show rate	Very High	High	Very High	1 — Very High
Previous no-show history	57.83% vs 46.30%; +11.53 pp	Very High	High	Very High	2 — Very High
Booking lead time	63.95% for 31–60 days vs 25.93% same-day	Very High	High	High	3 — Very High
Distance within high-risk segment	66.96% at 0–5 km vs 77.94% at >20–30 km	High	Moderate–High	High	4 — High
Reminder status	49.86% vs 54.63%; +4.77 pp	Moderate	Variable	Moderate	5 — Moderate
Appointment type within high-risk segment	72.76% Follow-up vs 67.09% Specialist	Moderate	Moderate	Moderate	6 — Moderate

The ranking does not imply that lower-ranked variables are irrelevant. Rather, it indicates that booking lead time and previous no-show history provide stronger and more consistent decision-support signals than reminder status or appointment type alone.

8.4 Finding 1 — High-Risk Segment: 31–60 Days and Previous No-Show History

The strongest Week 6 finding is the interaction between **booking lead time** and **previous no-show history**.

Appointments booked between 31 and 60 days in advance for patients with at least one previous no-show recorded a **70.52% observed no-show rate**.

For comparison, within the same 31–60-day booking window, appointments for patients with no previous no-shows had a **59.07% observed no-show rate**.

This represents an **11.45 percentage-point difference** within the same booking-lead segment.

Table 8.2—Booking Lead Time and Previous No-Show History

Booking Lead Time	No Previous No-Shows	1+ Previous No-Shows
0 days	18.75%	36.36%
1–3 days	17.74%	39.13%
4–7 days	27.32%	39.52%
8–14 days	33.06%	38.79%
15–30 days	41.09%	51.27%
31–60 days	59.07%	70.52%

The finding is particularly important because the high-risk segment is not defined by a single characteristic. It combines two observed risk indicators:

- a relatively long booking lead time; and
- a documented history of previous no-shows.

This makes the segment more useful for decision support than either characteristic considered independently.

Decision-support interpretation

HealthConnect can use this combination as a **priority identification rule** for attendance-support planning.

The finding should not be interpreted as evidence that long booking lead time or previous no-show history causes missed appointments. It identifies an observed segment in which missed appointments are substantially more common.

Business implication

Appointments meeting both conditions should be considered candidates for enhanced attendance-support processes, subject to operational feasibility and subsequent validation.

8.5 Finding 2 — Previous No-Show History Is a Strong Behavioral Indicator

Previous no-show history remains one of the strongest individual differentiators in the analysis.

Patients with at least one previous no-show had an observed no-show rate of:

57.83%

compared with:

46.30%

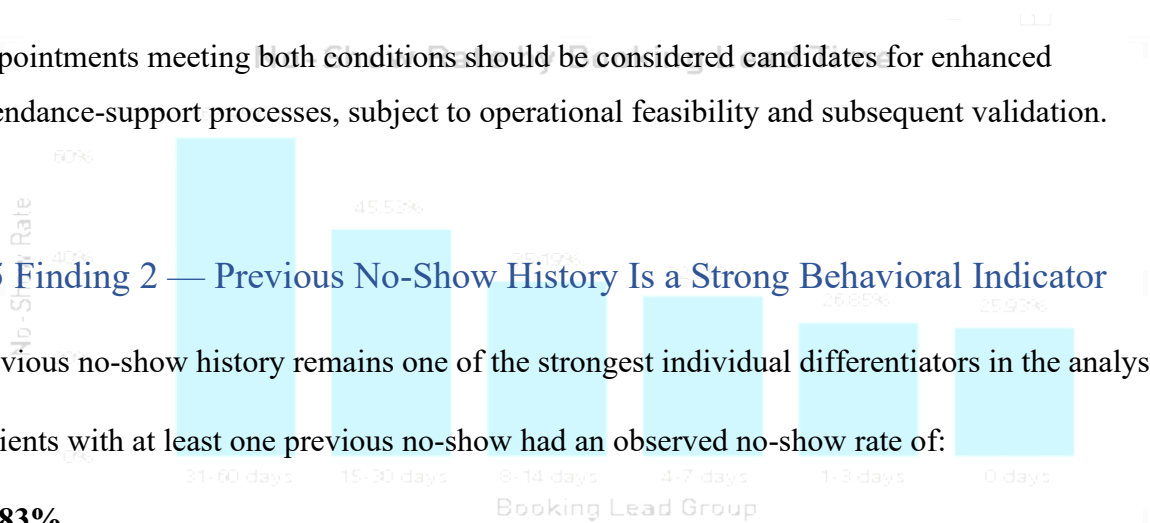
for patients with no previous no-shows.

The difference is therefore:

+11.53 percentage points.

This finding remained meaningful when previous no-show history was examined alongside booking lead time and reminder status.

For example, among appointments where a reminder was sent:



- patients with no previous no-shows had a 45.31% no-show rate;
- patients with 1+ previous no-shows had a 56.22% no-show rate.

This represents a **10.91 percentage-point difference** even within the reminder-sent group.

Decision-support interpretation

Previous no-show history should therefore be treated as a **core attendance-risk indicator** rather than merely a descriptive historical variable.

Business implication

HealthConnect could consider previous no-show history when prioritising appointments for additional attendance-support measures.

Any such use should be implemented as a support and prioritization mechanism rather than as a punitive classification.

8.6 Finding 3 — Booking Lead Time Is a Major Attendance-Risk Indicator

Booking lead time also remains a major differentiator.

Week 5 identified a strong increase in observed no-show rates as booking lead time increased. Week 6 confirmed that this pattern remained important when combined with previous no-show history.

At the overall level:

- Same-day bookings recorded a **25.93%** no-show rate.
- Appointments booked 31–60 days in advance recorded a **63.95%** no-show rate.

This represents a difference of approximately **38.02 percentage points**.

The joint analysis further demonstrated that the 31–60-day booking window remains the highest-risk lead-time category across both previous-no-show groups.

Decision-support interpretation

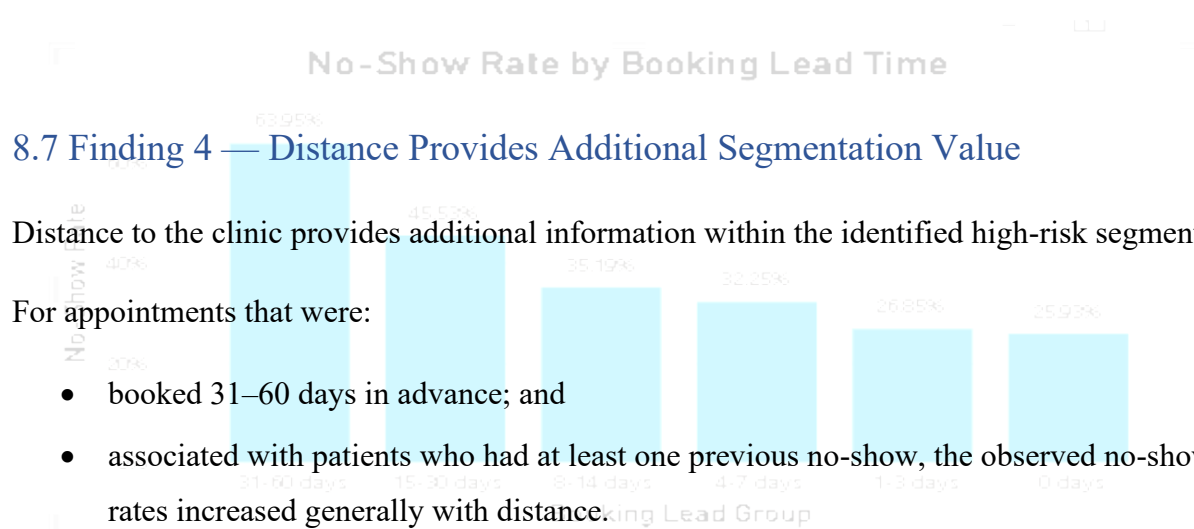
Long booking lead time should be treated as an important indicator when identifying appointments that may require additional attendance-support attention.

The finding also suggests that the timing of attendance-support activity may be important for long-lead appointments, although the available dataset does not establish which intervention or timing would be most effective.

Business implication

HealthConnect could consider an **earlier confirmation or follow-up workflow** for appointments booked substantially in advance.

This is a candidate operational response, not a proven intervention effect.



8.7 Finding 4 — Distance Provides Additional Segmentation Value

Distance to the clinic provides additional information within the identified high-risk segment.

For appointments that were:

- booked 31–60 days in advance; and
- associated with patients who had at least one previous no-show, the observed no-show rates increased generally with distance.

Table 8.3—Distance to Clinic and No-show Rate

Distance to Clinic	Eligible Appointments	No-Show Rate
0–5 km	224	66.96%
>5–10 km	339	69.32%
>10–15 km	218	72.02%
>15–20 km	90	73.33%
>20–30 km	68	77.94%
>30 km	16	75.00%

The highest observed rate was 77.94% among appointments in the >20–30 km group.

The >30 km group recorded 75.00%, but this result is based on only **16 eligible appointments** and should therefore not be treated as stable evidence.

Decision-support interpretation

Distance appears to provide useful secondary segmentation within the already identified high-risk population.

The results suggest that travel distance may help HealthConnect prioritize which high-risk appointments require greater attention.

Business implication

Distance could be incorporated as a **secondary prioritization variable**, particularly where operational resources for attendance support are limited.

However, distance should not replace the stronger indicators of previous no-show history and booking lead time.

8.8 Finding 5 — Reminder Status Shows a Modest but Variable Association

Week 5 showed a lower observed no-show rate among eligible appointments where a reminder was recorded as sent:

- Reminder sent: **49.86%**
- No reminder: **54.63%**

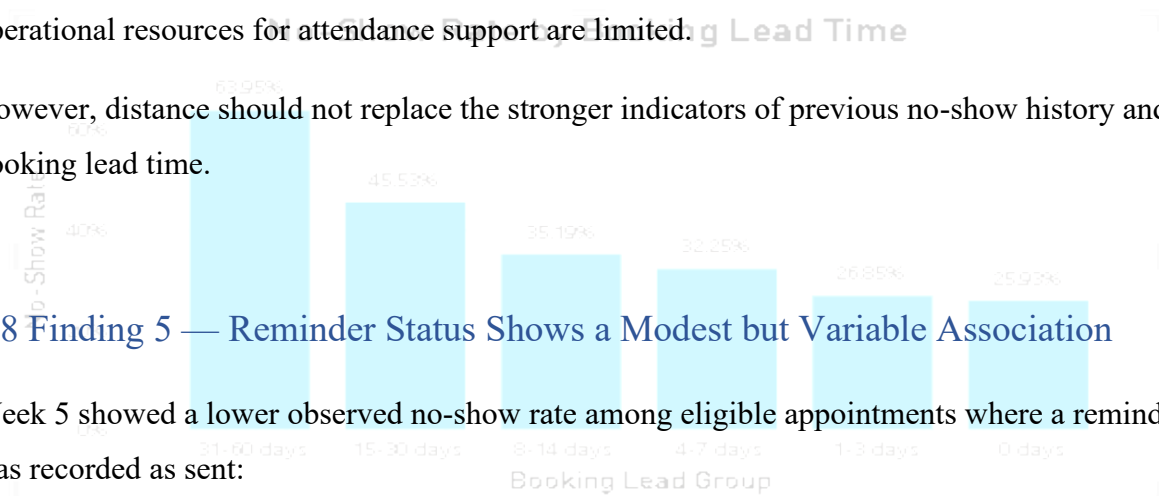
The difference is therefore:

4.77 percentage points.

Week 6 demonstrated that this association is not uniform across booking-lead segments.

For example:

- 15–30 days: 43.91% with reminder vs 49.72% without;
- 31–60 days: 62.58% with reminder vs 67.63% without.



However, several shorter booking windows showed little difference, while the same-day comparison produced a much larger difference based on a small no-reminder group.

The three-way high-risk analysis produced:

- **69.21%** with reminder, n = 708;
- **73.98%** without reminder, n = 269.

This represents a 4.77 percentage-point difference within the high-risk segment.

Decision-support interpretation

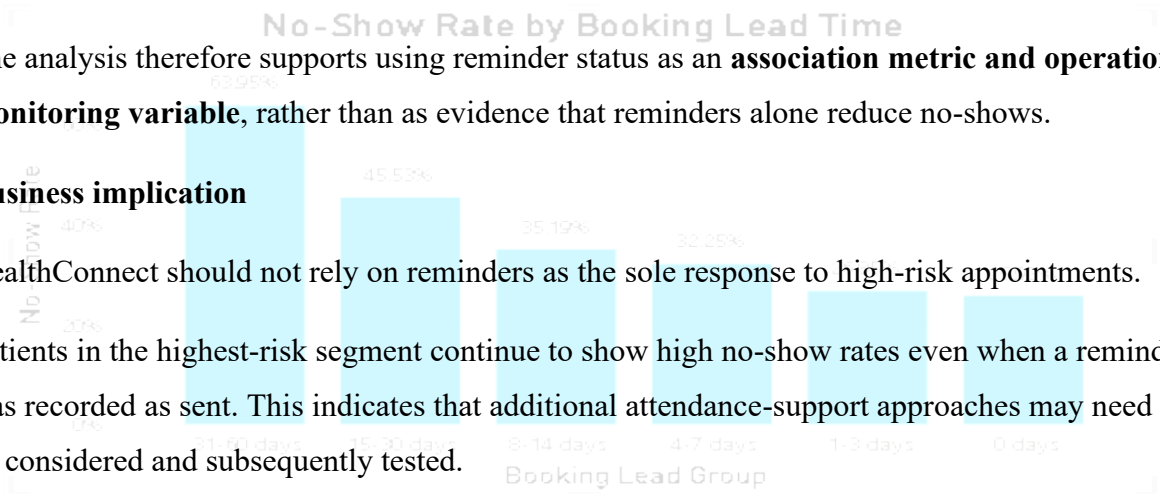
Reminder status is associated with lower observed no-show rates in several comparisons, but the relationship is not sufficiently uniform to describe reminders as a proven intervention effect.

The analysis therefore supports using reminder status as an **association metric and operational monitoring variable**, rather than as evidence that reminders alone reduce no-shows.

Business implication

HealthConnect should not rely on reminders as the sole response to high-risk appointments.

Patients in the highest-risk segment continue to show high no-show rates even when a reminder was recorded as sent. This indicates that additional attendance-support approaches may need to be considered and subsequently tested.



8.9 Finding 6 — Appointment Type Is a Secondary Differentiator

Appointment type was also examined within the high-risk segment.

For appointments booked 31–60 days in advance for patients with at least one previous no-show:

Table 8.4 —No-Show Rate by Appointment Type

Appointment Type	Eligible Appointments	No-Show Rate
Follow-up	279	72.76%
Diagnostic Test	131	71.76%

Appointment Type	Eligible Appointments	No-Show Rate
General Consultation	409	69.93%
Specialist Consultation	158	67.09%

Follow-up appointments recorded the highest observed no-show rate at 72.76%, while Specialist Consultations recorded the lowest at 67.09%.

The difference between these two categories is approximately **5.67 percentage points**.

Although this variation is relevant, it is considerably less pronounced than the differences associated with booking lead time and previous no-show history.

Decision-support interpretation

Appointment type should therefore be treated as a **secondary segmentation variable**, rather than the primary basis for attendance-risk prioritization.

Business implication

HealthConnect may use appointment type to refine support within an already identified high-risk group, but should avoid using appointment type alone to determine attendance risk.

8.10 Overall Business-Impact Ranking

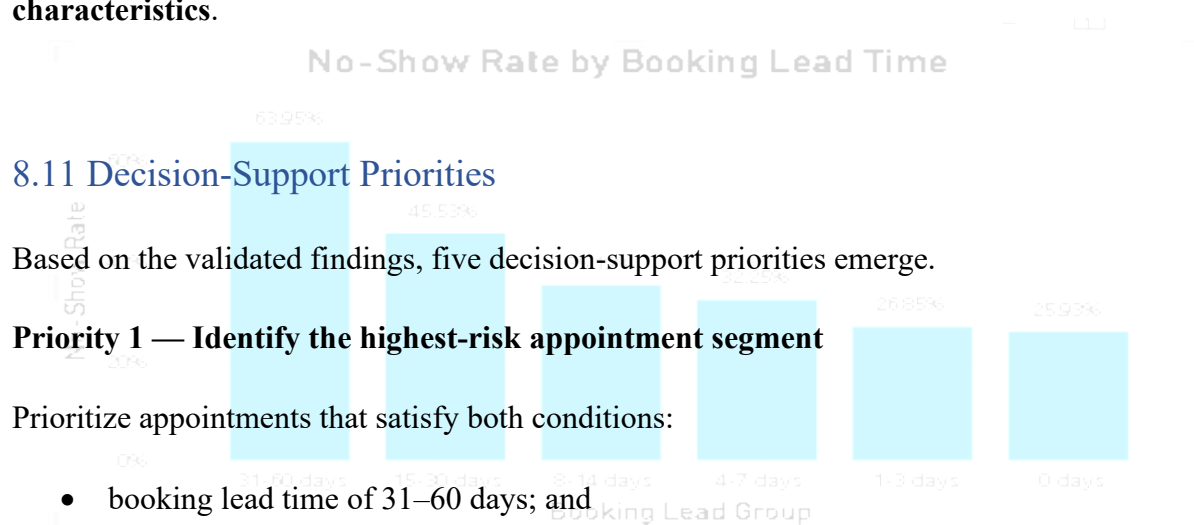
The validated findings can be summarized as follows:

Table 8.5 — Summarized Validated Findings

Rank	Finding	Key Evidence	Business Interpretation
1	High-risk segment	70.52% no-show	Strongest identified decision-support segment
2	Previous no-show history	+11.53 pp	Strong behavioral indicator
3	Booking lead time	63.95% vs 25.93%	Major appointment-level indicator

Rank	Finding	Key Evidence	Business Interpretation
4	Distance within high-risk segment	66.96%–77.94%	Useful secondary prioritization variable
5	Reminder status	+4.77 pp overall	Moderate and variable association
6	Appointment type	5.67 pp within high-risk segment	Secondary segmentation variable

The ranking indicates that HealthConnect should move away from a broad, one-size-fits-all attendance approach toward **targeted support based on identifiable appointment characteristics**.



8.11 Decision-Support Priorities

Based on the validated findings, five decision-support priorities emerge.

Priority 1 — Identify the highest-risk appointment segment

Prioritize appointments that satisfy both conditions:

- booking lead time of 31–60 days; and
- at least one previous no-show.

This segment has an observed no-show rate of **70.52%**.

Priority 2 — Use previous no-show history as a core targeting variable

Previous no-show history provides a strong and consistent differentiation between appointment groups.

It should therefore form part of any attendance-support prioritization logic and future modelling work.

Priority 3 — Pay particular attention to long-lead appointments

Appointments booked substantially in advance should receive additional consideration because no-show rates increase markedly at longer booking horizons.

Priority 4 — Use distance as a secondary prioritization variable

Distance can provide additional information within high-risk appointments, particularly where HealthConnect needs to distinguish between otherwise similar high-risk appointments.

Priority 5 — Treat reminder status as supportive evidence, not proof of effectiveness

Reminder status should continue to be monitored, but the available observational data does not justify concluding that reminders cause lower no-show rates.

8.12 Specific HealthConnect Actions

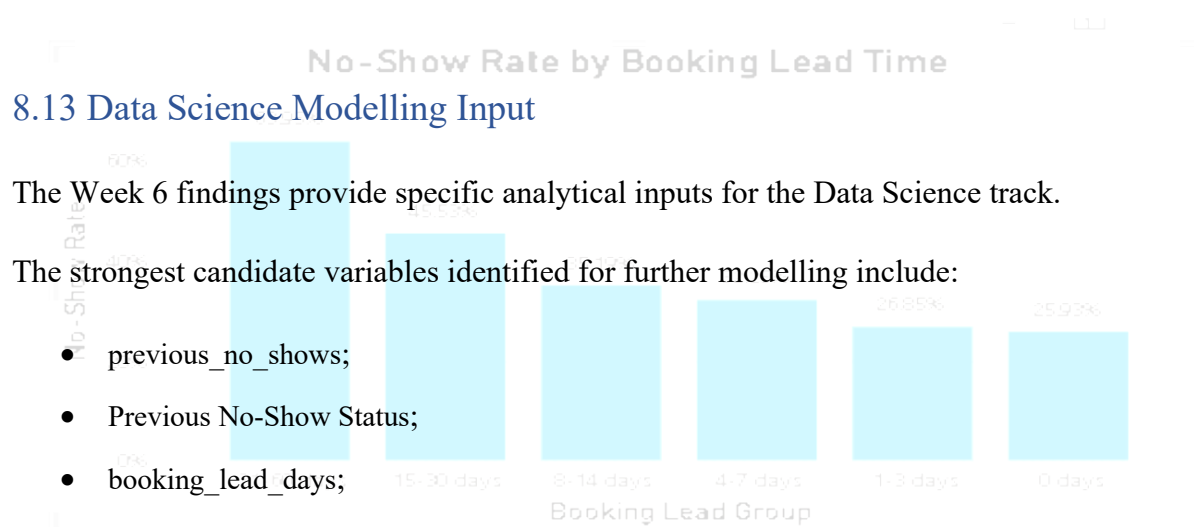
The analytical evidence supports the following candidate actions for HealthConnect.

Table 8.6 — Actions Supported by Analytical Evidence

Analytical Finding	Candidate Action	Evidence Basis	Validation Required
70.52% high-risk segment	Flag high-risk appointments for enhanced attendance support	Joint booking lead + previous no-show analysis	Week 7 testing
1+ previous no-shows = 57.83%	Include previous no-show history in prioritization	+11.53 pp difference	Operational feasibility
31–60-day lead = 63.95%	Consider earlier confirmation/follow-up	Strong lead-time gradient	Test timing and workload
Greater distance within high-risk group	Use distance for secondary prioritization	Increasing observed rates across most distance bands	Validate stability

Analytical Finding	Candidate Action	Evidence Basis	Validation Required
Reminder association = 4.77 pp	Continue monitoring reminder status	Lower observed rate overall	Controlled/validated testing
Appointment type differences	Use type only for secondary segmentation	Moderate variation	Validate sample stability

These actions should be regarded as **candidate operational responses** rather than proven interventions. The analysis establishes evidence for prioritization, but not causal effectiveness.



8.13 Data Science Modelling Input

The Week 6 findings provide specific analytical inputs for the Data Science track.

The strongest candidate variables identified for further modelling include:

- previous_no_shows;
- Previous No-Show Status;
- booking_lead_days;
- Booking Lead Group;
- distance_to_clinic_km;
- Distance Group;
- appointment_type;
- reminder_sent;
- appointment timing characteristics; and
- interactions between booking lead time and previous no-show history.

The most important interaction identified by the analysis is:

Booking Lead Time × Previous No-Show History

This interaction should be considered during feature engineering and model development because the combined segment produced a substantially higher observed no-show rate than either factor considered independently.

The analysis also identifies a potential three-way analytical structure involving:

Booking Lead Time × Previous No-Show History × Reminder Status

The highest observed subgroup was:

- 31–60 days;
- 1+ previous no-shows;
- no reminder;
- **73.98% no-show rate; n = 269.**

The corresponding subgroup where a reminder was sent recorded:

- **69.21% no-show rate; n = 708.**

These results can inform modelling and error-analysis decisions, but they should not be interpreted as evidence that reminder status causes the difference.

Modelling safeguards

Before these variables are used in predictive modelling, their timing and availability must be verified.

In particular:

- appointment_outcome must not be used as a predictor because it is the target outcome;
- waiting_time_minutes should not be used as a pre-appointment predictor unless its measurement timing is confirmed;
- reminder_sent should be assessed for potential leakage depending on when the prediction is intended to occur;
- derived high-risk segments should be treated as decision-support indicators rather than causal labels; and

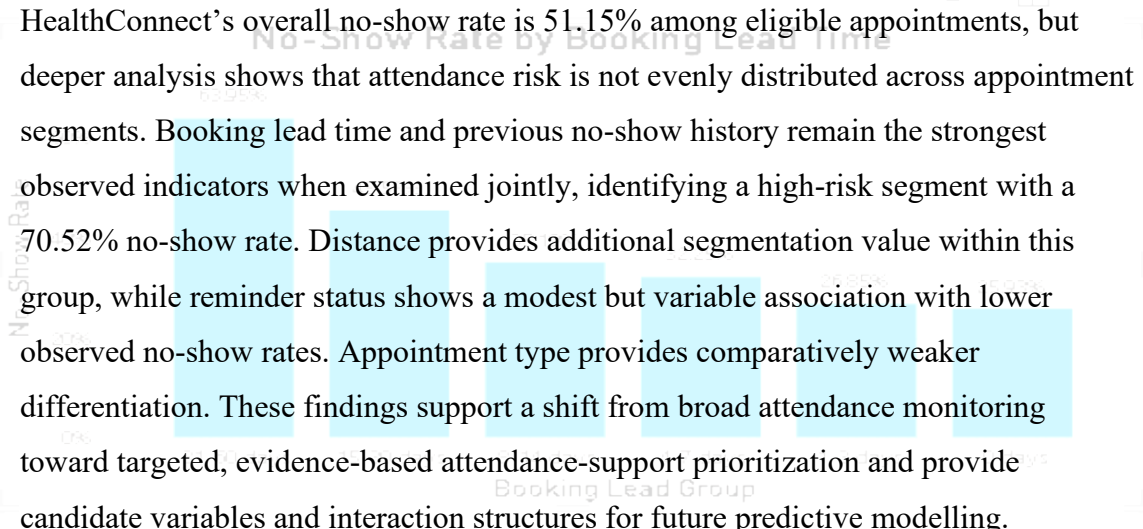
- subgroup performance should be evaluated during subsequent model validation.

The Week 6 analytical results therefore provide **candidate modelling inputs and interaction hypotheses**, rather than evidence that a specific predictive model has already been improved.

8.14 Refined Week 5 Conclusion

The Week 5 conclusion was that HealthConnect has a high overall no-show rate, with booking lead time and previous no-show history emerging as important indicators of attendance risk.

Week 6 provides a more precise conclusion.



HealthConnect's overall no-show rate is 51.15% among eligible appointments, but deeper analysis shows that attendance risk is not evenly distributed across appointment segments. Booking lead time and previous no-show history remain the strongest observed indicators when examined jointly, identifying a high-risk segment with a 70.52% no-show rate. Distance provides additional segmentation value within this group, while reminder status shows a modest but variable association with lower observed no-show rates. Appointment type provides comparatively weaker differentiation. These findings support a shift from broad attendance monitoring toward targeted, evidence-based attendance-support prioritization and provide candidate variables and interaction structures for future predictive modelling.

This refined conclusion is stronger than the Week 5 conclusion because it identifies not only the existence of a no-show problem, but also a specific segment where the problem is substantially more concentrated.

8.15 Decision-Support Interpretation and Caution

The findings in this chapter should be interpreted as **observed associations within the HealthConnect dataset**.

The analysis does not establish that:

- ✓ long booking lead time causes no-shows;
- ✓ previous no-shows cause future no-shows;
- ✓ distance causes missed appointments;
- ✓ reminders cause lower no-show rates; or
- ✓ any proposed intervention will reduce missed appointments.

Instead, the findings identify characteristics and combinations of characteristics associated with different levels of observed no-show risk.

This distinction is essential for responsible use of the analysis. The purpose of the Week 6 work is to provide a stronger evidence base for prioritization, operational planning, dashboard development, and future modelling—not to claim causal effects that the available observational data cannot establish.

8.16 Chapter Conclusion

The Week 6 advanced analysis substantially strengthens the analytical value of the HealthConnect project.

The most important finding is the identification of a **70.52% observed no-show segment among appointments booked 31–60 days in advance for patients with at least one previous no-show**. This segment provides a practical decision-support signal because it combines two independently meaningful indicators into a clearly identifiable appointment group.

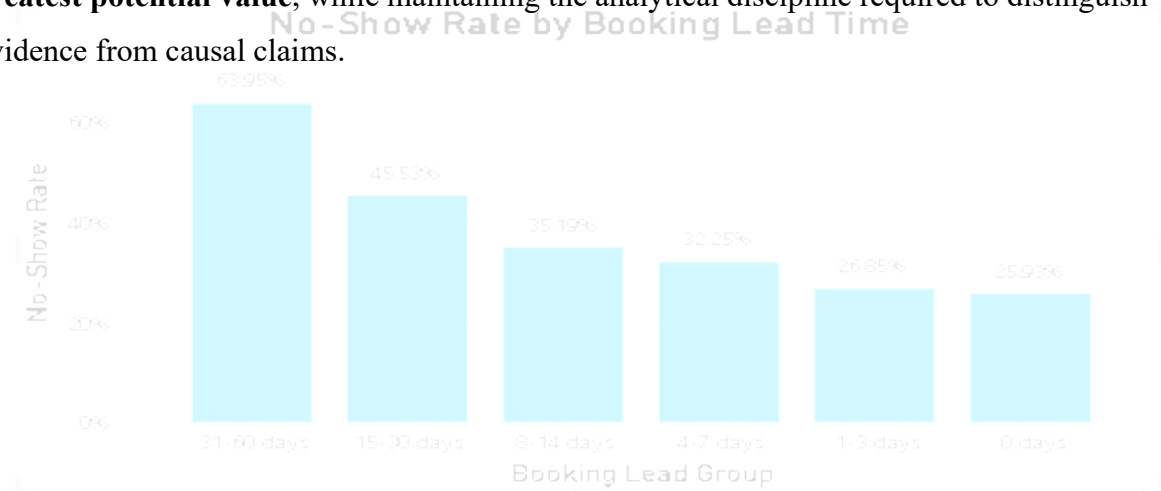
Previous no-show history and booking lead time remain the strongest individual indicators. Distance provides additional prioritization value within the high-risk segment, while reminder status and appointment type provide more limited and context-dependent information.

The resulting decision-support direction is therefore:

- ✓ **Identify high-risk appointments using validated appointment characteristics.**
- ✓ **Prioritize previous no-show history and booking lead time.**
- ✓ **Use distance as a secondary refinement variable.**

- ✓ **Treat reminders as an observational monitoring variable rather than a proven intervention.**
- ✓ **Use appointment type as secondary segmentation rather than the primary risk criterion.**
- ✓ **Provide the validated analytical patterns to Data Science as candidate modelling features and interaction hypotheses.**
- ✓ **Subject the decision-support indicators and proposed interventions to further testing in Week 7.**

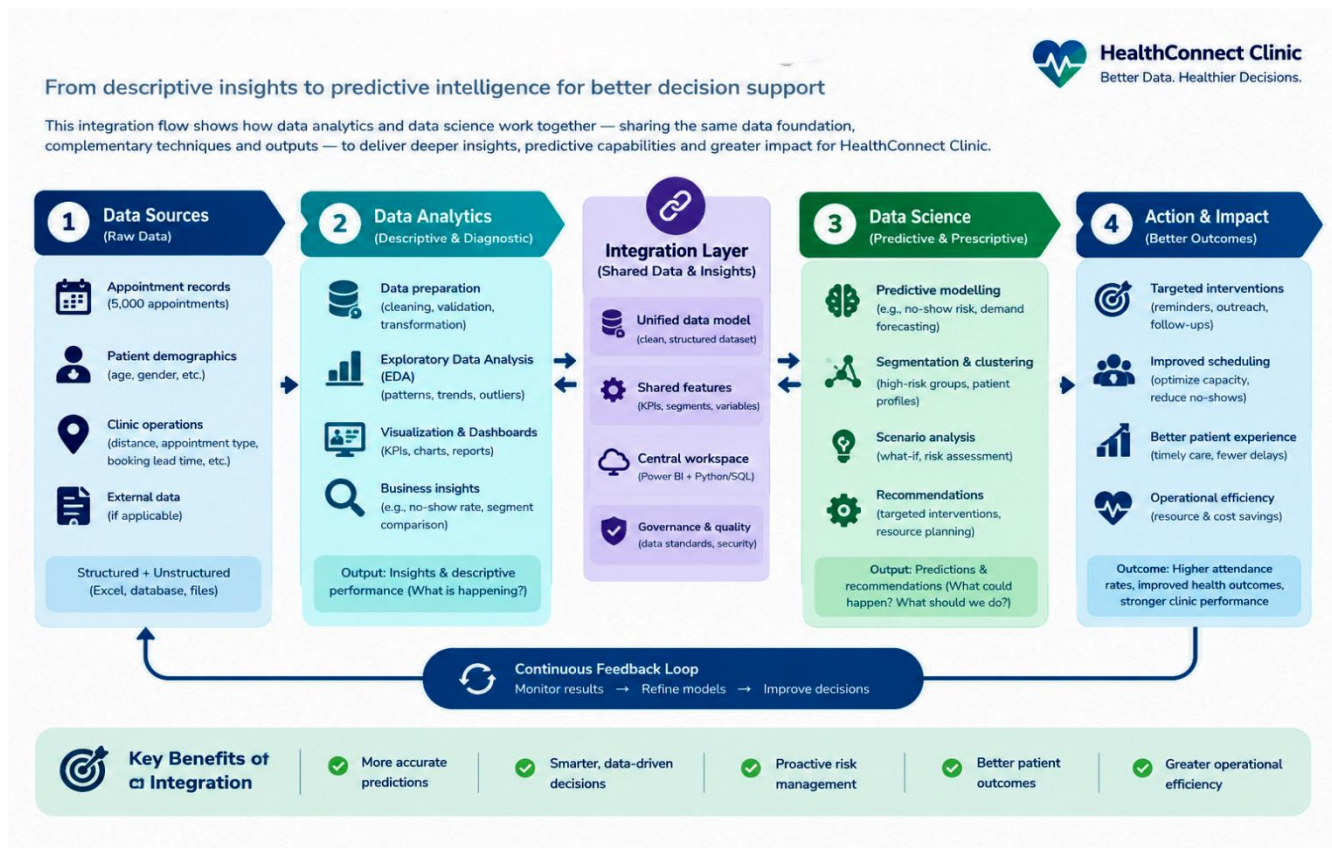
The Week 6 contribution therefore moves HealthConnect from **describing missed appointments** toward **identifying where attendance-support resources may have the greatest potential value**, while maintaining the analytical discipline required to distinguish evidence from causal claims.



Chapter 9

Cross-Track Integration

Figure 9.1 — Data Analytics → Data Science Integration Flow



9.1 Purpose

Week 6 requires the HealthConnect project tracks to move beyond parallel work and demonstrate meaningful integration between analytical findings and the development of the wider solution.

For the Data Analytics track, the primary cross-track dependency is the **Data Science track**. The purpose of this integration is to ensure that validated analytical findings are not treated as an

isolated reporting output, but are used to inform predictive modelling decisions, feature selection, interaction analysis, and subsequent model validation.

The Week 6 Data Analytics work therefore provides evidence about which appointment and patient characteristics are most strongly associated with observed no-show behavior and identifies candidate variables and interactions that can be investigated through predictive modelling.

The assignment requires the collaboration to produce a meaningful change, improvement, decision, integration, or validated output rather than communication alone.

9.2 Cross-Track Partner

Collaborating Track: Data Science

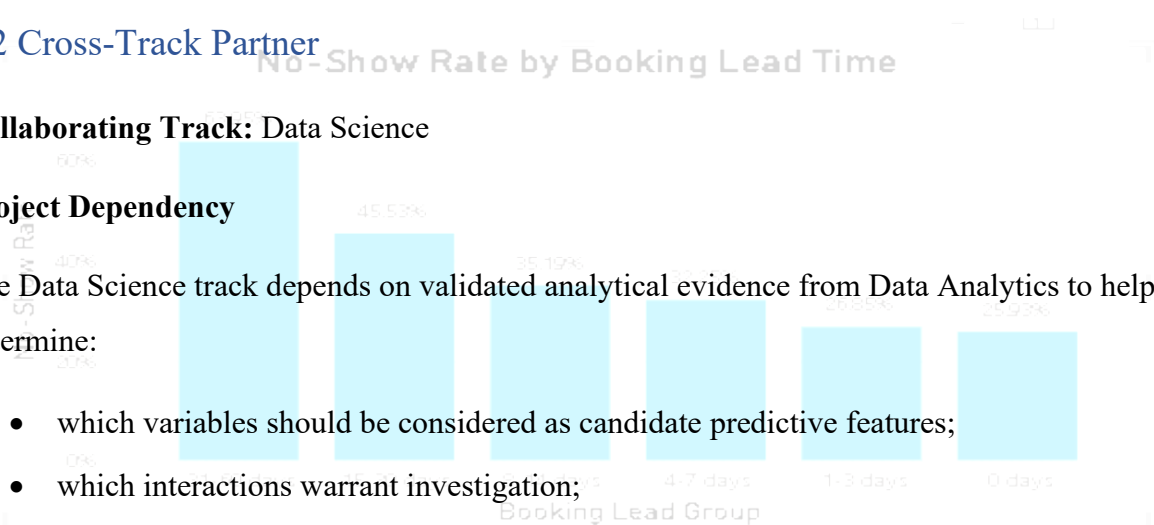
Project Dependency

The Data Science track depends on validated analytical evidence from Data Analytics to help determine:

- which variables should be considered as candidate predictive features;
- which interactions warrant investigation;
- which appointment segments require particular attention during model development;
- which variables may introduce leakage or timing problems;
- which findings should be tested through predictive modelling; and
- which analytical patterns should be examined during model validation and error analysis.

The Data Analytics track, in turn, benefits from Data Science modelling because predictive analysis can determine whether the observed patterns identified during descriptive and segmented analysis remain useful when multiple variables are considered simultaneously.

This creates a two-way dependency:



Data Analytics identifies and validates important patterns → Data Science tests those patterns through predictive modelling → modelling results can subsequently refine the analytical interpretation and decision-support framework.

9.3 Information and Outputs Provided to Data Science

The Data Analytics track provides the following validated findings as modelling inputs.

9.3.1 Overall Attendance Outcome

The eligible analytical population contains:

- **4,737 eligible appointments**
- **2,423 no-shows**
- **2,314 attended appointments**
- **51.15% no-show rate**
- **48.85% attendance rate**

The relatively balanced distribution between the two eligible outcome classes means that severe class imbalance is not an immediate concern at the descriptive stage.

However, model performance should still be assessed using appropriate classification metrics rather than accuracy alone.

9.3.2 Previous No-Show History

Previous no-show history was identified as a major behavioral indicator.

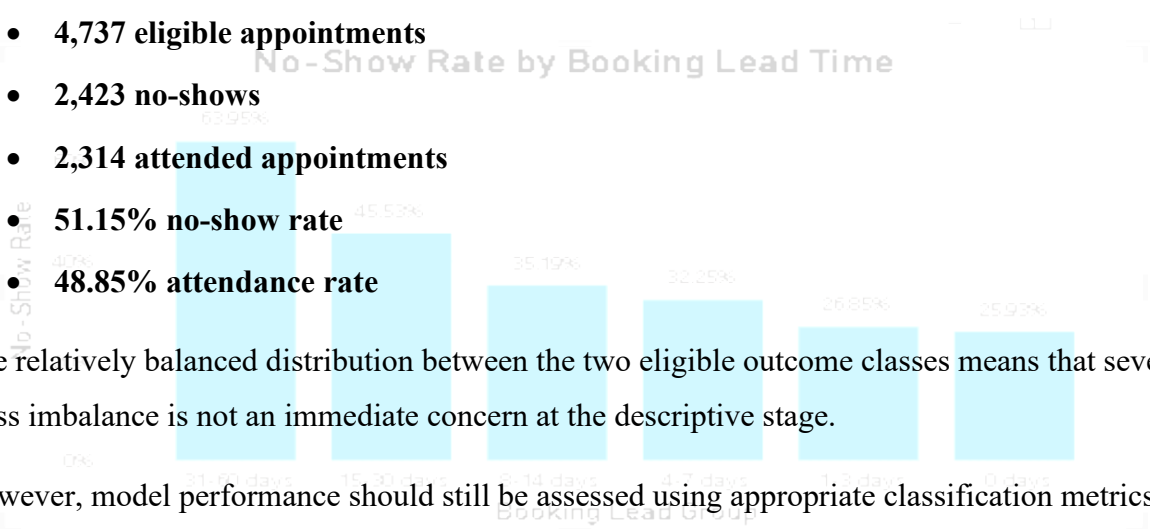


Table 9.1 —Previous no-show history and No-show Rate

Patient History	No-Show Rate
No previous no-shows	46.30%
1+ previous no-shows	57.83%
Difference	+11.53 pp

This variable should therefore be considered a candidate predictive feature.

The analysis also suggests that previous no-show history may interact with booking lead time.

9.3.3 Booking Lead Time

Booking lead time was another major observed indicator.

The overall analysis showed:

- Same-day bookings: **25.93%**
- 31–60-day bookings: **63.95%**

The segmented analysis further showed that the 31–60-day category remained particularly important when combined with previous no-show history.

This makes both:

- booking_lead_days; and
- the derived Booking Lead Group

potential modelling inputs, subject to the modelling design and intended prediction point.

9.4 Key Interaction Provided for Modelling

The most important interaction identified during Week 6 is:

Booking Lead Time × Previous No-Show History

The analysis demonstrated that these variables remain meaningful when examined jointly.

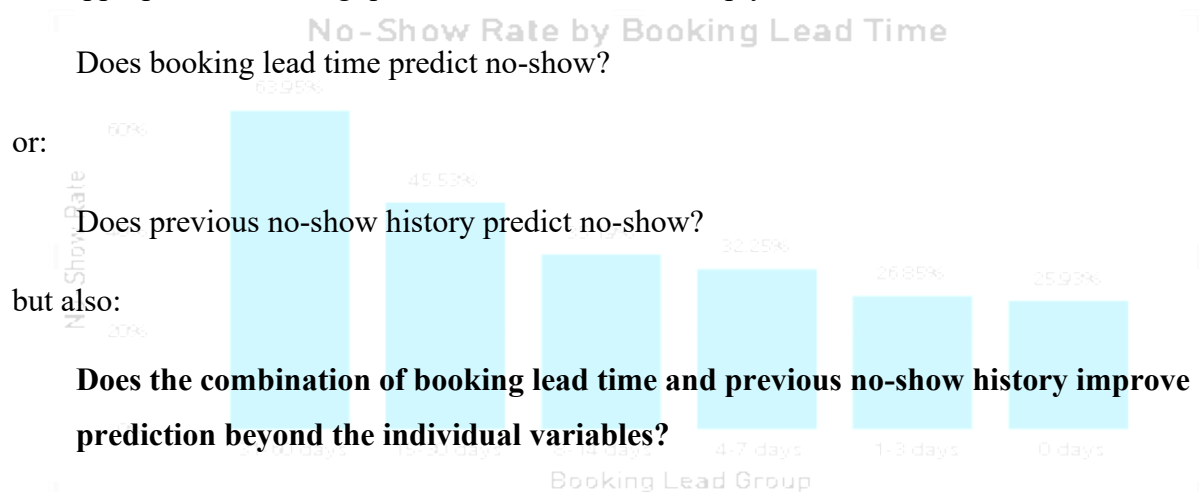
Within the 31–60-day booking window:

- patients with no previous no-shows had a **59.07%** no-show rate;
- patients with 1+ previous no-shows had a **70.52%** no-show rate.

The combined high-risk segment therefore recorded a **70.52% observed no-show rate**.

This finding is important for Data Science because it suggests that the relationship between the variables may contain useful interaction information that would not be captured by considering each variable independently.

The appropriate modelling question is therefore not simply:



This should be tested empirically rather than assumed.

9.5 Additional Candidate Features

The Week 6 analysis identified several additional variables that may contribute to predictive modelling.

Primary candidate features

- ✓ Previous no-show history
- ✓ Booking lead time
- ✓ Distance to clinic

- ✓ Appointment type

Secondary candidate features

5. Reminder status
6. Appointment day
7. Appointment time
8. Previous appointment volume
9. Age group
10. Gender, subject to appropriate analytical and ethical consideration

Candidate derived features

- Previous No-Show Status
- Booking Lead Group
- Distance Group
- High-Risk Segment Indicator
- Booking Lead Time $\times$ Previous No-Show History
- High-Risk Segment $\times$ Reminder Status

These are **candidate modelling inputs**, not confirmed final features.

Final feature selection should be determined through Data Science model development, validation, feature importance analysis, and timing/leakage assessment.

9.6 Reminder Status as a Modelling Consideration

Reminder status requires particular caution.

Week 6 found:

- Reminder sent: **49.86% no-show rate**
- No reminder: **54.63% no-show rate**
- Difference: **4.77 percentage points**

Within the high-risk segment:

- Reminder sent: **69.21%**
- No reminder: **73.98%**

These findings make reminder status analytically relevant, but they do not establish that reminders cause lower no-show rates.

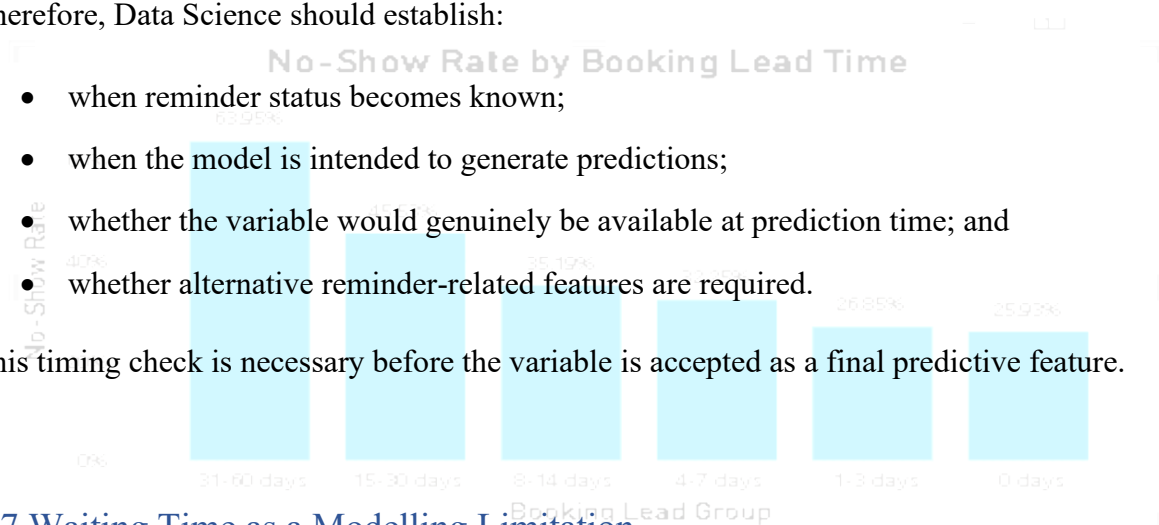
More importantly, the timing of the reminder variable must be confirmed.

If the intended predictive model makes its prediction **at the time of booking**, but reminder_sent is only known later, using that variable could introduce information leakage.

Therefore, Data Science should establish:

- when reminder status becomes known;
- when the model is intended to generate predictions;
- whether the variable would genuinely be available at prediction time; and
- whether alternative reminder-related features are required.

This timing check is necessary before the variable is accepted as a final predictive feature.



9.7 Waiting Time as a Modelling Limitation

waiting_time_minutes was intentionally excluded from the primary no-show-risk modelling recommendations.

The reason is that the available documentation does not establish with sufficient certainty that waiting time is measured before the appointment outcome is determined.

If waiting time is recorded after attendance or as part of the service experience, using it to predict whether a patient will attend would introduce temporal leakage.

The Week 6 analytical decision is therefore:

Waiting time should be treated primarily as a service-experience variable unless its measurement timing is confirmed to occur before the no-show outcome.

Data Science should confirm this timing before considering the variable for predictive modelling.

9.8 Outcome Variable Protection

The appointment_outcome field contains the observed result:

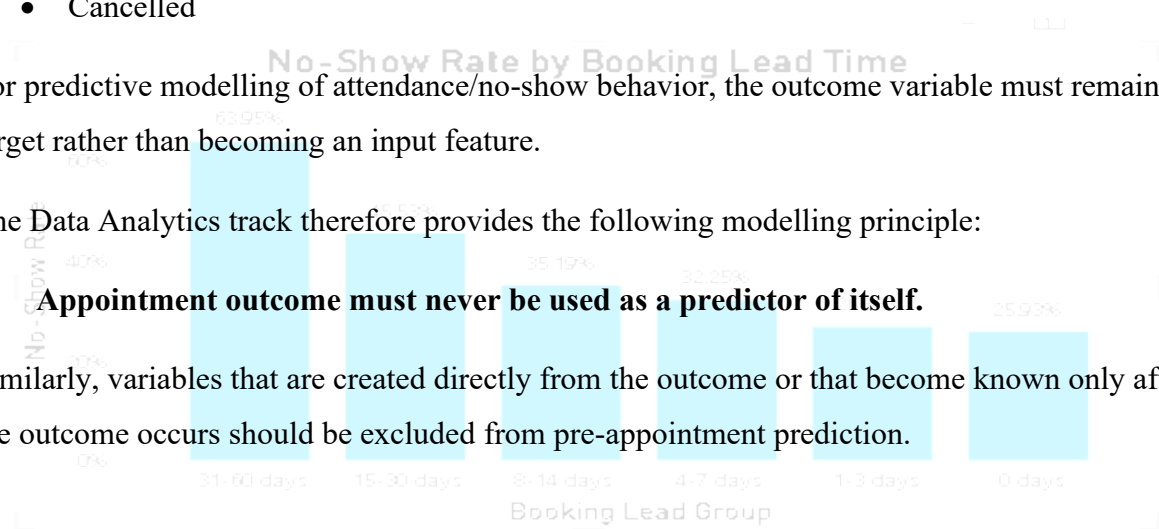
- Attended
- No-Show
- Cancelled

For predictive modelling of attendance/no-show behavior, the outcome variable must remain the target rather than becoming an input feature.

The Data Analytics track therefore provides the following modelling principle:

Appointment outcome must never be used as a predictor of itself.

Similarly, variables that are created directly from the outcome or that become known only after the outcome occurs should be excluded from pre-appointment prediction.



9.9 High-Risk Segment as a Decision-Support Input

The 70.52% high-risk segment is useful as a decision-support indicator:

31–60-day booking lead time + 1+ previous no-shows

However, it should not automatically be treated as a definitive predictive classification.

Its role in the Week 6 workflow is to:

- ✓ identify a high-observed-risk segment;
- ✓ provide a benchmark for predictive modelling;
- ✓ support feature and interaction selection;

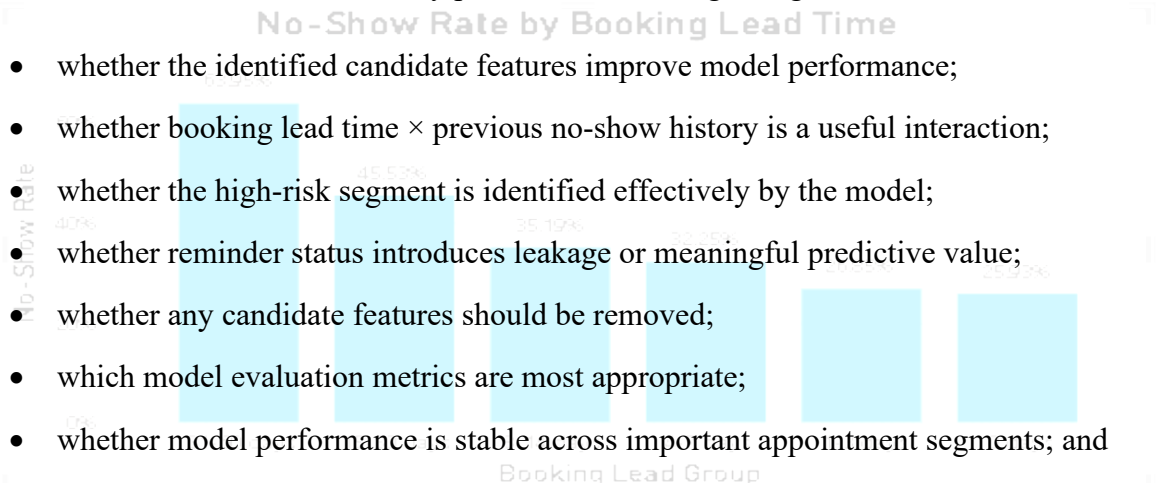
- ✓ provide a candidate operational prioritization rule; and
- ✓ create a reference group for subsequent model performance analysis.

Data Science can then test whether a predictive model can identify similar high-risk appointments more accurately or more flexibly than this simple rule.

9.10 Information Expected from Data Science

The cross-track integration also establishes the information that Data Analytics requires from Data Science.

The Data Science track should ideally provide evidence regarding:

- 
- whether the identified candidate features improve model performance;
 - whether booking lead time $\times$ previous no-show history is a useful interaction;
 - whether the high-risk segment is identified effectively by the model;
 - whether reminder status introduces leakage or meaningful predictive value;
 - whether any candidate features should be removed;
 - which model evaluation metrics are most appropriate;
 - whether model performance is stable across important appointment segments; and
 - whether model predictions are sufficiently useful for operational decision-making.

This information can subsequently be used to refine the Data Analytics dashboard, recommendations, and Week 7 testing plan.

9.11 Integration Activity

The principal integration activity for Week 6 is the conversion of validated analytical findings into a structured set of **candidate modelling inputs and interaction hypotheses**.

The Data Analytics track translated the Week 6 findings into:

Analytical input

Highest observed-risk segment

31–60 days + 1+ previous no-shows

→ **70.52% no-show rate**

Candidate individual features

- Previous no-show history
- Booking lead time
- Distance
- Appointment type
- Reminder status, subject to timing validation

Candidate interaction

Booking Lead Time × Previous No-Show History

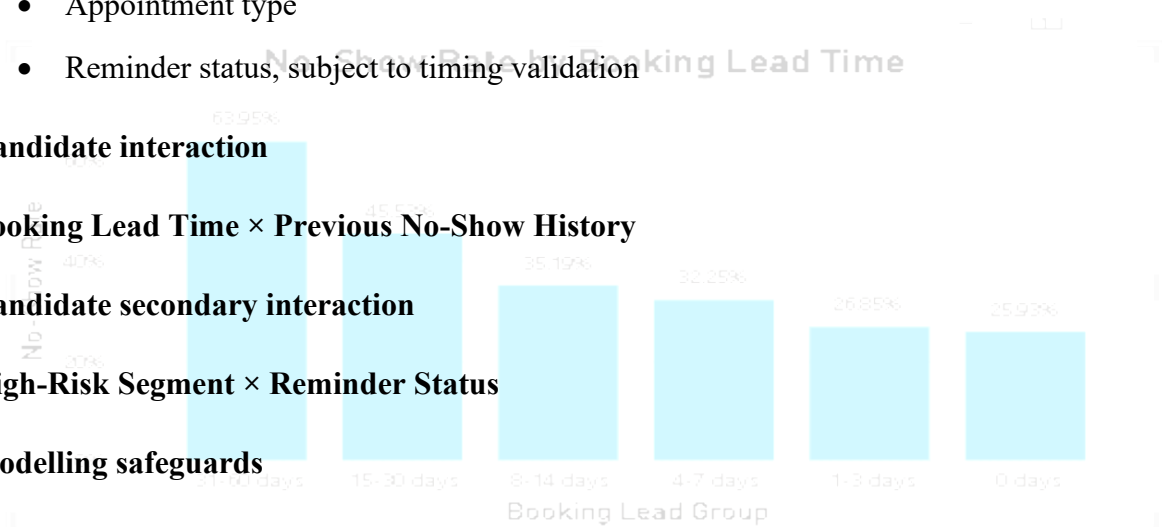
Candidate secondary interaction

High-Risk Segment × Reminder Status

Modelling safeguards

- prevent outcome leakage;
- verify variable timing;
- assess subgroup stability;
- validate predictive performance;
- distinguish prediction from causal explanation.

This represents a substantive analytical hand-off rather than a simple statement that the tracks communicated.



9.12 What Changed as a Result of the Integration

The Week 6 analytical work changed the cross-track direction in several important ways.

Change 1 — From individual variables to interactions

Week 5 primarily identified individual patterns.

Week 6 demonstrated that booking lead time and previous no-show history remain meaningful when examined jointly.

This creates a stronger basis for interaction testing in predictive modelling.

Change 2 — From general no-show analysis to high-risk segmentation

The analysis introduced a specific decision-support segment with a **70.52% observed no-show rate**.

This provides Data Science with a concrete high-risk population against which model predictions can be evaluated.

Change 3 — More disciplined feature selection

The deeper analysis showed that not every available variable should automatically become a model feature.

Reminder status requires timing validation, while waiting time may introduce temporal leakage.

This improves the quality of the modelling input.

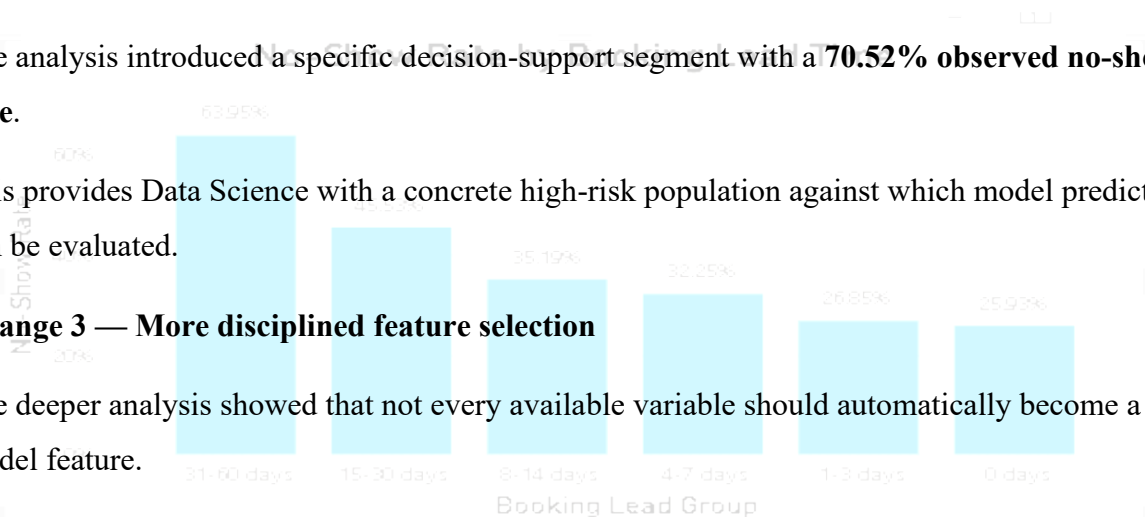
Change 4 — More precise interpretation of reminder data

The Week 5 reminder comparison could have been interpreted too broadly.

Week 6 demonstrated that the relationship varies by segment and therefore should be treated as observational rather than causal.

This prevents the modelling and business tracks from treating reminder status as proven intervention effectiveness.

Change 5 — Clearer Week 7 testing requirements



The integration creates specific testing questions for the next phase:

- Does the model reproduce the observed high-risk pattern?
- Does the booking lead $\times$ previous no-show interaction improve prediction?
- Are predictions stable across appointment segments?
- Does reminder status remain useful after other variables are considered?
- Are any candidate features unavailable at the intended prediction point?
- Does the model provide operationally useful prioritization?

9.13 Evidence of Cross-Track Integration

For the final Week 6 submission, the integration should be supported by concrete evidence rather than communication records alone.

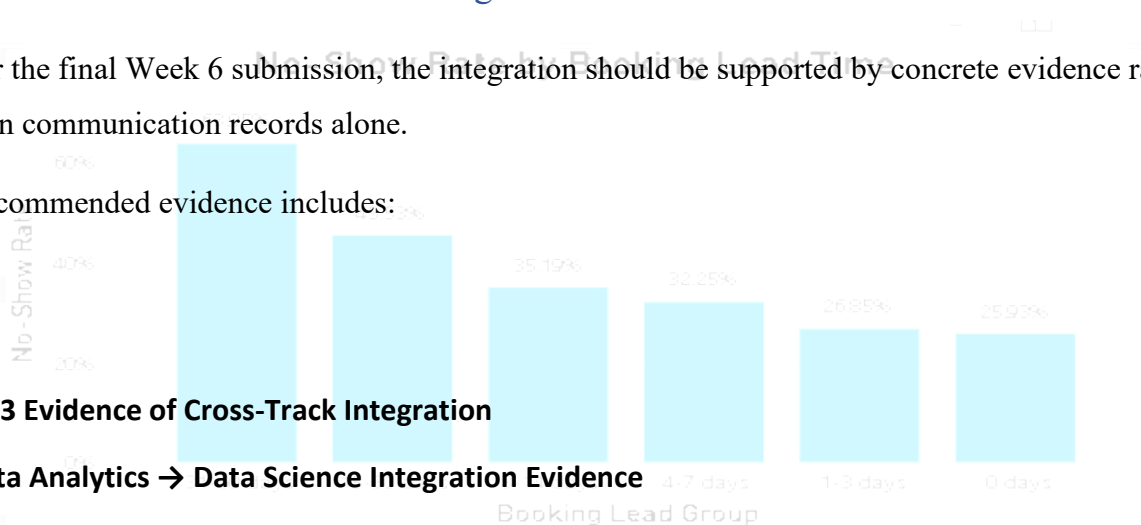
Recommended evidence includes:

9.13 Evidence of Cross-Track Integration

Data Analytics → Data Science Integration Evidence

During Week 6, the Data Analytics track translated validated analytical findings into structured modelling inputs for the Data Science track. The integration was based on the validated KPI framework, advanced segmentation analysis, and the identification of the interaction between booking lead time and previous no-show history.

Table 9.2 —Cross Track Recommendation Evidence

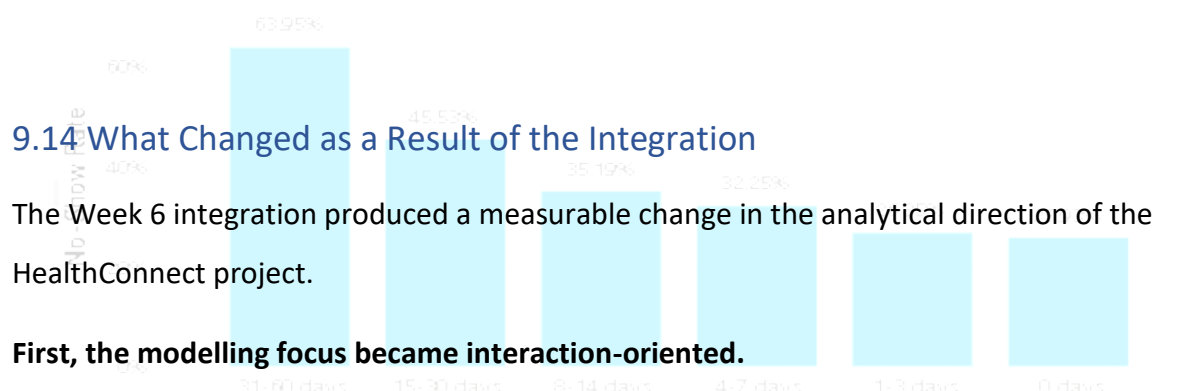


Integration Requirement	Week 6 Evidence
Track collaborated with	Data Science
Project dependency	Predictive modelling requires validated analytical findings, candidate features, interaction structures and clearly defined outcome variables.
Information/output received	The Data Analytics track worked from the validated HealthConnect appointment dataset and the KPI framework established from the Week 5 analysis.
Information/output provided	Validated risk patterns, the 70.52% high-risk segment, candidate modelling variables, the Booking Lead Time × Previous No-Show History interaction, and modelling cautions relating to timing and leakage.
Integration activity completed	Week 6 advanced segmentation was translated into a structured candidate-feature and interaction register for Data Science consideration.
What changed as a result	The analytical hand-off changed the modelling direction from consideration of individual variables alone toward testing interactions, particularly Booking Lead Time × Previous No-Show History. It also introduced explicit feature-timing, leakage and subgroup-stability checks.
Evidence retained	Validated KPI results, high-risk segment analysis, interaction analysis, candidate feature register, modelling safeguards and the Week 6 cross-track contribution documented in this report.

Integration Requirement	Week 6 Evidence
Downstream model change	Not yet evidenced in the available Week 6 materials. Confirmation of feature adoption, model modification or model-performance improvement is therefore deferred to Week 7 validation.

This wording is stronger because it directly answers all seven things AnalystLab requires: **who, dependency, received, provided, activity, change and evidence**. AnalystLab explicitly requires these elements and says communication alone is insufficient.

The existing report already provides the substantive analytical material supporting this: the 70.52% segment, candidate variables, interaction and modelling safeguards.



First, the modelling focus became interaction-oriented. Week 5 primarily examined individual attendance-risk indicators. Week 6 demonstrated that booking lead time and previous no-show history provide additional analytical value when considered jointly. The combination of a 31–60-day booking lead time and at least one previous no-show identified a segment with an observed no-show rate of **70.52%**.

Second, the candidate feature set became more disciplined.

Previous no-show history, booking lead time, distance, appointment type and reminder status were identified as potential modelling inputs, while reminder status remains subject to timing validation. Waiting time was also flagged as a potential temporal-leakage risk.

Third, the project gained an interpretable benchmark for predictive modelling.

The 70.52% high-risk segment provides Data Science with a clearly defined population against which predictive model performance can subsequently be evaluated.

Fourth, the interpretation of reminder information became more cautious.

The Week 6 analysis treats the relationship between reminder status and no-show behaviour as observational rather than causal. This prevents the project from incorrectly interpreting the observed difference as evidence that reminders cause improved attendance.

Finally, the integration established specific Week 7 validation requirements.

Data Analytics and Data Science must jointly determine whether the candidate features are available at the prediction point, whether the interaction improves predictive performance, whether the high-risk segment remains meaningful under modelling, and whether the resulting predictions can support operational decision-making.

Important status statement

The Week 6 report does not claim that the Data Science model was modified or improved. Such a claim will only be made when a corresponding Data Science artefact, model comparison, feature-selection record or documented modelling decision is available.

9.15 Evidence Availability and Outstanding Evidence

Evidence available within the Week 6 Data Analytics submission:

- Validated KPI calculations.
- Advanced segmentation results.
- 70.52% high-risk segment analysis.
- Booking Lead Time × Previous No-Show History interaction analysis.
- Candidate feature register.
- Feature-timing and leakage considerations.

- Updated dashboard/analytical reporting.
- Week 7 analytical testing matrix.
- Documented limitations and analytical decisions.

Evidence still required to confirm downstream Data Science integration:

- Data Science notebook or model version showing use of the analytical inputs.
- Feature-selection record showing which candidate variables were retained or rejected.
- Model comparison showing whether the proposed interaction or features affected model performance.
- Documented Data Science decision regarding the 70.52% high-risk segment.
- Cross-track meeting/decision record, where available.

Status: The Week 6 Data Analytics contribution is complete as an analytical hand-off, but downstream model adoption and performance improvement remain subject to confirmation.

9.16 Integration Status

At the point of documenting the Week 6 Data Analytics contribution, the analytical outputs and candidate modelling inputs are clearly defined.

However, the available evidence does not establish a completed Data Science model revision or a confirmed improvement in model performance.

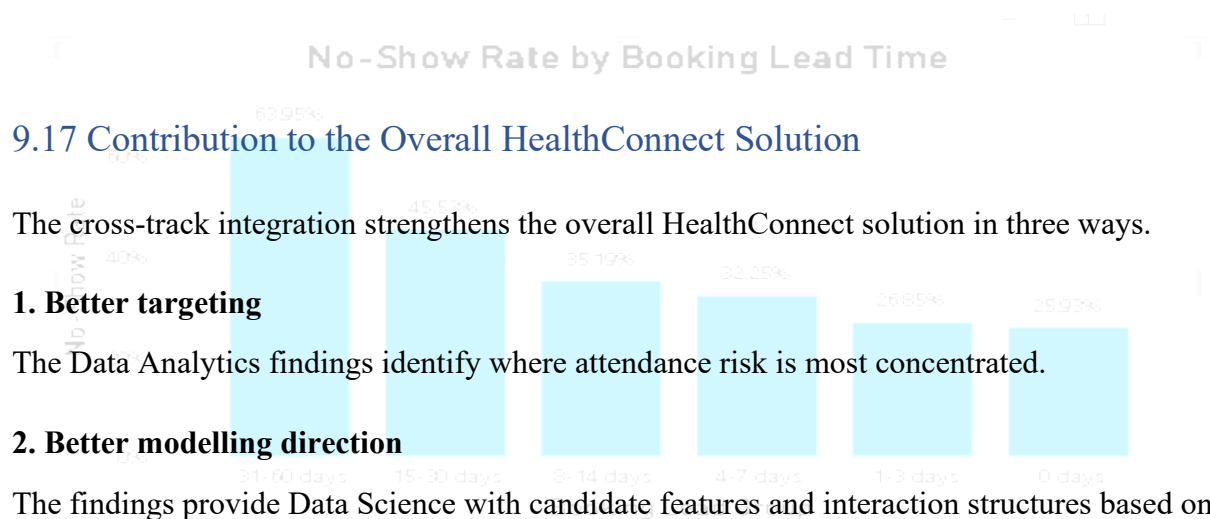
Therefore, the appropriate project status is:

Table 9.3 —Project Status

Integration Component	Status
Validated analytical findings	Completed
High-risk segment identified	Completed
Candidate modelling variables identified	Completed
Interaction hypothesis identified	Completed

Integration Component	Status
Leakage/timing risks identified	Completed
Analytical findings prepared as modelling inputs	Completed
Confirmed Data Science feature adoption	Requires evidence
Confirmed model modification	Requires evidence
Confirmed model-performance improvement	Requires evidence
Downstream modelling validation	Week 7 activity

This distinction is important because Week 6 requires evidence-based integration. It would be inappropriate to claim that the Data Science model changed or improved unless the corresponding artefact or documented decision exists.



9.17 Contribution to the Overall HealthConnect Solution

The cross-track integration strengthens the overall HealthConnect solution in three ways.

1. Better targeting

The Data Analytics findings identify where attendance risk is most concentrated.

2. Better modelling direction

The findings provide Data Science with candidate features and interaction structures based on validated evidence rather than arbitrary variable selection.

3. Better operational decision support

The high-risk segment provides a simple and interpretable benchmark that can eventually be compared with the output of a predictive model.

This creates a logical progression:

**Descriptive analysis → Segmentation → Validated risk indicators → Predictive modelling
→ Decision support → Testing and refinement**

The Data Analytics track therefore contributes not only a dashboard but also the analytical evidence required to connect descriptive business analysis with predictive and operational components of the HealthConnect solution.

9.18 Week 7 Cross-Track Requirements

The Week 6 integration establishes several requirements for Week 7.

Data Analytics and Data Science should jointly validate:

- ✓ whether the candidate features are available at the intended prediction point;
- ✓ whether the booking lead × previous no-show interaction improves predictive performance;
- ✓ whether the 70.52% high-risk segment remains meaningful under predictive modelling;
- ✓ whether model performance is stable across important operational segments;
- ✓ whether reminder status adds predictive value without introducing leakage;
- ✓ whether the model identifies a practical number of high-risk appointments;
- ✓ whether model thresholds are operationally realistic; and
- ✓ whether the resulting predictions can be incorporated into decision-support reporting.

These requirements connect directly to the Week 7 emphasis on testing, refinement, and end-to-end validation.

9.19 Cross-Track Integration Summary

The Week 6 Data Analytics contribution to Data Science can be summarized as follows:

Table 9.4— Week Six Contribution to Data Science

Integration Element	Data Analytics Contribution
Track collaborated with	Data Science

Integration Element	Data Analytics Contribution
Dependency	Predictive modelling requires validated analytical inputs
Key finding provided	31–60 days + 1+ previous no-shows = 70.52% no-show
Strong individual indicators	Booking lead time and previous no-show history
Key interaction	Booking Lead Time × Previous No-Show History
Additional variables	Distance, appointment type, reminder status
Modelling caution	Timing, leakage, subgroup stability
Outcome protection	Appointment outcome remains the target
Decision-support contribution	High-risk segment for prioritization
Week 7 dependency	Model validation and feature testing
Confirmed downstream model change	Not yet evidenced

9.20 Chapter Conclusion

The Week 6 Data Analytics work establishes a meaningful cross-track dependency with Data Science by converting validated descriptive and segmented findings into structured candidate modelling inputs.

The most important contribution is the identification of the interaction between **booking lead time and previous no-show history**, which produces a **70.52% observed no-show rate** for appointments booked 31–60 days in advance among patients with at least one previous no-show.

Additional analytical findings concerning distance, appointment type, and reminder status provide further candidate variables and segmentation structures.

At the same time, Week 6 establishes important modelling safeguards. Reminder status requires timing validation, waiting time may introduce temporal leakage, and appointment outcome must remain the target variable.

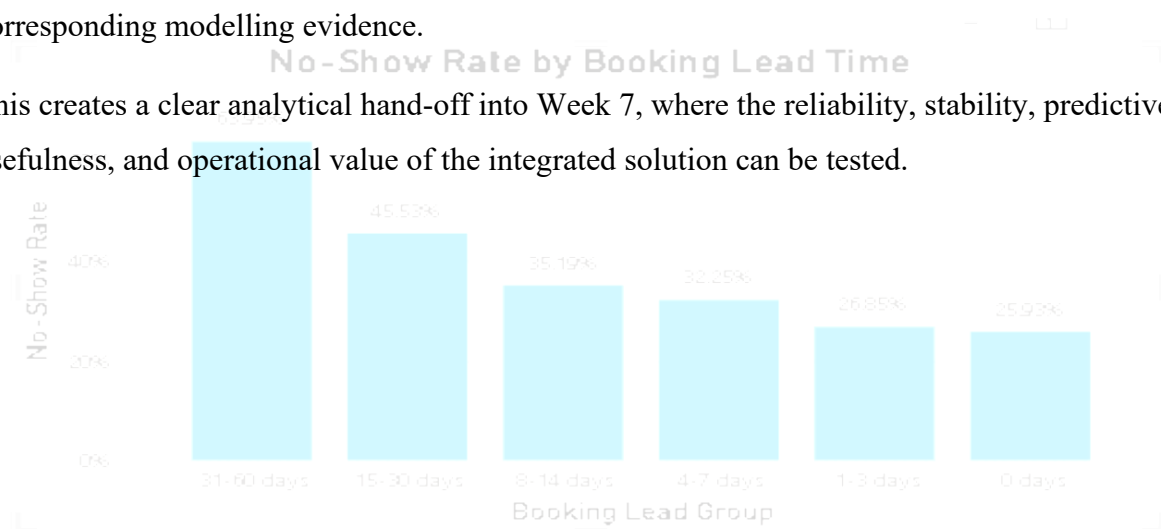
The integration therefore moves the HealthConnect project from:

“What patterns exist in the appointment data?” toward:

“Which validated patterns should be tested through predictive modelling and converted into reliable decision-support indicators?”

The Data Analytics contribution is therefore ready for downstream validation, while any claim that the Data Science model has already changed or improved should be supported by the corresponding modelling evidence.

This creates a clear analytical hand-off into Week 7, where the reliability, stability, predictive usefulness, and operational value of the integrated solution can be tested.



Chapter 10

Updated Dashboard and Analytical Reporting

Figure 10.1— HealthConnect Advanced Appointment Risk Analysis Dashboard

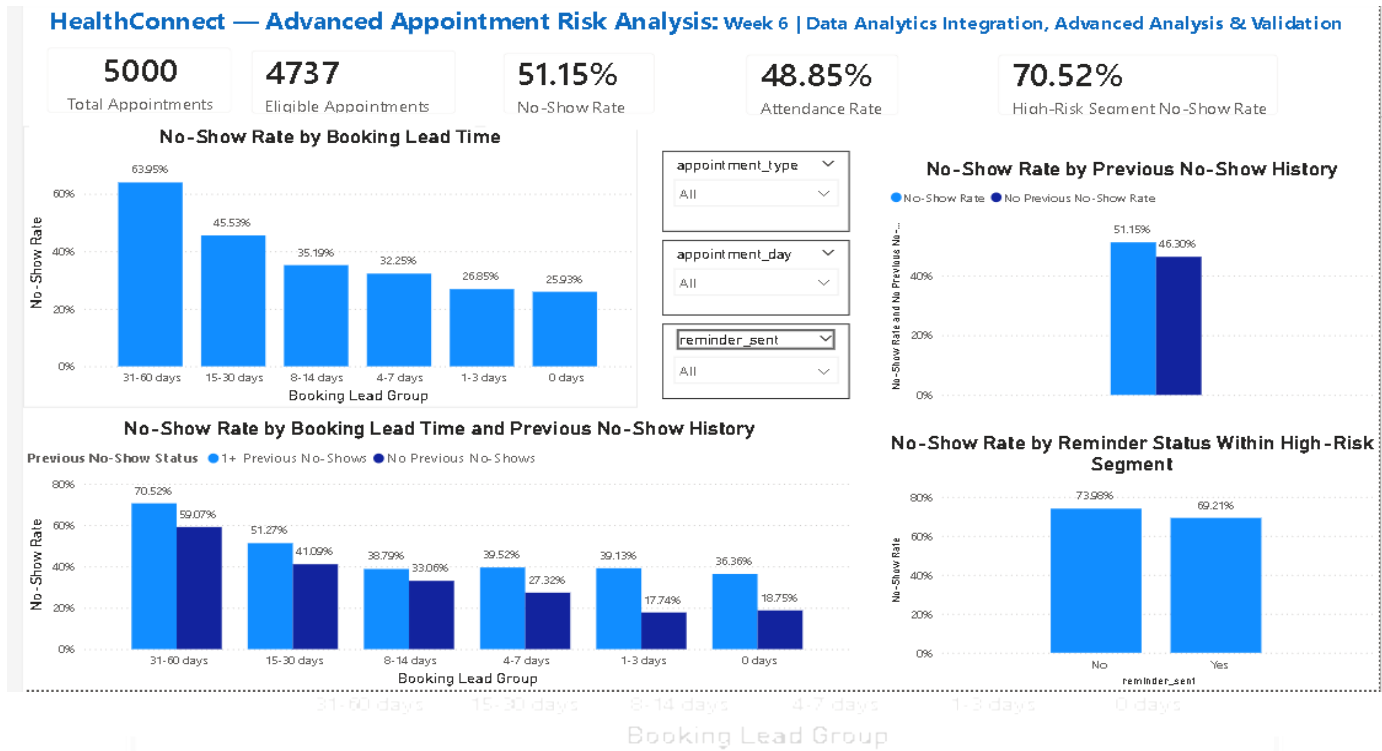
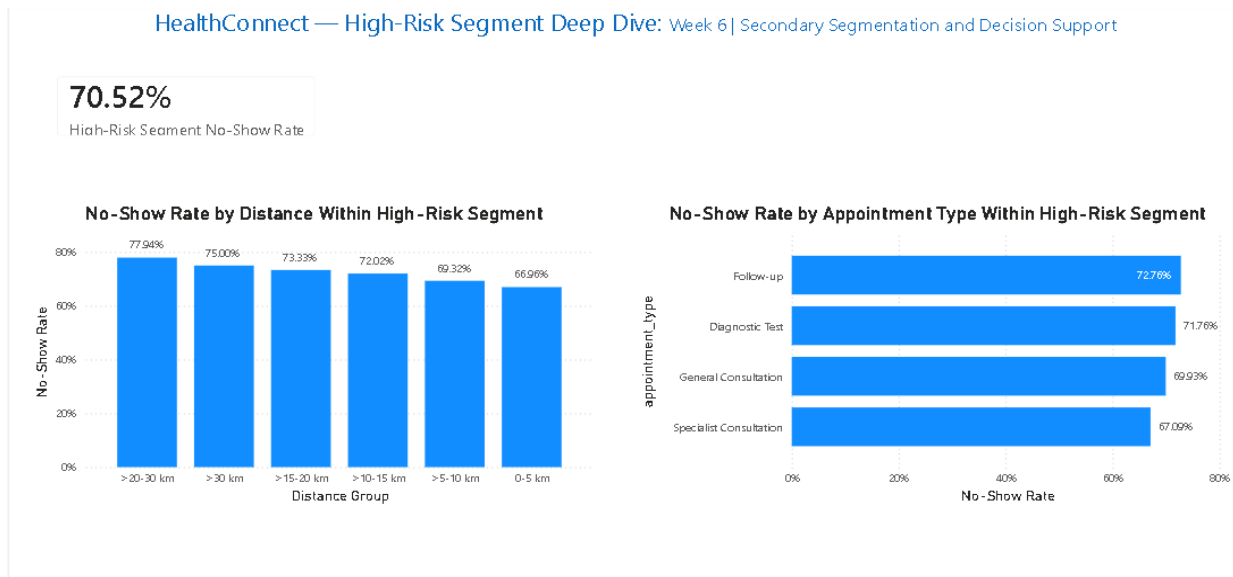


Figure 10.2— HealthConnect High-Risk Segment Deep Dive Dashboard



10.1 Purpose

The Week 6 dashboard update was designed to translate the advanced analytical findings into a more decision-oriented reporting interface.

The Week 5 dashboard provided an initial view of HealthConnect appointment attendance, no-show rates, and major appointment characteristics. It established the core KPIs and helped identify where no-show rates were higher.

Week 6 builds on that foundation rather than replacing it. The dashboard was refined to reflect the validated findings from the advanced analysis, particularly the identification of a high-risk appointment segment based on the combination of booking lead time and previous no-show history.

This approach is consistent with the Week 6 requirement to improve the existing dashboard or analytical report based on new findings rather than reproduce the Week 5 deliverable.

The central reporting question therefore changed from:

Where are no-show rates higher?

to:

Which appointment segments represent the greatest observed attendance risk, and where should HealthConnect Prioritize attendance-support attention?

10.2 Week 5 Dashboard Baseline

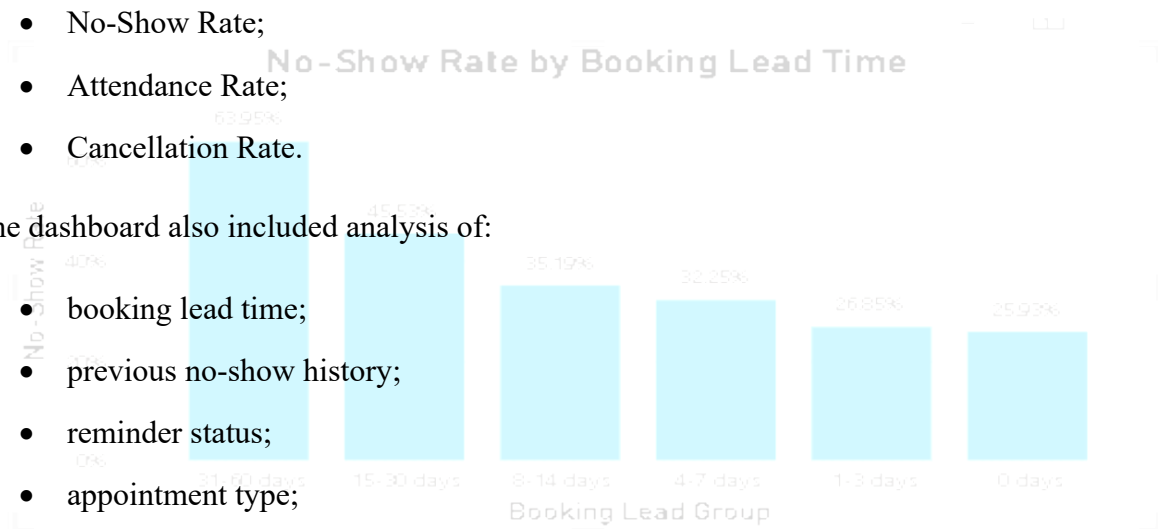
The Week 5 dashboard was designed primarily as an appointment attendance monitoring tool.

Its core metrics included:

- Total Appointments;
- Eligible Appointments;
- No-Show Rate;
- Attendance Rate;
- Cancellation Rate.

The dashboard also included analysis of:

- booking lead time;
- previous no-show history;
- reminder status;
- appointment type;
- distance to clinic;
- appointment day; and
- appointment time.



These visuals successfully established the overall attendance situation and highlighted important patterns.

However, the Week 5 dashboard had a limitation from a decision-support perspective: the most important risk indicators were primarily displayed independently.

For example:

- booking lead time could be examined separately;

- previous no-show history could be examined separately; and
- reminder status could be examined separately.

The deeper Week 6 analysis demonstrated that the combination of booking lead time and previous no-show history provides substantially stronger decision-support value than either variable considered alone.

The dashboard therefore needed to make this interaction visible.

10.3 Week 6 Dashboard Improvement Objective

The Week 6 dashboard improvement had four objectives:

- ✓ **Preserve** the validated Week 5 KPIs.
- ✓ **Add** the high-risk segment identified through advanced analysis.
- ✓ **Visualise** important interactions rather than only individual variables.
- ✓ **Improve** the dashboard's usefulness for operational decision-making and future predictive modelling.

The updated dashboard therefore represents an evolution from:

Monitoring → Segmentation → Decision Support

rather than a complete redesign.

This distinction is important because Week 6 is intended to demonstrate progression from Week 5 while maintaining continuity with the existing HealthConnect analytical work.

10.4 Refined KPI Structure

The Week 6 dashboard retains the core KPIs validated in Chapter 5.

Table 10.1 —Primary KPI Cards

KPI	Week 6 Value	Role
Total Appointments	5,000	Overall appointment volume
Eligible Appointments	4,737	Primary attendance-analysis denominator
No-Show Rate	51.15%	Core attendance-risk KPI
Attendance Rate	48.85%	Core attendance KPI
High-Risk Segment Rate	70.52%	Decision-support indicator

The Cancellation Rate of **5.26%** remains an important contextual metric and may be displayed where dashboard space permits.

The primary no-show and attendance KPIs continue to use eligible appointments, defined as appointments with an outcome of either **Attended** or **No-Show**. Cancelled appointments are excluded from these denominators.

10.5 New High-Risk Segment KPI

The most important Week 6 dashboard addition is the **High-Risk Segment Rate**.

Definition

The high-risk segment consists of appointments that meet both conditions:

- booking lead time = **31–60 days**; and
- patient has **1 or more previous no-shows**.

Observed result

High-Risk Segment No-Show Rate = 70.52%

This KPI provides a concise summary of the most important Week 6 finding.

It allows dashboard users to see immediately that the overall no-show rate of 51.15% does not represent the entire attendance-risk distribution.

A substantially higher rate is observed within a specific appointment segment.

Interpretation

The 70.52% value should be interpreted as a **decision-support indicator**, not as a causal or definitive prediction of individual patient behavior.

Its purpose is to identify an appointment segment that warrants closer operational attention and further validation.

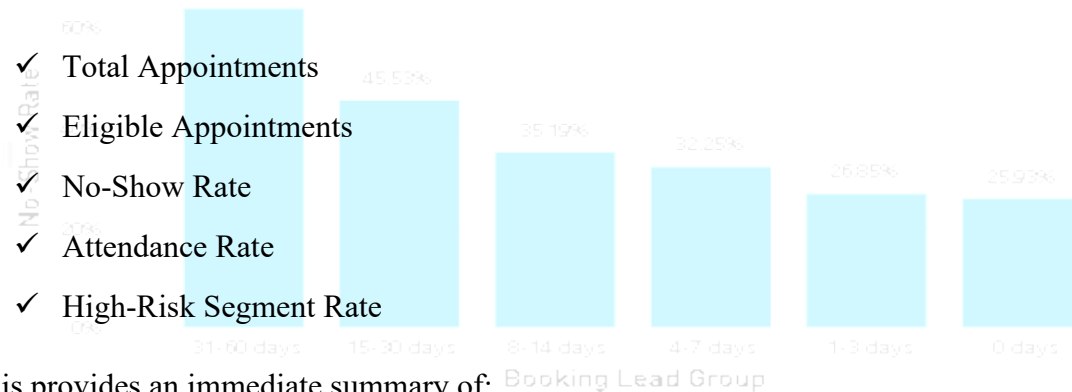
10.6 Updated Dashboard Visual Structure

The Week 6 dashboard is organized around a hierarchy of decision importance.

Row 1 — Executive KPI Summary

The first row contains:

- ✓ Total Appointments
- ✓ Eligible Appointments
- ✓ No-Show Rate
- ✓ Attendance Rate
- ✓ High-Risk Segment Rate



This provides an immediate summary of:

- appointment volume;
- analytical population;
- overall attendance performance; and
- the magnitude of the highest-priority observed risk segment.

Row 2 — Core Risk Indicators

The second row focuses on the two strongest individual indicators:

- **No-Show Rate by Booking Lead Time**

- **No-Show Rate by Previous No-Show History**

These visuals preserve important Week 5 analysis while establishing the foundation for the interaction analysis that follows.

Row 3 — Primary Decision-Support Analysis

The central visual is:

No-Show Rate by Booking Lead Time and Previous No-Show History

The visual compares no-show rates for:

- no previous no-shows; and
- 1+ previous no-shows

across the booking-lead categories.

The underlying results are:

Table 10.2 —No-Show Rate by Booking Lead Time and Previous No-Show History

Booking Lead Time	No Previous No-Shows	1+ Previous No-Shows
0 days	18.75%	36.36%
1–3 days	17.74%	39.13%
4–7 days	27.32%	39.52%
8–14 days	33.06%	38.79%
15–30 days	41.09%	51.27%
31–60 days	59.07%	70.52%

This is the principal Week 6 decision-support visual because it makes the interaction between the two strongest analytical indicators visible.

The visual allows users to see that the highest observed rate occurs at the intersection of:

31–60 days + 1+ previous no-shows

10.7 High-Risk Segment by Reminder Status

A second new Week 6 visual examines reminder status specifically within the high-risk segment.

Recommended title

No-Show Rate by Reminder Status Within High-Risk Segment

Results:

Reminder Status	Eligible Appointments	No-Show Rate
Reminder sent	708	69.21%
No reminder	269	73.98%

The difference is:

4.77 percentage points

in favour of the reminder-sent group.

The visual should not be labelled “**Reminder Effectiveness**” because the analysis is observational and does not establish that reminders caused the difference.

Dashboard interpretation

The visual communicates two important points simultaneously:

- ✓ reminder status is associated with a somewhat lower observed no-show rate within this segment; and
- ✓ no-show rates remain very high regardless of reminder status.

This supports the Week 6 conclusion that reminders should not be treated as a proven standalone solution for the highest-risk appointments.

10.8 High-Risk Segment by Distance

The updated dashboard also incorporates distance as a secondary segmentation variable.

Table 10.3 —No-Show Rate by Distance Within High-Risk Segment

Distance to Clinic	Eligible Appointments	No-Show Rate
0–5 km	224	66.96%
>5–10 km	339	69.32%
>10–15 km	218	72.02%
>15–20 km	90	73.33%
>20–30 km	68	77.94%
>30 km	16	75.00%

The visual demonstrates a generally increasing pattern across the distance groups up to the >20–30 km category.

The >30 km category should be accompanied by a caution because it contains only **16 eligible appointments**.

Dashboard note

A small-sample annotation should therefore be used rather than allowing the 75.00% value to appear as equally stable evidence.

This is an example of how the Week 6 dashboard moves beyond simply displaying numbers and incorporates analytical context into decision support.

10.9 High-Risk Segment by Appointment Type

Appointment type provides an additional but weaker layer of segmentation.

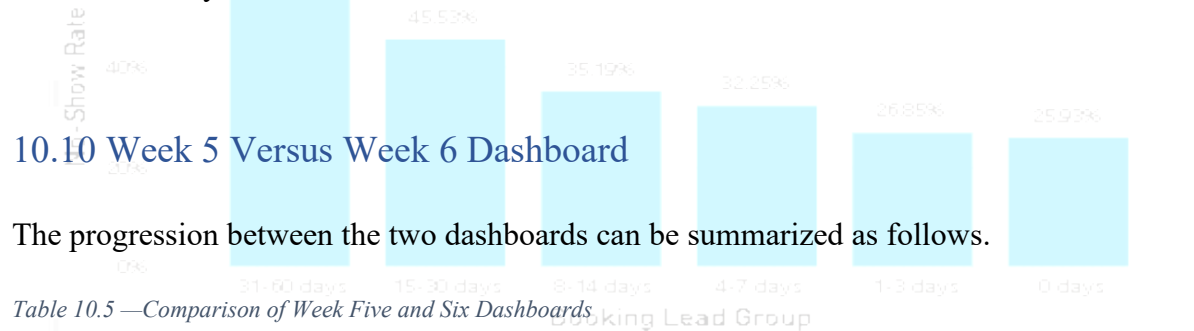
Table 10.4 —No-Show Rate by Appointment Type Within High-Risk Segment

Appointment Type	Eligible Appointments	No-Show Rate
Follow-up	279	72.76%
Diagnostic Test	131	71.76%
General Consultation	409	69.93%
Specialist Consultation	158	67.09%

The visual helps identify whether certain appointment types require additional attention within the already identified high-risk segment.

However, the dashboard should communicate that appointment type is a **secondary differentiator**.

The 5.67 percentage-point difference between the highest and lowest categories is relatively modest compared with the broader differences associated with booking lead time and previous no-show history.



10.10 Week 5 Versus Week 6 Dashboard

The progression between the two dashboards can be summarized as follows.

Table 10.5 —Comparison of Week Five and Six Dashboards

Area	Week 5	Week 6
Overall KPIs	Established	Validated and retained
No-show analysis	Broad segmentation	Deeper segmentation
Booking lead time	Individual analysis	Combined with previous no-show history
Previous no-show history	Individual analysis	Combined with booking lead time
Reminder status	Overall comparison	Segment-specific analysis
Distance	General segmentation	High-risk segment analysis
Appointment type	General segmentation	High-risk segment analysis
High-risk segment	Not explicitly defined	70.52% identified

Area	Week 5	Week 6
Decision support	General monitoring	Targeted prioritization
Data Science contribution	General findings	Candidate features and interaction hypothesis
Week 7 preparation	Initial	Specific testing requirements

This demonstrates that the Week 6 dashboard is an analytical progression rather than a duplicate of the Week 5 deliverable.

10.11 Dashboard Annotation for Decision Support

A concise explanatory annotation should be included on the dashboard to prevent misinterpretation of the high-risk indicator.

Recommended dashboard annotation

Week 6 Decision-Support Finding: Appointments booked 31–60 days in advance for patients with at least one previous no-show have the highest identified observed no-show rate at 70.52%. Within this segment, no-show rates remain high regardless of reminder status. These results represent observed associations and should not be interpreted as causal effects.

This annotation serves three purposes:

- ✓ identifies the main Week 6 finding;
- ✓ explains the operational relevance; and
- ✓ protects against causal overinterpretation.

10.12 Dashboard Validation

The updated dashboard was subject to analytical validation to ensure that the new decision-support elements remained consistent with the underlying calculations.

KPI validation

The following values must reconcile with the validated Week 6 calculations:

Table 10.6 —Key calculations for Week Six

Validation Item	Expected Result
Total Appointments	5,000
Eligible Appointments	4,737
No-Shows	2,423
Attended	2,314
Cancelled	263
No-Show Rate	51.15%
Attendance Rate	48.85%
Cancellation Rate	5.26%
Repeat No-Show Rate	57.83%
High-Risk Segment Rate	70.52%
High-Risk + Reminder	69.21%
High-Risk + No Reminder	73.98%

The core reconciliation is:

$$5,000 = 2,423 \text{ No-Shows} + 2,314 \text{ Attended} + 263 \text{ Cancelled}$$

and:

$$4,737 \text{ Eligible} = 2,423 \text{ No-Shows} + 2,314 \text{ Attended}$$

Therefore:

$$51.15\% + 48.85\% = 100\%$$

These checks ensure that the dashboard remains consistent with the validated KPI framework.

10.13 Filter and Interaction Validation

The dashboard also requires functional validation of its interactive components.

The following should be checked before final submission:

KPI cards

- KPI values display correctly;
- count cards use appropriate number formatting;
- percentage cards display two decimal places;
- no unintended visual-level filters alter the headline KPIs.

Booking Lead Time

- categories appear in the correct chronological order;
- 0 days appears first;
- 31–60 days appears after 15–30 days;
- no 61+ days category is incorrectly presented as populated.

Previous No-Show History

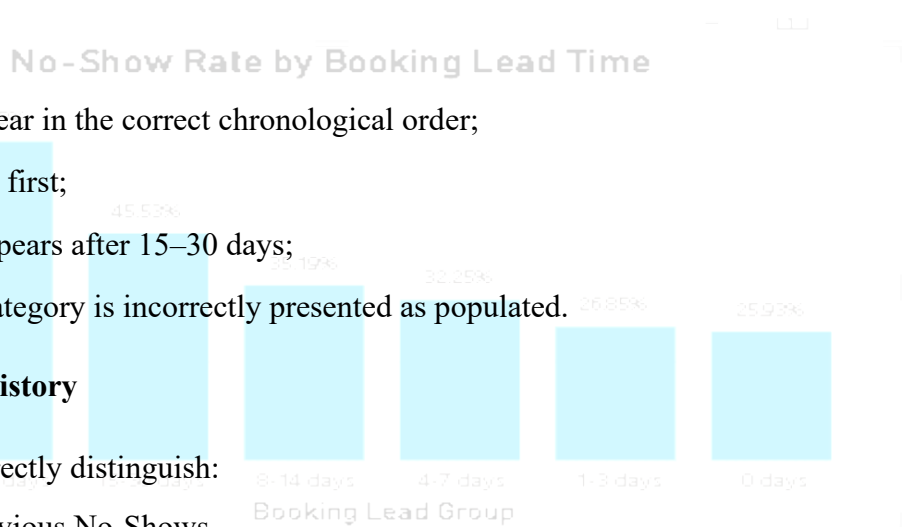
- categories correctly distinguish:
 - No Previous No-Shows
 - 1+ Previous No-Shows.

Distance

- categories are correctly ordered;
- Missing values are excluded from the analytical comparison where appropriate;
- 30 km is clearly treated as a small subgroup.

High-Risk Segment

- only appointments satisfying both conditions are included:
 - 31–60 days;



- 1+ previous no-shows.

Reminder analysis

- reminder comparisons use eligible appointments;
- reminder status is not presented as causal effectiveness.

These checks are important because an analytically correct measure can still produce misleading results if visual filters, sorting, or segmentation logic are incorrect.

10.14 Analytical Reporting Improvements

The Week 6 reporting layer was also improved beyond the dashboard itself.

The analytical reporting now distinguishes between:

Core performance measures

- No-Show Rate
- Attendance Rate

Behavioral risk indicator

- Repeat No-Show Rate

Association metric

- No-Show Rate by Reminder Status

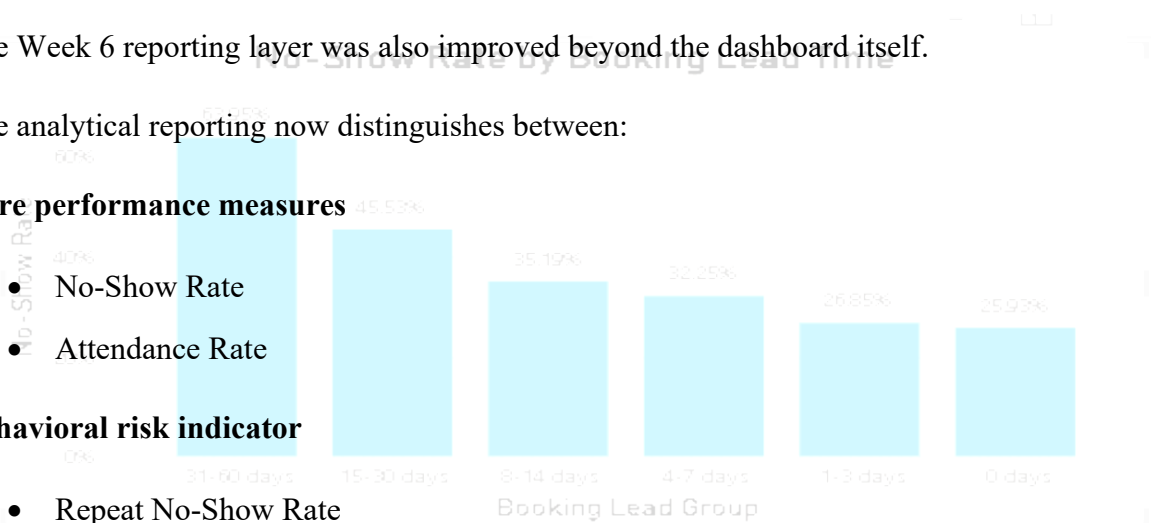
Decision-support indicator

- High-Risk Segment No-Show Rate

Contextual metric

- Cancellation Rate

This classification reduces the likelihood that all metrics will be interpreted as having the same analytical purpose.



For example, the 70.52% high-risk segment rate is not another general KPI equivalent to the 51.15% overall no-show rate. It is a **decision-support indicator** designed to identify a specific appointment segment.

10.15 Evidence-Based Dashboard Interpretation

The updated dashboard supports the following evidence-based conclusions.

Finding 1

Overall attendance performance remains poor, with a **51.15% no-show rate among eligible appointments**.

Finding 2

No-show risk is not evenly distributed across appointments.

Finding 3

The highest identified segment is appointments booked 31–60 days in advance for patients with previous no-shows, with a **70.52% observed no-show rate**.

Finding 4

Within this high-risk segment, the no-show rate remains high even when a reminder was sent.

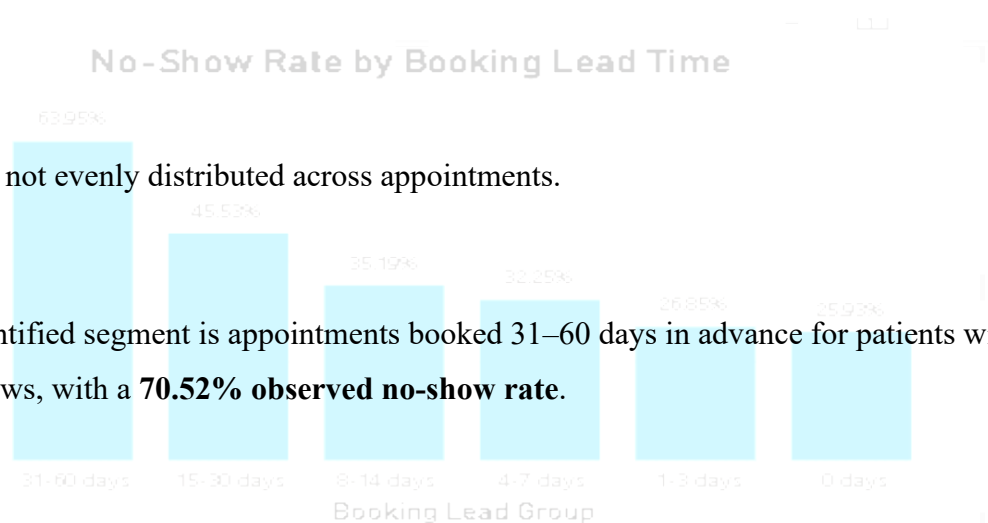
Finding 5

Distance provides additional segmentation value within the high-risk population.

Finding 6

Appointment type provides additional but comparatively weaker differentiation.

These findings make the dashboard more useful for decision support because they move the user from a general view of clinic performance toward specific areas requiring further attention.



10.16 Dashboard Implications for HealthConnect

The updated dashboard suggests a potential operational workflow in which appointment-support attention is prioritized according to observed risk characteristics.

A conceptual workflow is:

Appointment Created



Check Booking Lead Time



Check Previous No-Show History



Identify High-Risk Segment



Apply Secondary Segmentation

- Distance
- Appointment Type
- Reminder Status



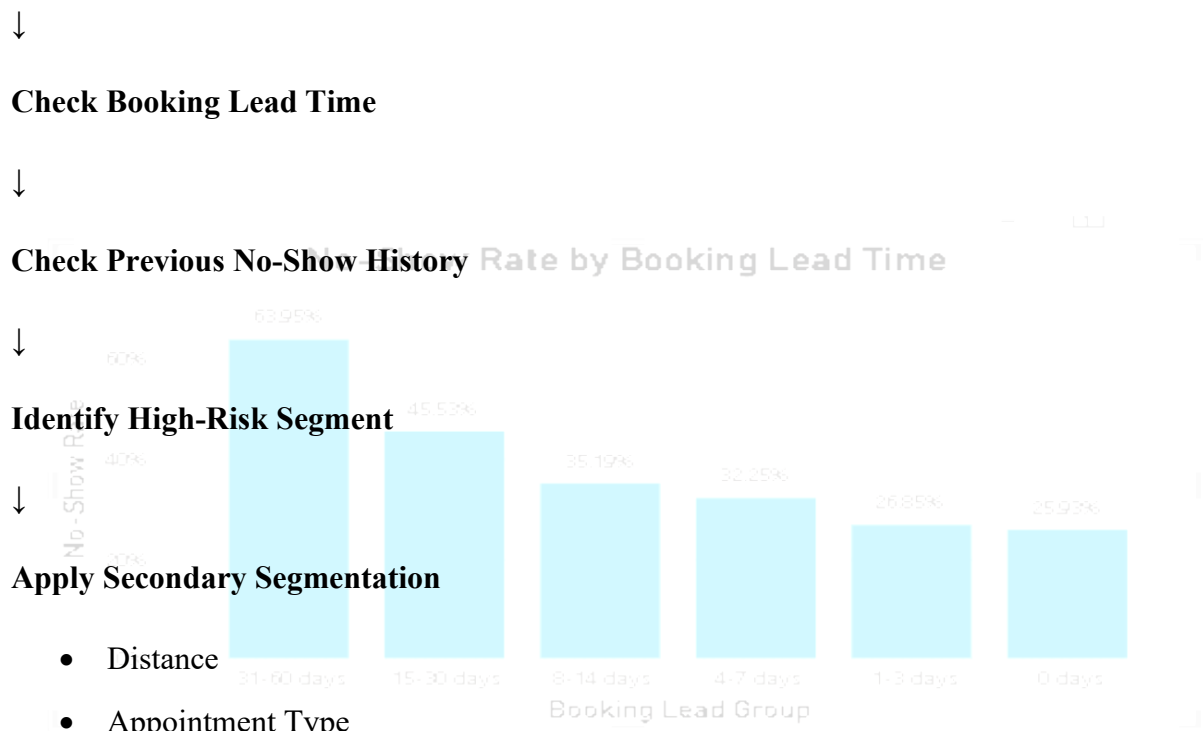
Prioritize Attendance-Support Attention



Monitor Attendance Outcome



Validate Results in Week 7



This should be treated as a **decision-support concept**, not as a proven operational intervention.

The dashboard identifies where HealthConnect may wish to focus attention; it does not establish which intervention will produce the best outcome.

10.17 Dashboard Contribution to Data Science

The updated dashboard also provides a visual reference for the Data Science track.

The high-risk segment can be used as a benchmark against which predictive model outputs may later be compared.

For example, Data Science can examine whether a predictive model:

- identifies the same high-risk population;
- identifies additional high-risk appointments missed by the simple rule;
- improves prioritization beyond the two-variable rule;
- performs consistently across appointment types and distance groups; and
- provides sufficiently reliable predictions for operational use.

This creates a useful relationship between the dashboard and predictive modelling.

The dashboard provides an **interpretable analytical benchmark**, while the predictive model can potentially provide a more flexible risk-identification mechanism.

10.18 Limitations of the Updated Dashboard

Several limitations remain relevant.

1. Observational analysis

The dashboard shows associations and observed differences. It does not establish causation.

2. Reminder timing

The dataset records whether a reminder was sent but does not establish sufficient detail about reminder timing, content, or delivery success.

3. Small subgroups

Some detailed segments contain relatively few observations, particularly the >30 km high-risk group with only 16 eligible appointments.

4. Over-segmentation

Adding too many segmentation dimensions can produce small and unstable groups.

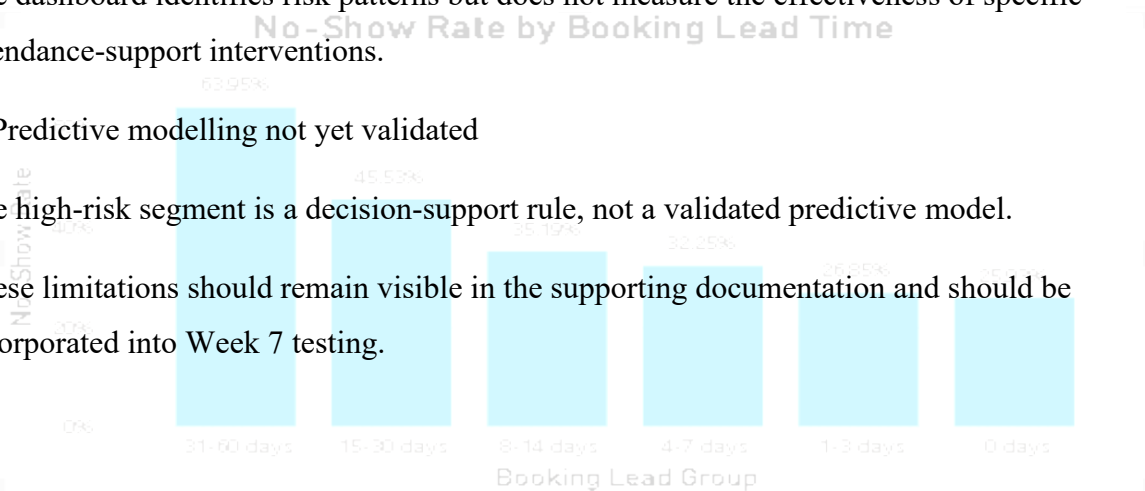
5. No intervention evaluation

The dashboard identifies risk patterns but does not measure the effectiveness of specific attendance-support interventions.

6. Predictive modelling not yet validated

The high-risk segment is a decision-support rule, not a validated predictive model.

These limitations should remain visible in the supporting documentation and should be incorporated into Week 7 testing.



10.19 Week 7 Dashboard Testing Requirements

The updated dashboard should be carried forward into Week 7 for broader testing.

The testing requirements include:

- ✓ Reconcile all KPI values against the source calculations.
- ✓ Confirm that the high-risk segment consistently returns 70.52%.
- ✓ Verify the 69.21% versus 73.98% reminder comparison within the high-risk segment.
- ✓ Test dashboard filters and cross-filtering.
- ✓ Validate sorting of all grouped categories.
- ✓ Confirm that cancellations remain excluded from primary attendance denominators.

- ✓ Test small-sample indicators and annotations.
- ✓ Confirm that no stale Week 5 values remain in the updated dashboard.
- ✓ Test whether the high-risk rule is operationally useful.
- ✓ Compare the simple high-risk rule with future predictive model outputs where available.
- ✓ Confirm that dashboard language does not imply causality.
- ✓ Document any discrepancies or refinements identified during testing.

These requirements support the Week 7 progression toward testing, refinement, and end-to-end validation.

10.20 Dashboard Evidence to Retain

To demonstrate the Week 5 → Week 6 improvement, the following evidence should be retained with the final submission:

Table 10.7 — Dashboard Evidence to be retained from Week 5

Evidence	Purpose
Week 5 dashboard	Establishes baseline
Week 6 updated dashboard	Demonstrates improvement
KPI calculation table	Validates headline metrics
High-risk segment calculation	Supports 70.52% indicator
Booking Lead × Previous No-Show analysis	Supports primary decision-support visual
High-Risk × Reminder analysis	Supports reminder visual
High-Risk × Distance analysis	Supports distance visual
High-Risk × Appointment Type analysis	Supports appointment-type visual
Dashboard validation checklist	Demonstrates quality control
Screenshot/export of final dashboard	Provides presentation evidence
Power BI file/report	Provides reproducible analytical artefact

The assignment specifically recognises dashboards and supporting analytical outputs as appropriate evidence for the Week 6 submission.

10.21 Week 5 → Week 6 Dashboard Change Summary

The dashboard evolution can be summarized as:

Week 5 established the attendance-monitoring dashboard; Week 6 converted it into a more targeted decision-support dashboard by validating the original KPIs, introducing the 70.52% high-risk segment, visualizing interactions between booking lead time and previous no-show history, and adding high-risk segmentation by reminder status, distance, and appointment type.

The principal improvement is therefore not simply the addition of more charts.

It is the shift in analytical purpose:

From describing appointment attendance → to supporting targeted attendance-risk prioritization.

10.22 Chapter Conclusion

The Week 6 dashboard represents a meaningful improvement over the Week 5 analytical reporting layer.

The core Week 5 KPIs were retained and validated, ensuring continuity and preventing unnecessary reconstruction of the original work. The major Week 6 analytical finding—the **70.52% observed no-show rate among appointments booked 31–60 days in advance for patients with at least one previous no-show**—was incorporated as the central decision-support indicator.

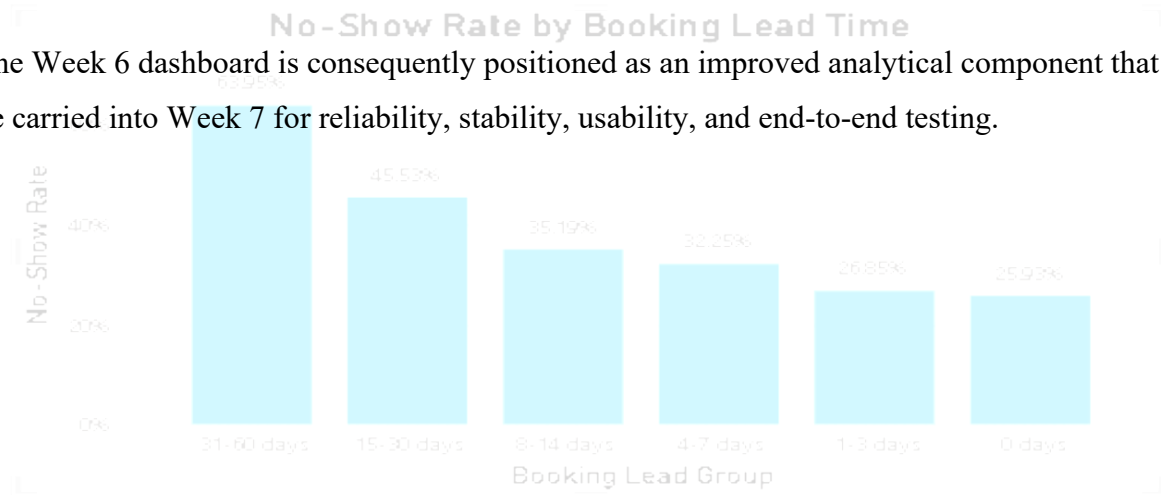
The dashboard was also strengthened through interaction analysis and high-risk segmentation across:

- booking lead time;
- previous no-show history;
- reminder status;
- distance to clinic; and
- appointment type.

The resulting dashboard is therefore better aligned with the operational question of **where HealthConnect should prioritize attendance-support attention.**

At the same time, the reporting maintains appropriate analytical caution. Reminder status is not labelled as proven effectiveness, small subgroups are flagged, and the high-risk segment is treated as an observed decision-support signal rather than a causal or predictive certainty.

The Week 6 dashboard is consequently positioned as an improved analytical component that can be carried into Week 7 for reliability, stability, usability, and end-to-end testing.



Chapter 11

Limitations, Risks and Analytical Decisions

11.1 Purpose

The purpose of this chapter is to document the analytical limitations, risks, dependencies, and methodological decisions identified during Week 6. The analysis builds directly on the Week 5 findings and focuses on deeper validation rather than repeating the original exploratory analysis.

Week 6 strengthened the analytical evidence by examining important findings across multiple segments, validating the principal KPIs, refining the interpretation of reminder-related results, and identifying a high-risk appointment segment. The analysis also identified limitations that must be addressed before the findings are used for predictive modelling, operational testing, or broader decision-making.

The Week 6 assignment requires interns to review issues from Weeks 4 and 5, identify what has been resolved, what remains unresolved, and what new issues or dependencies have emerged. It also requires the potential impact and recommended mitigation to be documented.



11.2 Week 6 Validation Context

Week 6 was designed as a progression from the Week 5 analytical baseline:

Week 5 Output → Improve → Integrate → Validate → Document → Prepare for Week 7

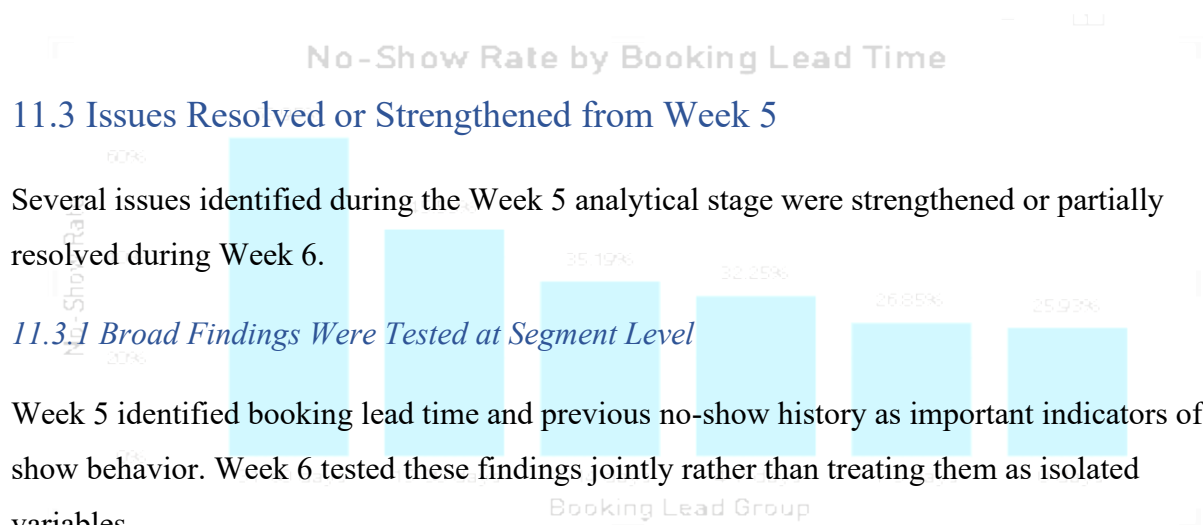
The Week 5 analysis established the overall no-show rate, attendance rate, reminder-related differences, repeat no-show patterns, appointment-segment differences, and the initial dashboard. Week 6 then subjected the most important findings to additional segmentation and validation.

The principal areas investigated were:

- Booking lead time.
- Previous no-show history.

- Reminder status.
- Interactions between booking lead time and previous no-show history.
- Interactions between reminder status and high-risk segments.
- Distance to the clinic within the high-risk segment.
- Appointment type within the high-risk segment.
- KPI consistency and denominator discipline.
- Implications for Data Science modelling.

This approach is consistent with the Week 6 requirement to deepen and validate Week 5 findings and use them to support project decisions.



11.3 Issues Resolved or Strengthened from Week 5

Several issues identified during the Week 5 analytical stage were strengthened or partially resolved during Week 6.

11.3.1 Broad Findings Were Tested at Segment Level

Week 5 identified booking lead time and previous no-show history as important indicators of no-show behavior. Week 6 tested these findings jointly rather than treating them as isolated variables.

The combined analysis showed that the findings remained meaningful. Appointments booked 31–60 days in advance had a 70.52% no-show rate among patients with at least one previous no-show, compared with 59.07% among patients without previous no-shows.

This provided stronger evidence that the Week 5 findings were not simply isolated one-variable observations.

11.3.2 Previous No-Show History Was Validated as a Priority Indicator

The Week 5 Repeat No-Show Rate was 57.83%, compared with 46.30% among patients without previous no-shows, a difference of 11.53 percentage points.

Week 6 confirmed that previous no-show history continued to differentiate attendance behavior across booking lead-time segments. It was therefore retained as a priority analytical indicator.

11.3.3 Booking Lead Time Was Further Validated

The Week 5 finding that longer booking lead times were associated with higher observed no-show rates remained meaningful after segmentation.

The 31–60-day group remained the most important booking-lead segment, with a 63.95% overall no-show rate and substantially higher rates when combined with previous no-show history.

11.3.4 Reminder Interpretation Was Refined

Week 5 showed a 49.86% no-show rate when a reminder was sent compared with 54.63% when no reminder was recorded.

Week 6 demonstrated that this difference was not uniform across booking-lead groups or patient-history groups. As a result, the analysis was reframed from a potentially causal interpretation of reminder effectiveness to a neutral **No-Show Rate by Reminder Status** association.

This is a methodological improvement because the available data demonstrate an association but do not establish that reminders caused the observed difference.

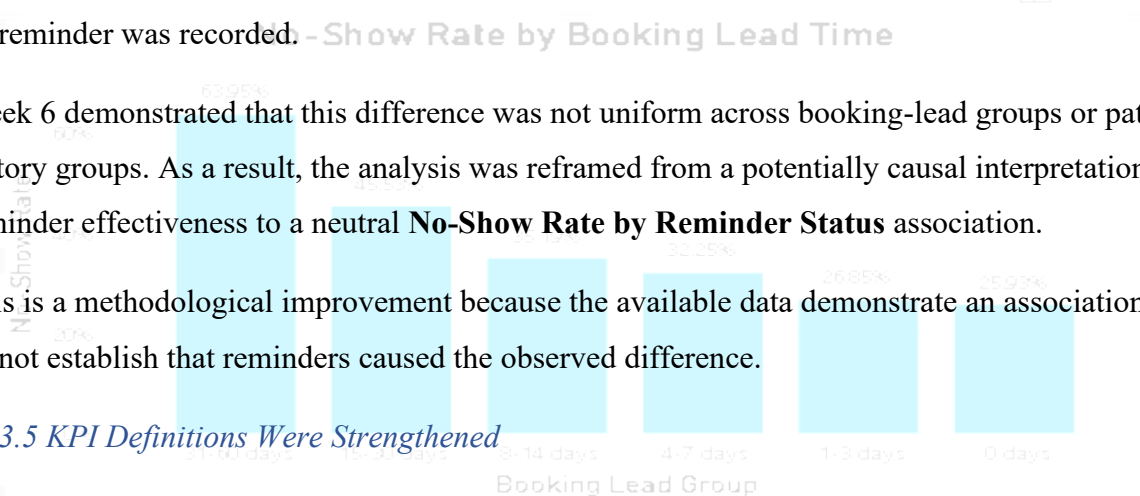
11.3.5 KPI Definitions Were Strengthened

Week 6 confirmed the use of 4,737 eligible appointments as the denominator for the primary No-Show Rate and Attendance Rate. Cancelled appointments were excluded from these measures because they do not represent either an attended or missed appointment.

This maintained consistency between the analytical calculations, dashboard, and written interpretation.

11.4 Unresolved Analytical Limitations

Despite the additional validation, several limitations remain.



11.4.1 Observational Data Does Not Establish Causality

The dataset is observational. Therefore, differences between groups should be interpreted as associations rather than causal effects.

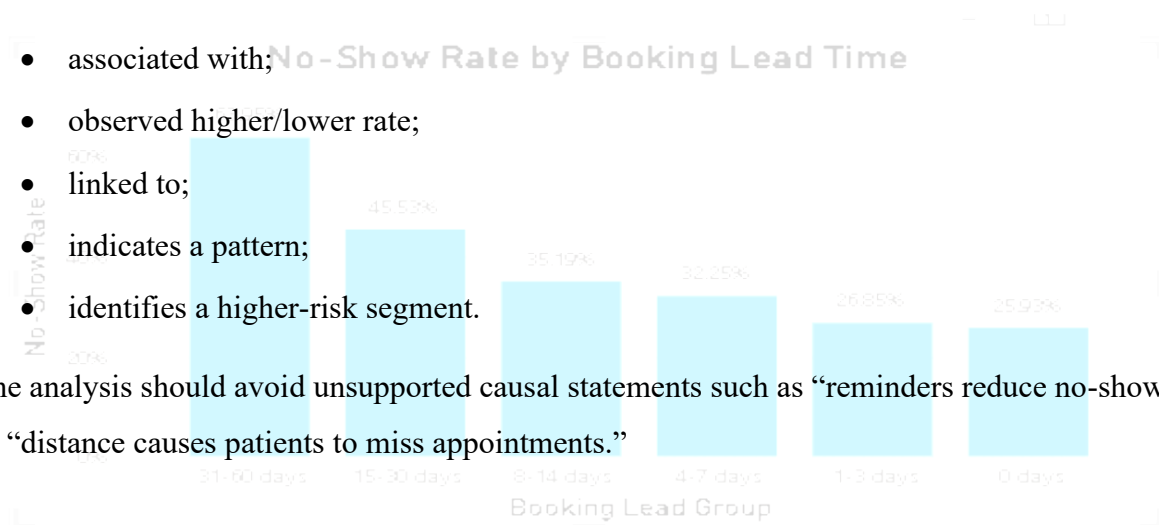
For example, the lower observed no-show rate among appointments with reminders does not establish that sending a reminder caused the lower rate. Other factors may influence both reminder assignment and attendance behavior.

Similarly, the relationship between distance and no-show behavior does not prove that distance causes missed appointments.

All Week 6 conclusions should therefore use language such as:

- associated with;
- observed higher/lower rate;
- linked to;
- indicates a pattern;
- identifies a higher-risk segment.

The analysis should avoid unsupported causal statements such as “reminders reduce no-shows” or “distance causes patients to miss appointments.”



11.5 New Risk: Over-Segmentation

Week 6 introduced multiple levels of segmentation to investigate interactions between variables. This improved analytical depth but also created a risk of producing increasingly small subgroups.

For example, the three-way analysis of booking lead time, previous no-show history, and reminder status produced some small cells. The same-day appointment group containing patients with previous no-shows and no reminder had only **8 eligible observations**.

Such small groups can produce unstable percentages and should not be used as major decision-making evidence.

The analytical decision is therefore to Prioritize patterns that demonstrate:

- ✓ Sufficient sample size.
- ✓ Consistency across related segments.
- ✓ Meaningful differences.
- ✓ Clear business relevance.

This prevents isolated small-cell percentages from being treated as robust findings.

11.6 Data Quality and Data-Structure Risks

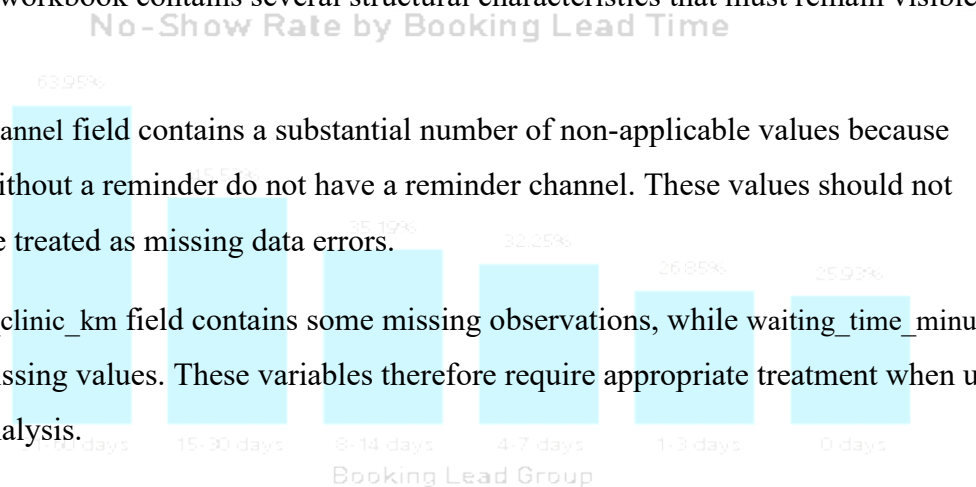
The underlying workbook contains several structural characteristics that must remain visible in interpretation.

The reminder_channel field contains a substantial number of non-applicable values because appointments without a reminder do not have a reminder channel. These values should not automatically be treated as missing data errors.

The distance_to_clinic_km field contains some missing observations, while waiting_time_minutes also contains missing values. These variables therefore require appropriate treatment when used in segmented analysis.

The high-risk distance analysis also illustrates the importance of sample-size awareness. Within the high-risk segment, the >30 km group contained only **16 eligible appointments**. Although its observed no-show rate was 75.00%, this group is too small to support a strong general conclusion.

The mitigation is to retain the group for exploratory analysis while clearly flagging its small sample size and avoiding strong operational decisions based on it alone.



11.7 Reminder Data Limitations

The dataset records whether a reminder was sent and, where applicable, the reminder channel. However, the available analytical fields do not establish:

- the exact time the reminder was sent;
- how many reminders were sent;
- the content of the reminder;
- whether the patient opened or read the reminder;
- whether the patient responded;
- whether the reminder was sent sufficiently close to the appointment;
- whether reminder assignment was random or operationally targeted.

Consequently, reminder status can currently be analyzed as an observational characteristic but not as a controlled intervention.

The Week 6 analysis found:

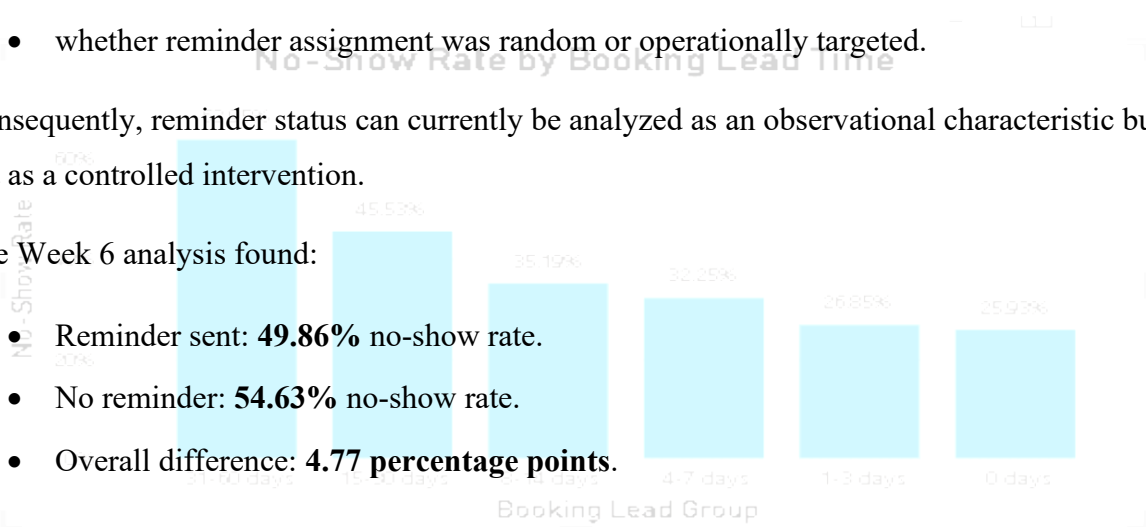
- Reminder sent: **49.86%** no-show rate.
- No reminder: **54.63%** no-show rate.
- Overall difference: **4.77 percentage points**.

Within the high-risk segment:

- Reminder sent: **69.21%**, n=708.
- No reminder: **73.98%**, n=269.

These results support further investigation but do not establish reminder effectiveness.

A future controlled or quasi-experimental evaluation would be required before making a causal claim about reminder effectiveness.



11.8 Waiting-Time Timing Risk

waiting_time_minutes was not promoted to a primary no-show predictor because the dataset does not establish sufficiently clearly that waiting time was measured before the appointment outcome was determined.

If waiting time is recorded after a patient attends the clinic, using it to predict whether the patient will attend would introduce temporal leakage.

For this reason, Week 6 continues to treat waiting time primarily as a **service-experience measure**, rather than a pre-appointment no-show predictor.

This distinction is important for both business interpretation and future predictive modelling.

11.9 Small-Sample and Stability Risk

The deeper segmentation performed during Week 6 provides more detailed information but increases the possibility of unstable estimates.

The most important examples include:

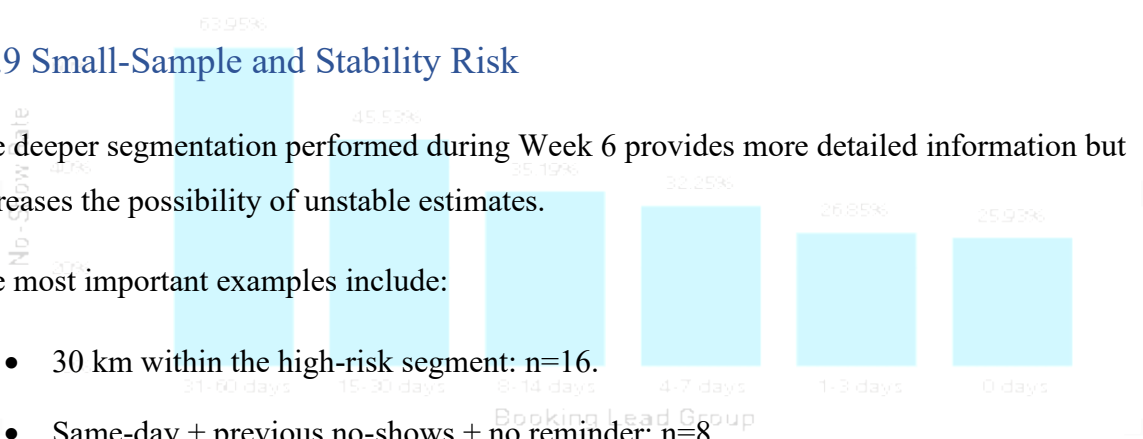
- 30 km within the high-risk segment: n=16.
- Same-day + previous no-shows + no reminder: n=8.
- Other small reminder/lead-time combinations with relatively low counts.

The mitigation is to use sample size alongside percentage differences when evaluating findings.

A high percentage with a very small denominator should not automatically outrank a slightly smaller percentage supported by a substantially larger population.

For Week 7, segment stability should therefore be tested before the high-risk segmentation is operationalised.

No-Show Rate by Booking Lead Time



11.10 Modelling and Feature-Leakage Risks

The Week 6 analytical findings provide candidate inputs for Data Science, but not every available variable should automatically become a predictive-model feature.

Potential candidate variables include:

- Previous no-show history.
- Previous appointment count.
- Booking lead time.
- Distance to clinic.
- Appointment type.
- Appointment day.
- Appointment time.
- Age group.
- Reminder status, subject to timing validation.
- Interactions between booking lead time and previous no-show history.

However, feature timing must be established before modelling.

Reminder Status

If the model is intended to predict no-show risk at the time of booking, reminder_sent may contain information that becomes available only after booking. In that situation, using it could create temporal leakage.

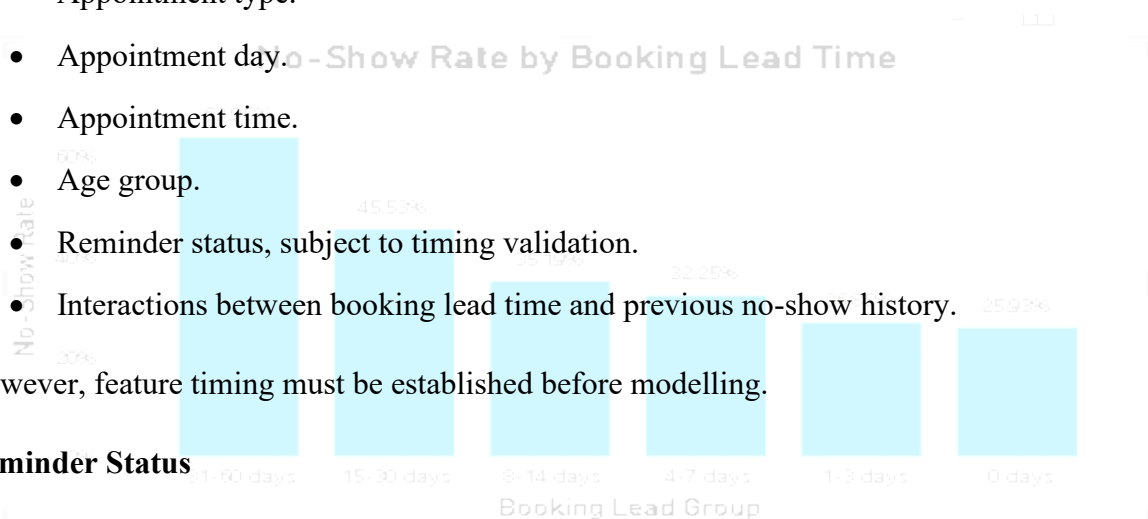
Waiting Time

waiting_time_minutes should not be used as a pre-appointment predictor unless its measurement timing is confirmed to occur before the prediction point.

Appointment Outcome

appointment_outcome is the target variable and must never be used as a predictive feature.

High-Risk Segment



The high-risk segment can be used as a derived decision-support indicator, but it should not be interpreted as a causal label. Data Science should test whether the underlying variables and their interaction provide predictive value independently and robustly.

11.11 Cross-Track Integration Risk

Week 6 requires actual cross-track integration rather than communication alone. The assignment explicitly states that collaboration must result in a meaningful change, improvement, decision, integration, or validated output.

The primary integration point for Data Analytics is Data Science.

The analytical findings provided to Data Science include:

- Previous no-show history as a strong attendance-risk indicator.
- Booking lead time as an important predictor candidate.
- The interaction between booking lead time and previous no-show history.
- Distance as a secondary segmentation variable.
- Reminder status as an observational variable requiring timing validation.
- Appointment type as a weaker secondary differentiator.

A remaining integration risk is that the analytical track cannot claim that a modelling change occurred unless evidence of that change exists.

Therefore, the report distinguishes between:

Analytical contribution completed:

Validated findings and candidate modelling inputs were produced.

Downstream modelling change:

Requires evidence from the Data Science track showing that the findings were incorporated into feature engineering, model development, error analysis, or model validation.

This distinction protects the report from overstating the extent of cross-track integration.

11.12 Analytical Decisions Made in Week 6

The following decisions were made as a result of the Week 6 analysis.

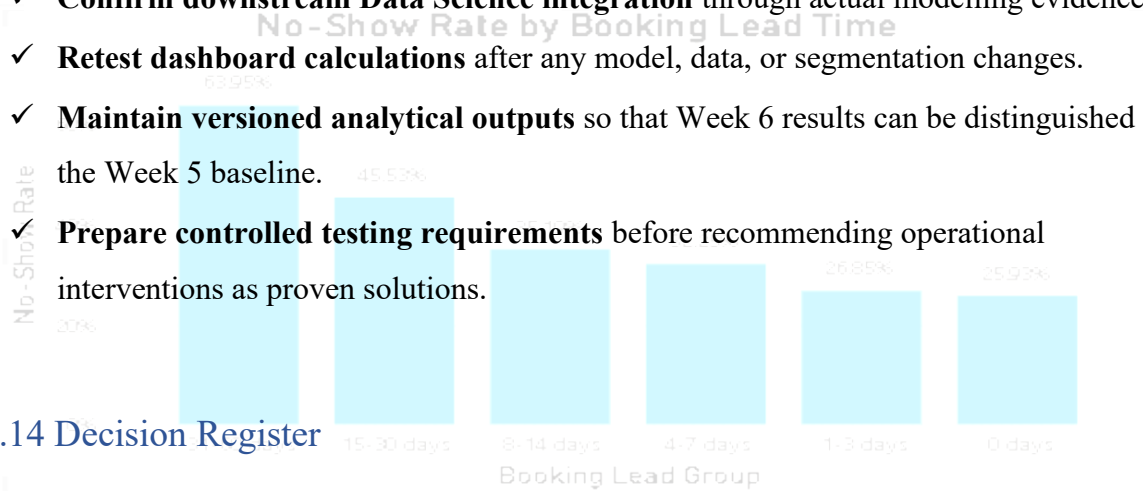
Table 11.1 —Week Six analytical Decision

Decision	Week 6 decision
Overall No-Show Rate	Retain at 51.15%
Attendance Rate	Retain at 48.85%
Previous No-Show History	Retain as a priority risk indicator
Booking Lead Time	Retain as a major segmentation variable
High-Risk Segment	Add 70.52% as a decision-support indicator
Reminder Analysis	Reframe as No-Show Rate by Reminder Status rather than causal effectiveness
Distance	Retain as a secondary segmentation variable
Appointment Type	Retain as a secondary, lower-priority differentiator
Cancellation Rate	Retain as contextual metric at 5.26%
Waiting Time	Treat primarily as service-experience information
Small subgroups	Interpret cautiously and avoid basing major decisions on very small cells
Predictive modelling	Test booking lead time $\times$ previous no-show history as a candidate interaction
Causal claims	Do not make causal claims from the observational dataset

11.13 Risk Mitigation Actions

The following controls should be carried into Week 7:

- ✓ **Maintain denominator discipline** by using 4,737 eligible appointments for primary no-show and attendance measures.
- ✓ **Keep cancellations separate** from attendance/no-show calculations.
- ✓ **Display sample sizes** where subgroup findings may otherwise be misleading.
- ✓ **Flag small segments**, particularly the >30 km and very granular three-way cells.
- ✓ **Use neutral association language** when interpreting reminder, distance, and other observational relationships.
- ✓ **Validate feature timing** before variables are incorporated into predictive models.
- ✓ **Exclude outcome variables from model features** to prevent target leakage.
- ✓ **Test interaction effects** rather than assuming that individual variables explain all risk.
- ✓ **Confirm downstream Data Science integration** through actual modelling evidence.
- ✓ **Retest dashboard calculations** after any model, data, or segmentation changes.
- ✓ **Maintain versioned analytical outputs** so that Week 6 results can be distinguished from the Week 5 baseline.
- ✓ **Prepare controlled testing requirements** before recommending operational interventions as proven solutions.



11.14 Decision Register

The Week 6 decision register can be summarized as follows:

Table 11.2—Week Six Decision Register

Area	Evidence	Decision	Risk Controlled
No-show KPI	51.15%	Retain	KPI inconsistency
Attendance KPI	48.85%	Retain	Denominator errors
Prior no-shows	57.83% vs 46.30%	Prioritize	Weak targeting
Booking lead	63.95% at 31–60 days	Prioritize	Broad intervention
Joint segment	70.52%	Add high-risk indicator	Failure to identify priority group

Area	Evidence	Decision	Risk Controlled
Reminder	4.77 pp overall difference	Treat as association	Causal overclaim
High-risk + reminder	73.98% vs 69.21%	Investigate further	Assuming reminders solve risk
Distance	Increasing pattern within high-risk group	Secondary indicator	Overstating distance effect
Appointment type	Limited variation within high-risk group	Secondary indicator	Over-targeting by appointment type
Waiting time	Timing uncertain	Exclude from pre-appointment prediction	Temporal leakage
Small cells	Some very low n	Apply caution	Unstable estimates
Data Science integration	Candidate features/interactions identified	Validate downstream adoption	Unsupported integration claim

11.15 Implications for Week 7

The Week 6 findings establish several requirements for the next testing stage.

Analytical Testing

Week 7 should test whether the high-risk segment remains stable under additional validation and whether its observed rate remains materially higher than the broader appointment population.

Segment Stability

The 31–60-day and previous-no-show interaction should be evaluated for stability rather than relying on a single observed percentage.

Reminder Analysis

The reminder association should be tested further, particularly within the higher-risk groups, while continuing to avoid causal interpretation.

Sample-Size Validation

Small subgroups should be reviewed before they are used for operational targeting.

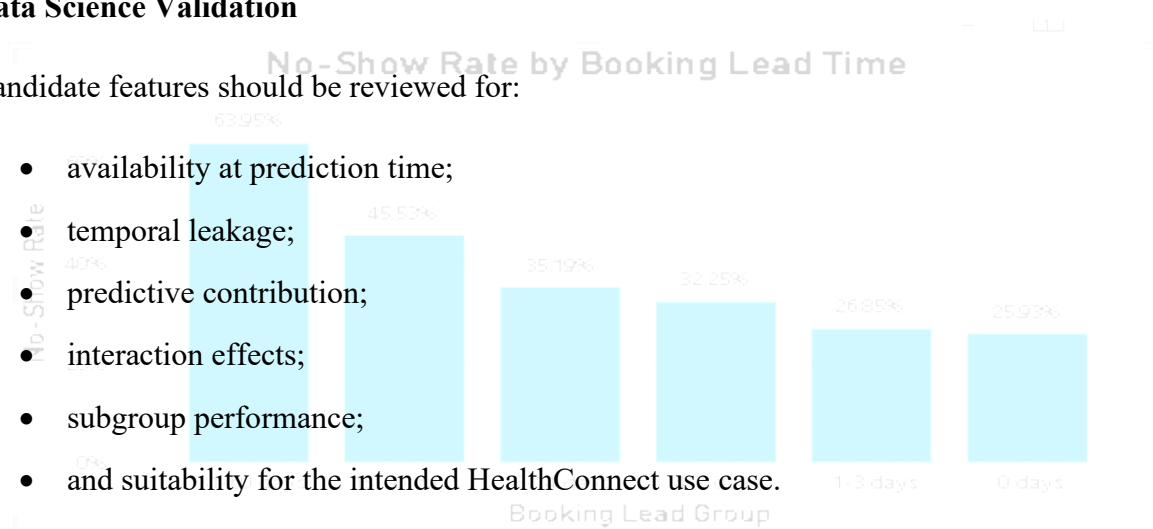
Dashboard Testing

All updated dashboard measures, filters, sorting, denominators, and subgroup calculations should be retested to ensure that the Week 6 improvements do not introduce calculation or interaction errors.

Data Science Validation

Candidate features should be reviewed for:

- availability at prediction time;
- temporal leakage;
- predictive contribution;
- interaction effects;
- subgroup performance;
- and suitability for the intended HealthConnect use case.



Decision-Support Testing

The practical usefulness of the high-risk segment should be evaluated. Week 7 should consider whether identifying a high-risk appointment creates an actionable opportunity for HealthConnect staff rather than simply producing another analytical statistic.

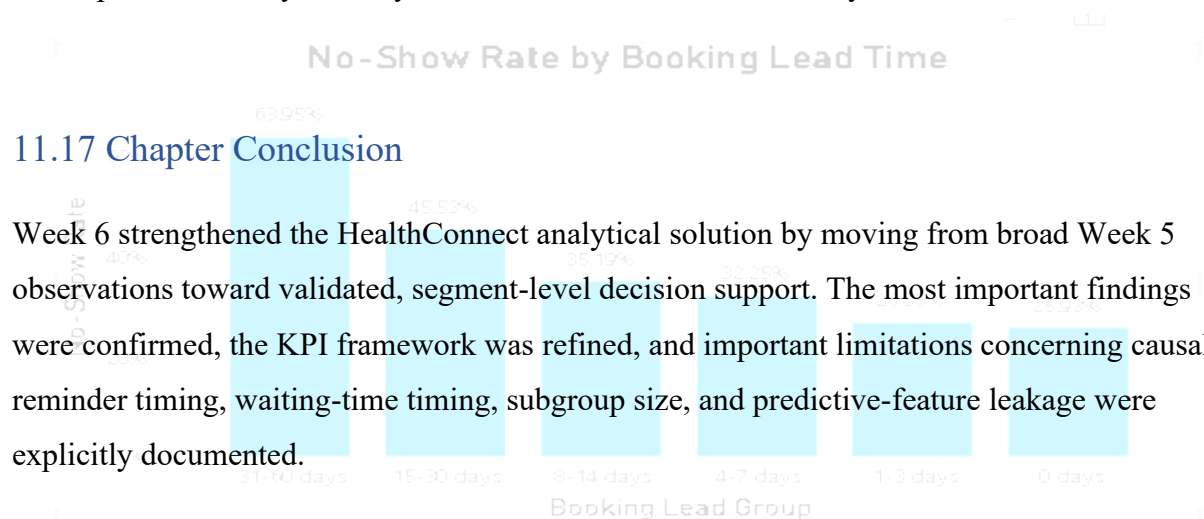
11.16 Overall Analytical Risk Position

The Week 6 analysis provides stronger evidence than the Week 5 baseline, but it does not eliminate the limitations of the underlying observational dataset.

The most reliable conclusion is that certain appointment characteristics are associated with substantially different observed no-show rates. In particular, booking lead time and previous no-show history remain important indicators, and their combination identifies a high-risk segment with a **70.52% observed no-show rate**.

The highest observed three-way subgroup—appointments booked 31–60 days in advance for patients with previous no-shows and no recorded reminder—had a **73.98% no-show rate among 269 eligible appointments**. However, this should be interpreted as a targeting signal rather than evidence that the absence of a reminder caused the higher rate.

The findings therefore support targeted investigation, decision support, and predictive modelling development, but they do not yet establish the effectiveness of any intervention.



11.17 Chapter Conclusion

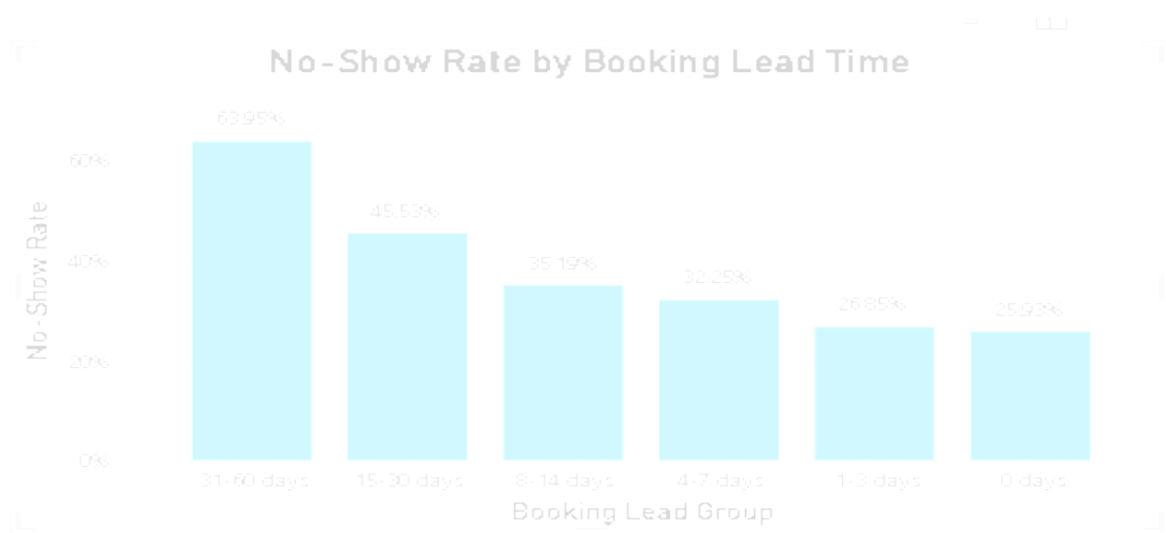
Week 6 strengthened the HealthConnect analytical solution by moving from broad Week 5 observations toward validated, segment-level decision support. The most important findings were confirmed, the KPI framework was refined, and important limitations concerning causality, reminder timing, waiting-time timing, subgroup size, and predictive-feature leakage were explicitly documented.

The analysis identifies **booking lead time and previous no-show history as the strongest current attendance-risk indicators**, with their combination producing a high-risk segment of **70.52% no-show rate**. Distance provides additional segmentation value within this group, while appointment type appears to be a weaker differentiator.

The principal analytical decision is therefore to move HealthConnect toward **targeted attendance-risk identification rather than broad, undifferentiated intervention**.

At the same time, the analysis establishes clear boundaries around what can currently be claimed. The findings represent observed associations, not causal effects. Further testing is required before interventions such as enhanced reminders or targeted support can be described as effective.

These limitations and decisions provide the analytical foundation for Week 7, where the focus should move toward testing, stability assessment, dashboard validation, feature-timing checks, and further cross-track validation. This is consistent with the Week 6 progression from development and integration toward evidence that the work is ready for the next testing stage.



Chapter 12

Week 7 Testing Requirements

12.1 Purpose

Week 7 will move the HealthConnect Data Analytics work from advanced analysis and validation into structured testing, refinement, and end-to-end validation.

The purpose of this chapter is to define the analytical tests required to determine whether the findings, KPIs, dashboard, segmentation logic, and cross-track analytical inputs developed during Week 6 remain reliable and suitable for the next stage of the HealthConnect project.

The Week 6 progression requires the project to move from **Integration** → **Advanced Development** → **Validation** toward **Testing** → **Refinement** → **End-to-End Validation** in Week 7.

The Week 7 testing plan therefore focuses on validating the work already developed rather than starting a new analysis or selecting a new dataset.

12.2 Week 6 Outputs Requiring Testing

The principal Week 6 analytical outputs requiring testing are:

- ✓ The validated No-Show Rate of **51.15%**.
- ✓ The Attendance Rate of **48.85%**.
- ✓ The Repeat No-Show Rate of **57.83%**.
- ✓ The No-Show Rate by Reminder Status.
- ✓ The High-Risk Segment No-Show Rate of **70.52%**.
- ✓ The interaction between booking lead time and previous no-show history.
- ✓ The three-way high-risk analysis involving booking lead time, previous no-show history, and reminder status.
- ✓ The distance analysis within the high-risk segment.

- ✓ The appointment-type analysis within the high-risk segment.
- ✓ The refined KPI framework.
- ✓ The updated HealthConnect Appointment Attendance Dashboard.
- ✓ The analytical inputs provided to the Data Science track.

These outputs should be treated as Week 7 test subjects rather than assumed to be final.

12.3 KPI Reconciliation Testing

The first Week 7 requirement is to confirm that all primary KPIs continue to reconcile with the underlying dataset.

The following values should be retested:

Table 12.1 — KPIs to be retested

KPI	Week 6 Baseline	Week 7 Test Requirement
Total Appointments	5,000	Reconcile with source records
Eligible Appointments	4,737	Reconcile with Attended + No-Show
No-Shows	2,423	Reconcile with outcome records
Attended	2,314	Reconcile with outcome records
Cancelled	263	Reconcile with outcome records
No-Show Rate	51.15%	Recalculate independently
Attendance Rate	48.85%	Recalculate independently
Cancellation Rate	5.26%	Recalculate against all appointments
Repeat No-Show Rate	57.83%	Recalculate by patient history
High-Risk Segment Rate	70.52%	Recalculate from segment definition

The primary no-show and attendance measures must continue to use **eligible appointments only**, defined as appointments with an outcome of Attended or No-Show.

Cancellations must remain excluded from these two denominators.

12.4 High-Risk Segment Stability Testing

The most important Week 7 analytical test will be whether the identified high-risk segment remains meaningfully elevated when subjected to additional validation.

The current high-risk segment is defined as:

Appointments booked 31–60 days in advance for patients with at least one previous no-show.

The Week 6 observed no-show rate for this segment was:

70.52%

Week 7 should test:

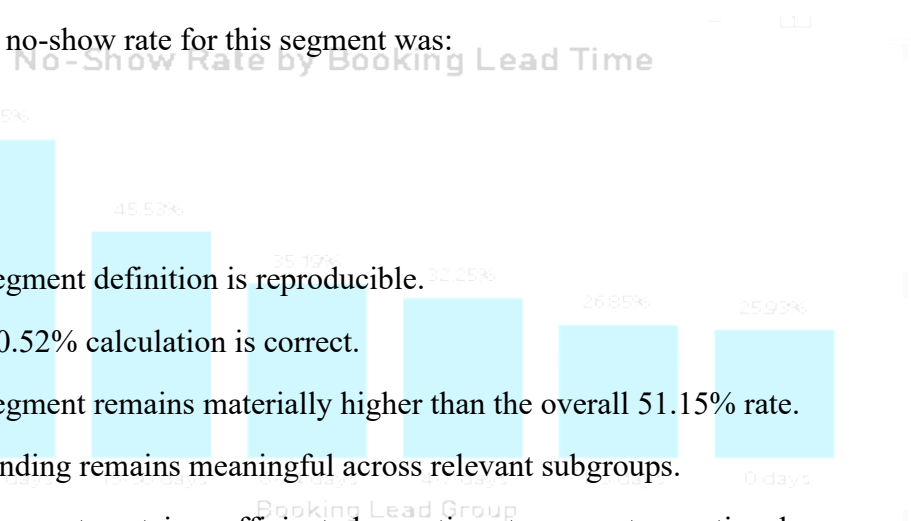
- Whether the segment definition is reproducible.
- Whether the 70.52% calculation is correct.
- Whether the segment remains materially higher than the overall 51.15% rate.
- Whether the finding remains meaningful across relevant subgroups.
- Whether the segment contains sufficient observations to support operational use.
- Whether minor data or filtering changes materially alter the result.

The objective is not to prove that 70.52% is a permanent population parameter. Rather, it is to determine whether the segment represents a sufficiently stable and useful analytical signal for further testing.

12.5 Booking Lead Time Testing

Week 5 and Week 6 both identified booking lead time as a major attendance-risk indicator.

Week 7 should therefore test whether the observed relationship remains consistent.



Particular attention should be given to:

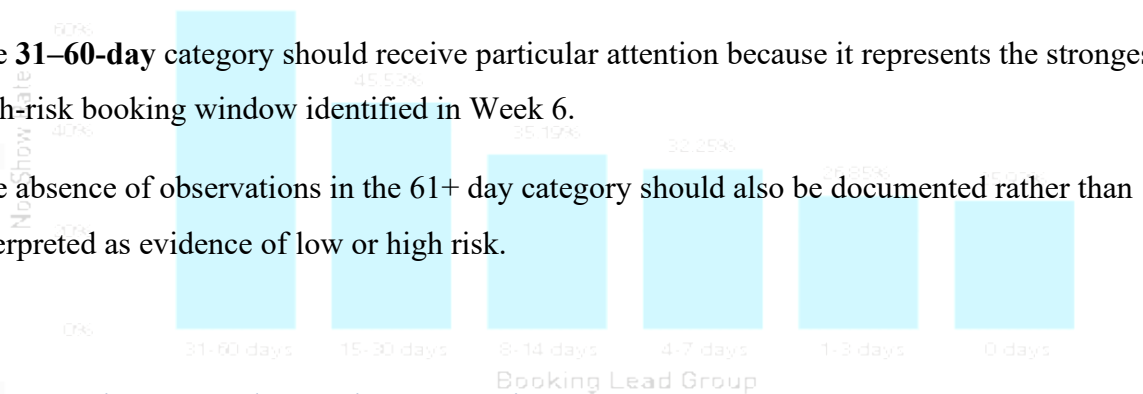
- Same-day appointments.
- 1–3 days.
- 4–7 days.
- 8–14 days.
- 15–30 days.
- 31–60 days.

The primary question will be:

Does the increase in observed no-show rate associated with longer booking lead time remain sufficiently consistent to support continued use of booking lead time as a decision-support variable?

The **31–60-day** category should receive particular attention because it represents the strongest high-risk booking window identified in Week 6.

The absence of observations in the 61+ day category should also be documented rather than interpreted as evidence of low or high risk.



12.6 Previous No-Show History Testing

Week 6 identified previous no-show history as one of the strongest differentiators of attendance behavior.

The baseline comparison was:

- ✓ 1+ previous no-shows: **57.83%**
- ✓ No previous no-shows: **46.30%**
- ✓ Difference: **11.53 percentage points**

Week 7 should test:

- ✓ Whether these rates reconcile with the underlying records.

- ✓ Whether the difference remains meaningful after additional segmentation.
- ✓ Whether the relationship remains visible within different booking-lead groups.
- ✓ Whether the result is sufficiently stable to justify its continued use as a priority indicator.
- ✓ Whether the variable is available at the intended prediction or intervention point.

This final point is particularly important for Data Science integration because a useful predictor must be available at the time the prediction is intended to be made.

12.7 Interaction Testing: Booking Lead Time × Previous No-Show History

The Week 6 analysis identified an important interaction between booking lead time and previous no-show history.

The strongest observed segment was:

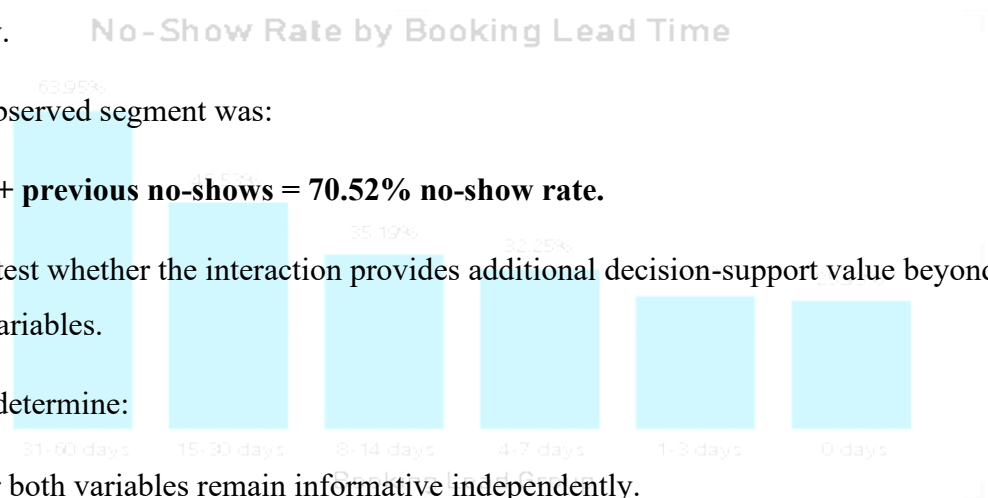
31–60 days + 1+ previous no-shows = 70.52% no-show rate.

Week 7 should test whether the interaction provides additional decision-support value beyond the individual variables.

Testing should determine:

- Whether both variables remain informative independently.
- Whether their combination consistently identifies a higher-risk group.
- Whether the interaction remains meaningful across relevant subgroups.
- Whether the high-risk classification is stable.
- Whether the interaction should be provided to Data Science as a candidate modelling feature or interaction term.

The objective is not to assume that the interaction is predictive simply because it is associated with a high observed no-show rate.



12.8 Reminder Association Testing

Reminder status requires additional validation because Week 6 established an association rather than a causal effect.

The overall Week 6 results were:

- ✓ Reminder sent: **49.86%**
- ✓ No reminder: **54.63%**
- ✓ Difference: **4.77 percentage points**

Within the high-risk segment:

- ✓ Reminder sent: **69.21%**, n=708.
- ✓ No reminder: **73.98%**, n=269.

Week 7 should test whether these differences remain reasonably stable across relevant segments.

Testing should include:

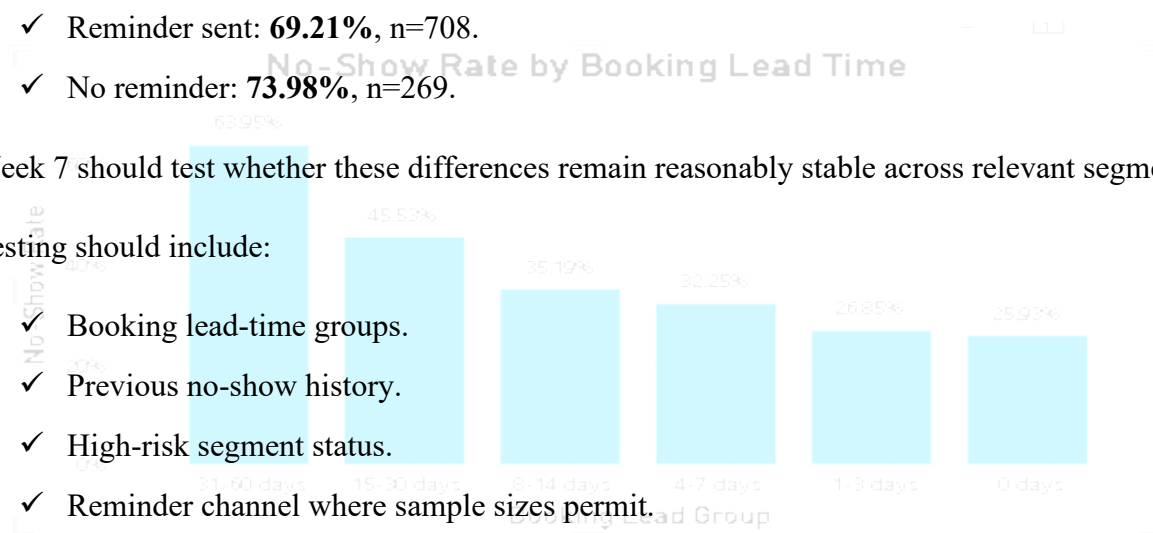
- ✓ Booking lead-time groups.
- ✓ Previous no-show history.
- ✓ High-risk segment status.
- ✓ Reminder channel where sample sizes permit.
- ✓ Sample sizes supporting each comparison.

The wording should remain “**No-Show Rate by Reminder Status**” rather than “Reminder Effectiveness” unless future testing provides evidence capable of supporting a causal interpretation.

12.9 Reminder Channel Testing

Where sample sizes are adequate, Week 7 may further examine the three reminder channels:

- ✓ SMS.



- ✓ Email.
- ✓ WhatsApp.

The Week 5 observed no-show rates were approximately:

- ✓ SMS: **47.98%**
- ✓ Email: **50.99%**
- ✓ WhatsApp: **52.74%**

These differences should not be interpreted as evidence that one channel is inherently better.

Week 7 testing should instead determine whether the observed differences are sufficiently consistent and adequately supported by sample size to justify further investigation.

The analysis should also consider whether patients are systematically assigned to different channels, since non-random channel assignment could influence observed differences.

12.10 Distance-to-Clinic Testing

Distance showed an increasing pattern within the highest-risk segment.

The Week 6 results were:

Table 12.2—Distance to clinic Testing Results

Distance	No-Show Rate
0–5 km	66.96%
>5–10 km	69.32%
>10–15 km	72.02%
>15–20 km	73.33%
>20–30 km	77.94%
>30 km	75.00%

Week 7 should test whether this pattern remains sufficiently consistent to justify continued use of distance as a secondary prioritization variable.

Particular caution is required for the **>30 km** category because it contained only **16 eligible appointments**.

The test should therefore distinguish between:

- evidence of a general distance-related pattern; and
- evidence supporting a specific threshold such as >30 km.

The latter requires substantially stronger evidence than the former.

12.11 Appointment-Type Testing

Appointment type produced comparatively smaller differences within the high-risk segment:

- Follow-up: **72.76%**
- Diagnostic Test: **71.76%**
- General Consultation: **69.93%**
- Specialist Consultation: **67.09%**

Week 7 should determine whether these differences remain meaningful after validation.

The current analytical position is that appointment type should remain a **secondary segmentation variable**, rather than the primary basis for identifying attendance risk.

Testing should confirm whether this decision remains appropriate.

12.12 Small-Sample Stability Testing

Week 7 should explicitly test the stability of findings generated from small subgroups.

Examples include:

- 30 km distance group within the high-risk segment: n=16.
- Same-day + previous no-shows + no reminder: n=8.

- Other highly granular combinations involving booking lead time, patient history, and reminder status.

The testing requirement is:

No operational recommendation should depend primarily on a subgroup whose sample size is too small to provide a reasonably stable estimate.

Where a small group remains analytically interesting, it should be retained as an exploratory finding and clearly labelled as requiring further validation.

12.13 Dashboard Validation Testing

The Week 6 dashboard introduced new decision-support components and therefore requires functional testing in Week 7.

Testing should verify:

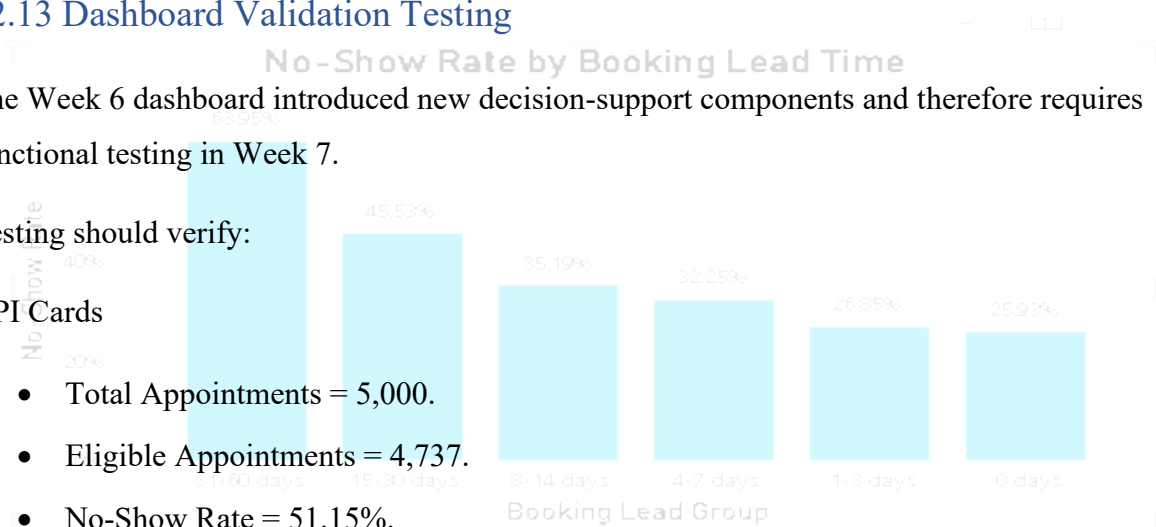
KPI Cards

- Total Appointments = 5,000.
- Eligible Appointments = 4,737.
- No-Show Rate = 51.15%.
- Attendance Rate = 48.85%.
- High-Risk Segment Rate = 70.52%.

Visual Calculations

The dashboard should reproduce the validated analytical results for:

- Booking Lead Time.
- Previous No-Show History.
- Booking Lead $\times$ Previous No-Show History.
- Reminder Status.
- High-Risk $\times$ Reminder Status.



- High-Risk × Distance.
- High-Risk × Appointment Type.

Filters and Interactions

Week 7 should test whether:

- Slicers produce expected changes.
- Cross-filtering does not distort KPI denominators.
- Cancellations remain excluded from primary attendance calculations.
- Sorting remains correct.
- High-risk calculations respond correctly to filters.
- No visual unexpectedly changes the denominator of another KPI.

The previously identified filter-context issue that caused a total to display **1,366** rather than 5,000 should be specifically regression-tested to ensure it does not recur.

12.14 Data Quality Regression Testing

Week 7 should confirm that the analytical dataset remains structurally consistent.

The following checks should be repeated:

- Appointment IDs remain unique.
- No unexpected duplicate rows are introduced.
- Booking dates do not occur after appointment dates.
- Booking lead days remain consistent with booking and appointment dates.
- Previous no-shows do not exceed previous appointments.
- Outcome categories remain valid.
- Structural None values in reminder channel remain correctly interpreted.
- Missing distance and waiting-time values remain appropriately handled.
- Derived fields continue to calculate correctly.

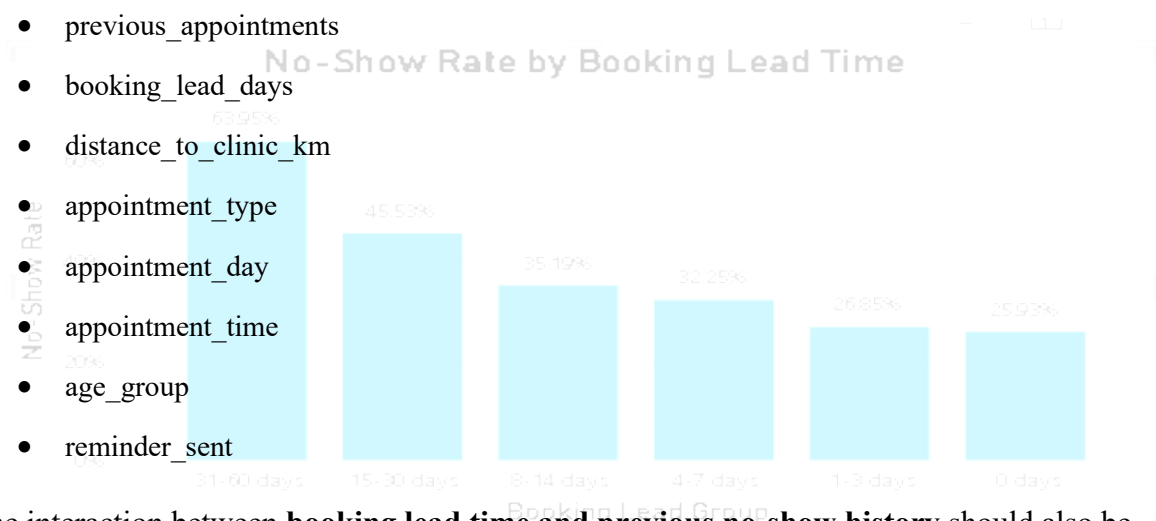
Any transformation or derived dataset used in Week 7 should remain separate from the original project resource, consistent with the project requirement not to overwrite original resources.

12.15 Data Science Feature Validation

The Week 6 analytical findings were provided as candidate modelling inputs. Week 7 should verify whether these inputs are appropriate for predictive modelling.

Candidate variables requiring review include:

- previous_no_shows
- previous_appointments
- booking_lead_days
- distance_to_clinic_km
- appointment_type
- appointment_day
- appointment_time
- age_group
- reminder_sent



The interaction between **booking lead time** and **previous no-show history** should also be considered.

For each candidate feature, Data Science should confirm:

- ✓ Whether the variable is available at prediction time.
- ✓ Whether it contains target leakage.
- ✓ Whether it contributes useful predictive information.
- ✓ Whether it remains stable across relevant groups.
- ✓ Whether it is appropriate for the intended HealthConnect use case.

The Week 6 assignment specifically identifies Data Analytics → Data Science as a valid integration pathway in which analytical findings can inform modelling decisions.

12.16 Waiting-Time Feature Validation

Before any future modelling activity uses `waiting_time_minutes`, Week 7 should establish when this variable is recorded.

If waiting time is recorded after attendance or during the service experience, it should remain excluded from a model intended to predict pre-appointment no-show risk.

The testing requirement is therefore:

Confirm the temporal position of waiting-time measurement before considering it as a predictive feature.

Until this is established, waiting time should remain a service-experience variable rather than a pre-appointment risk predictor.

12.17 Causal-Interpretation Testing

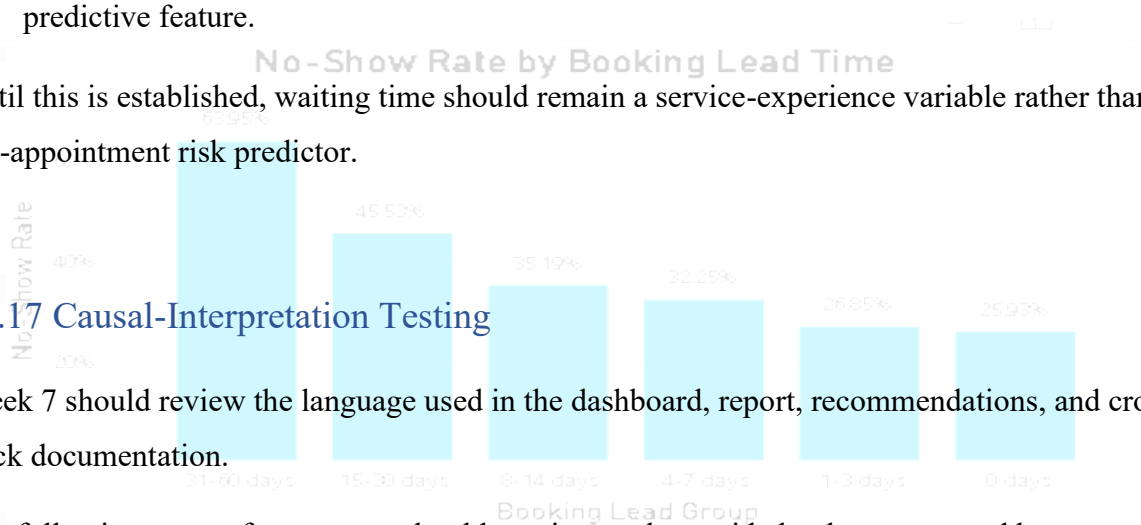
Week 7 should review the language used in the dashboard, report, recommendations, and cross-track documentation.

The following types of statements should continue to be avoided unless supported by an appropriate experimental or quasi-experimental design:

- “Reminders reduce no-shows.”
- “Distance causes missed appointments.”
- “Long booking lead time causes no-shows.”
- “Previous no-shows cause future no-shows.”

Preferred terminology should remain:

- “associated with”;
- “observed higher no-show rate”;



- “higher-risk segment”;
- “attendance-risk indicator”;
- “candidate intervention group”;
- “requires further testing.”

This is an important quality-control requirement because analytical accuracy includes not only correct calculations but also correct interpretation.

12.18 Decision-Support Testing

The Week 6 dashboard has moved beyond descriptive reporting toward decision support.

Week 7 should therefore test whether the identified high-risk segment is practically useful.

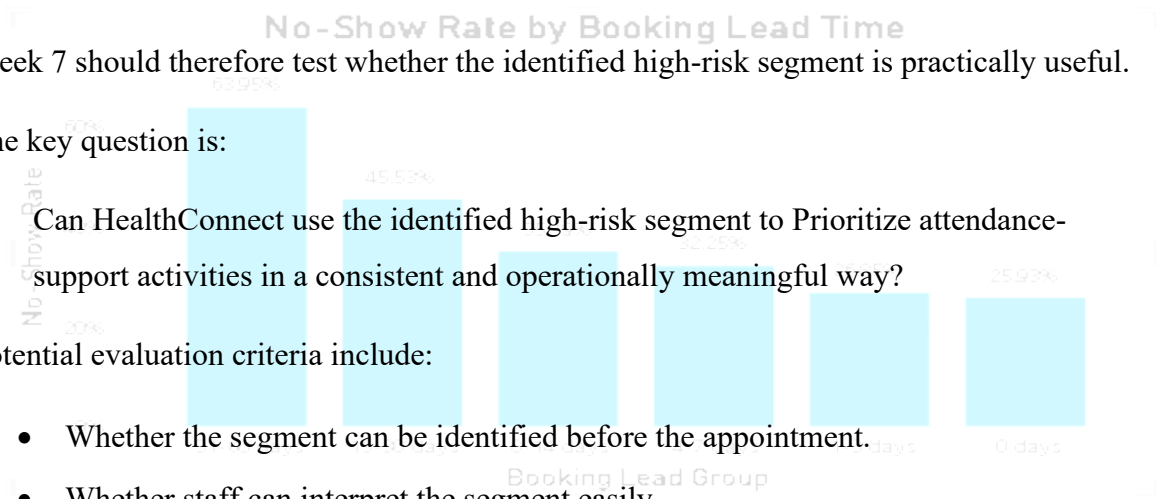
The key question is:

Can HealthConnect use the identified high-risk segment to Prioritize attendance-support activities in a consistent and operationally meaningful way?

Potential evaluation criteria include:

- Whether the segment can be identified before the appointment.
- Whether staff can interpret the segment easily.
- Whether the number of appointments identified is manageable.
- Whether the information can support a defined operational action.
- Whether the segment remains sufficiently specific to justify prioritisation.
- Whether the resulting action can later be evaluated.

The purpose is not to assume that an intervention will work, but to establish whether the analytical output is suitable for controlled operational testing.



12.19 Cross-Track Testing Requirements

The Week 6 assignment requires meaningful cross-track integration and evidence of what was exchanged and what changed as a result.

For the Data Analytics track, Week 7 should therefore verify:

- ✓ Which analytical findings were received by Data Science.
- ✓ Which candidate variables were considered.
- ✓ Whether the booking-lead $\times$ previous-no-show interaction was incorporated or tested.
- ✓ Whether any feature was excluded because of leakage or timing concerns.
- ✓ Whether modelling results support or challenge the Week 6 analytical conclusions.
- ✓ Whether the Data Science findings require revision of the analytical dashboard or report.

This creates a two-way validation loop:

Data Analytics → Data Science → Model Results → Data Analytics Refinement

Such a loop will provide stronger evidence of genuine multidisciplinary integration than communication alone.

12.20 Week 7 Evidence to Retain

The following evidence should be retained as part of the Week 7 analytical record:

- ✓ KPI reconciliation results.
- ✓ Updated analytical calculations.
- ✓ Segment stability analysis.
- ✓ High-risk segment validation.
- ✓ Small-sample checks.
- ✓ Reminder-status validation.
- ✓ Distance validation.
- ✓ Updated dashboard screenshots or file version.

- ✓ Dashboard interaction testing evidence.
- ✓ Data-quality validation results.
- ✓ Data Science feature-review evidence.
- ✓ Modelling feedback relevant to analytical conclusions.
- ✓ Updated decision register.
- ✓ Updated risk register where applicable.
- ✓ Version-controlled analytical files.
- ✓ GitHub commit evidence where applicable.

The Week 6 submission requirements already emphasise supporting reports, dashboards, notebooks, code, models, testing evidence, documentation, and project-management artefacts where applicable.

12.21 Week 7 Analytical Test Matrix

Table 12.3 —Proposed Week Seven Analytical Test

Test Area	Test Objective	Expected Outcome
KPI reconciliation	Confirm all core measures	Values reconcile with source data
Denominator validation	Protect eligible-appointment logic	4,737 remains primary denominator
High-risk segment	Validate 70.52% finding	Segment remains reproducible
Booking lead time	Test pattern stability	Long-lead pattern remains meaningful or is revised
Previous no-shows	Validate 11.53 pp difference	Priority indicator remains supported or is revised
Lead × history interaction	Test combined value	Interaction demonstrates sufficient decision value
Reminder status	Validate association	Association remains accurately characterised
Reminder channels	Explore channel differences	Only stable, adequately supported differences retained
Distance	Test secondary segmentation	General pattern validated cautiously

Test Area	Test Objective	Expected Outcome
Appointment type	Test secondary role	Remains secondary unless evidence changes
Small subgroups	Assess estimate stability	Unstable cells excluded from major decisions
Dashboard	Test calculations and interactions	Visuals reconcile with validated results
Data quality	Regression test source/derived data	No unexpected structural errors
Feature timing	Prevent leakage	Only appropriately timed variables proceed
Data Science integration	Validate analytical hand-off	Evidence of actual downstream use
Decision support	Assess operational usefulness	High-risk output is actionable/testable
Documentation	Confirm traceability	Week 7 changes are documented

12.22 Week 7 Exit Criteria

The Data Analytics work should be considered ready to progress beyond Week 7 testing when:

- ✓ Core KPIs reconcile with the underlying dataset.
- ✓ The high-risk segment can be reproduced consistently.
- ✓ Major findings have been tested for stability.
- ✓ Small-sample findings are appropriately controlled.
- ✓ Reminder findings remain correctly framed as observational.
- ✓ Dashboard calculations and interactions pass validation.
- ✓ Data-quality checks remain satisfactory.
- ✓ Predictive-feature timing and leakage risks have been reviewed.
- ✓ Cross-track analytical integration has supporting evidence.
- ✓ Any material changes to Week 6 conclusions are documented.
- ✓ Remaining limitations and risks are clearly recorded.

- ✓ The analytical outputs are sufficiently validated for final integration and presentation preparation.

These criteria support the Week 6 requirement that project outputs be prepared for the next testing stage rather than treated as final production deliverables. The programme explicitly states that Week 6 should demonstrate readiness for Week 7 testing and continued project progression.

12.23 Chapter Conclusion

Week 7 will not represent a restart of the HealthConnect analytical work. It will provide a structured testing and refinement stage for the outputs developed during Weeks 5 and 6.

The principal testing priority is the **70.52% high-risk segment**, supported by the combined relationship between booking lead time and previous no-show history. This finding must be tested for reproducibility, stability, sample adequacy, and practical usefulness before being treated as an operational targeting mechanism.

The core KPIs, reminder association, distance segmentation, appointment-type analysis, and dashboard functionality will also require validation.

The most important methodological safeguards are to maintain denominator discipline, avoid causal claims, control small-sample interpretation, prevent feature leakage, and verify the actual downstream use of analytical findings by Data Science.

Successful completion of these tests will allow HealthConnect to move from validated analytical findings toward a more reliable, integrated, and testable decision-support solution. This directly supports the programme's required progression from **Week 6 Integration and Validation to Week 7 Testing, Refinement, and End-to-End Validation**.

Chapter 13

Week 6 Summary and Overall Analytical Contribution

Requirement	Week 6 Summary
1. Planned accomplishment	Advance the Week 5 HealthConnect Data Analytics work through deeper segmentation, KPI validation, interaction analysis, dashboard improvement, cross-track integration and preparation for Week 7 testing.
2. Work completed	Advanced segmentation, KPI validation and refinement, high-risk segment identification, interaction analysis, decision-support analysis, dashboard refinement, modelling-input preparation, risk assessment and Week 7 test planning were completed.
3. Improvement from Week 5	The analysis progressed from broad attendance measurement and individual-variable exploration toward deeper segmentation and interaction analysis.
4. Integration completed	Validated analytical findings were converted into candidate modelling inputs for the Data Science track, including the Booking Lead Time × Previous No-Show History interaction.
5. Track collaborated with	Data Science.
6. Information exchanged	Data Analytics provided validated risk patterns, KPI findings, segment definitions, candidate variables, interaction hypotheses and modelling safeguards. The analytical work used the validated appointment dataset and KPI framework as its foundation.
7. Resulting change	The project moved from identifying individual attendance-risk indicators toward testing combined risk characteristics and establishing a high-risk benchmark for predictive modelling.

Requirement	Week 6 Summary
8. Key findings	Overall no-show rate was 51.15%. Previous no-show history and booking lead time remained important observed differentiators. Their combination produced a 70.52% observed no-show rate for appointments booked 31–60 days in advance among patients with at least one previous no-show. Distance provided additional segmentation value, while appointment type was a weaker secondary differentiator.
9. Major challenges	Key challenges included observational data, uncertain reminder timing, uncertain waiting-time timing, small subgroup sizes, potential feature leakage and the absence of downstream evidence confirming Data Science model changes.
10. Important decisions	Reminder status was retained as an observational association rather than a causal measure. Waiting time was treated cautiously because its timing relative to the outcome was not established. Appointment outcome was protected as the modelling target and not treated as a predictor.
11. Remaining issues	The stability of the high-risk segment, predictive usefulness of candidate features, interaction performance, reminder timing, potential leakage, subgroup stability, dashboard validation and actual downstream Data Science use remain to be tested.
12. Contribution to HealthConnect	The Data Analytics track transformed the broad appointment attendance problem into a more focused risk-segmentation problem and produced validated analytical evidence that can support decision-making and predictive modelling.
13. Week 7 focus	Week 7 will focus on KPI reconciliation, high-risk segment stability, interaction testing, dashboard validation, data-quality testing, feature-timing and leakage review, Data Science validation, decision-support testing and documentation of any changes to the Week 6 conclusions.

13.1 Purpose

This chapter summarizes the Data Analytics work completed during Week 6 of the HealthConnect project and evaluates its overall contribution to the project objective:

How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?

Week 6 built directly on the Week 5 analytical baseline. Rather than repeating the initial exploratory analysis, the work focused on deeper investigation, validation, integration, refinement, and preparation for Week 7 testing.

The result is a more focused analytical solution that moves HealthConnect from broad measurement of appointment attendance toward identification of specific appointment segments associated with substantially higher observed no-show rates.

13.2 Week 5 to Week 6 Progression

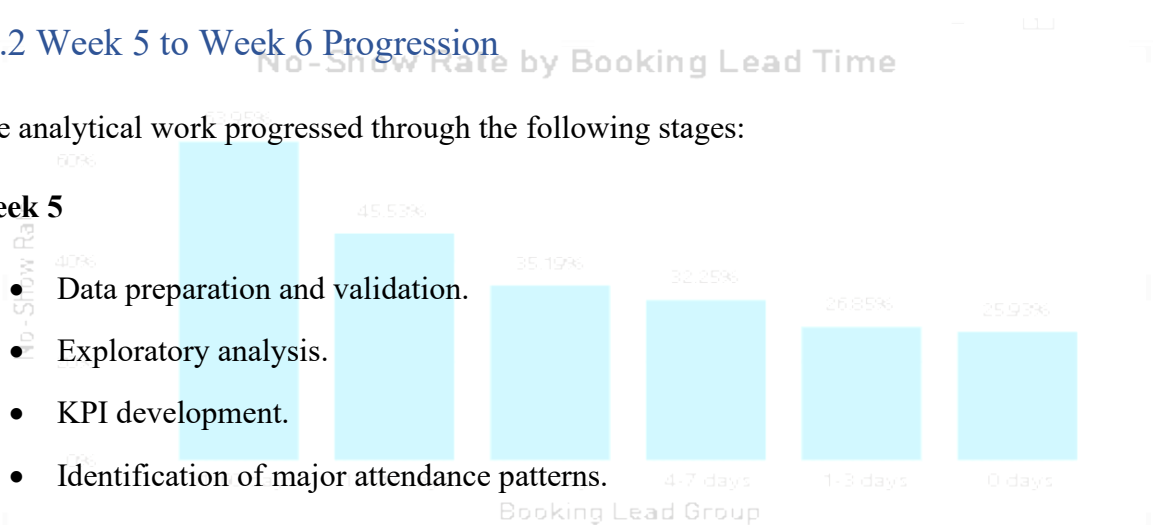
The analytical work progressed through the following stages:

Week 5

- Data preparation and validation.
- Exploratory analysis.
- KPI development.
- Identification of major attendance patterns.
- Initial dashboard development.
- Initial business recommendations.

Week 6

- Deeper investigation of Week 5 findings.
- Cross-segment analysis.
- Interaction analysis.
- KPI validation and refinement.
- Identification of a high-risk appointment segment.
- Improved dashboard and decision-support reporting.
- Cross-track analytical contribution to Data Science.



- Documentation of analytical limitations and risks.
- Definition of Week 7 testing requirements.

This progression demonstrates that Week 6 extended the existing solution rather than creating a separate analytical product.

13.3 Major Analytical Findings

The Week 6 analysis confirmed and strengthened several findings from Week 5.

13.3.1 Overall No-Show Rate

The overall no-show rate among eligible appointments remained:

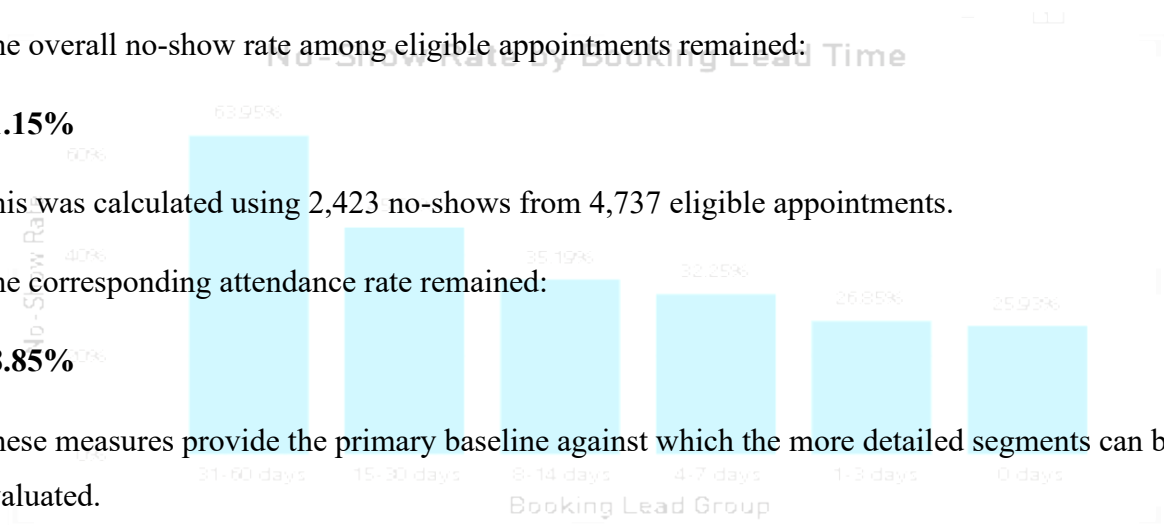
51.15%

This was calculated using 2,423 no-shows from 4,737 eligible appointments.

The corresponding attendance rate remained:

48.85%

These measures provide the primary baseline against which the more detailed segments can be evaluated.



13.3.2 Previous No-Show History

Previous no-show history remained one of the strongest differentiators identified in the analysis.

Patient History	No-Show Rate
No previous no-shows	46.30%
1+ previous no-shows	57.83%
Difference	11.53 pp

This finding supports the continued use of previous no-show history as a priority attendance-risk indicator.

13.3.3 Booking Lead Time

Booking lead time also remained strongly associated with observed no-show behavior.

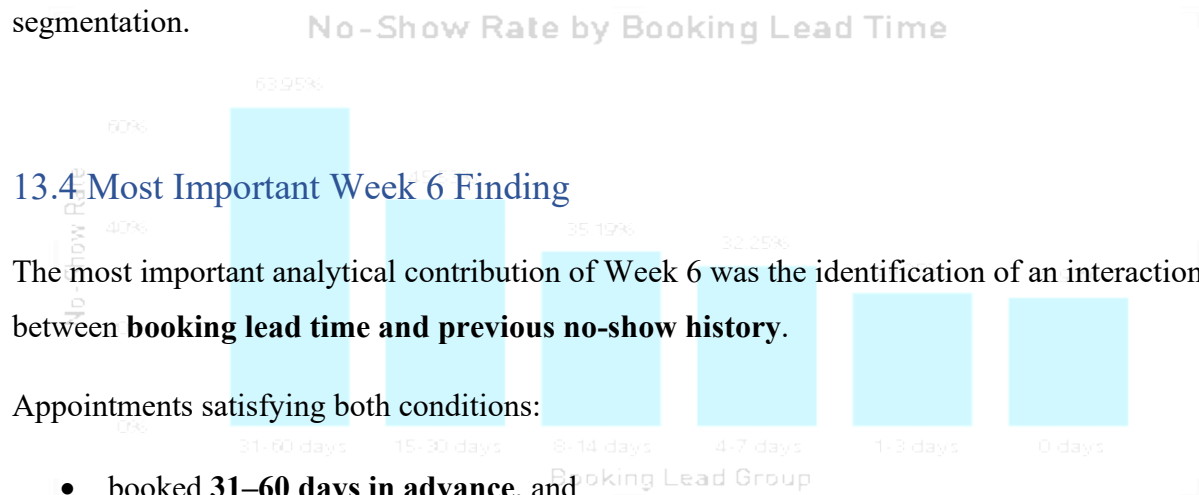
The overall no-show rate increased substantially across longer booking windows, reaching:

63.95% for appointments booked 31–60 days in advance.

By comparison, same-day appointments had an observed no-show rate of:

25.93%

This indicates that booking lead time remains an important variable for attendance-risk segmentation.



13.4 Most Important Week 6 Finding

The most important analytical contribution of Week 6 was the identification of an interaction between **booking lead time and previous no-show history**.

Appointments satisfying both conditions:

- booked **31–60 days in advance**, and
- associated with patients having **at least one previous no-show**

had an observed no-show rate of:

70.52%

This is substantially above the overall eligible-appointment no-show rate of 51.15%.

The finding is important because neither variable needs to be considered in isolation. Their combination identifies a more specific population in which attendance risk is considerably higher.

The result should nevertheless be interpreted as an observed association rather than proof that either characteristic cause missed appointments.

13.5 Highest Three-Way Risk Segment

Week 6 extended the analysis further by incorporating reminder status.

Within the high-risk segment:

Reminder Status	No-Show Rate	Eligible n
Reminder sent	69.21%	708
No reminder	73.98%	269

The highest observed subgroup was therefore:

**31–60-day booking lead + 1+ previous no-shows + no recorded reminder =
73.98% no-show rate (n=269).**

This finding provides an important decision-support signal, but it should not be interpreted as evidence that failure to send a reminder caused the higher no-show rate.

The Week 6 evidence instead suggests that reminder status may provide additional segmentation information within an already high-risk population.

13.6 Additional High-Risk Segment Findings

Week 6 also examined the high-risk segment by distance and appointment type.

Distance

The observed no-show rate increased across most distance categories:

- 0–5 km: 66.96%
- 5–10 km: 69.32%

- 10–15 km: 72.02%
- 15–20 km: 73.33%
- 20–30 km: 77.94%
- 30 km: 75.00%

The pattern suggests that distance can provide additional segmentation value within the high-risk population.

However, the >30 km group contained only 16 eligible appointments and therefore requires cautious interpretation.

Appointment Type

Within the same high-risk segment:

- Follow-up: 72.76%
- Diagnostic Test: 71.76%
- General Consultation: 69.93%
- Specialist Consultation: 67.09%

The variation was relatively modest compared with the differences associated with booking lead time and previous no-show history.

Consequently, appointment type was retained as a **secondary** rather than primary risk indicator.

13.7 KPI Validation and Refinement

Week 6 confirmed that the principal Week 5 KPIs remained analytically valid.

The refined framework is:

Table 13.1 —Week Five KPIs Confirmation

KPI / Indicator	Week 6 Result	Role
No-Show Rate	51.15%	Core KPI

KPI / Indicator	Week 6 Result	Role
Attendance Rate	48.85%	Core KPI
Repeat No-Show Rate	57.83%	Behavioral KPI
No-Show Rate by Reminder Status	49.86% vs 54.63%	Association metric
High-Risk Segment Rate	70.52%	Decision-support indicator
Cancellation Rate	5.26%	Contextual metric

The major refinement was the reframing of reminder analysis.

Rather than treating the difference between reminder and no-reminder groups as proof of reminder effectiveness, Week 6 formally treats it as an observational association.

This improves the methodological quality of the KPI framework.

13.8 Dashboard Contribution

The Week 6 dashboard evolved from a primarily descriptive reporting tool into a more targeted decision-support dashboard.

The Week 5 dashboard primarily answered:

Where are no-show rates higher?

The Week 6 dashboard extends this question to:

Which combinations of appointment characteristics identify the highest observed attendance-risk segments, and where should HealthConnect Prioritize further investigation or support?

The dashboard therefore adds:

- High-Risk Segment KPI.
- Booking Lead × Previous No-Show analysis.
- High-Risk × Reminder analysis.

- High-Risk × Distance analysis.
- High-Risk × Appointment Type analysis.
- Clearer distinction between core KPIs and supporting indicators.
- Explicit caution around observational findings.

This represents a substantive improvement in the analytical usefulness of the dashboard.

13.9 Business Decision-Support Contribution

The Week 6 analysis changes the practical interpretation of the HealthConnect problem.

Instead of treating all appointments as having approximately the same attendance risk, the analysis indicates that attendance risk is concentrated within identifiable segments.

The strongest current targeting signal is:

Patients with at least one previous no-show whose appointments are booked 31–60 days in advance.

This segment has an observed no-show rate of **70.52%**.

The analysis therefore supports a potential shift from broad attendance interventions toward **targeted attendance-support strategies**.

Possible operational applications include:

- ✓ Earlier identification of high-risk appointments.
- ✓ Additional confirmation or follow-up for selected high-risk appointments.
- ✓ Greater attention to patients with previous no-show history.
- ✓ Earlier engagement with appointments booked substantially in advance.
- ✓ Use of distance as a secondary prioritisation factor.
- ✓ Further testing of reminder strategies rather than assuming their effectiveness.

These are candidate decision-support actions, not proven interventions. Their effectiveness should be established through subsequent testing.

13.10 Contribution to Data Science

The Data Analytics track provided Data Science with evidence that can inform predictive modelling decisions.

The most relevant candidate variables and relationships include:

- Previous no-show history.
- Previous appointment history.
- Booking lead time.
- Distance to clinic.
- Appointment type.
- Appointment day and time.
- Reminder status, subject to timing validation.
- The interaction between booking lead time and previous no-show history.

The strongest analytical contribution is the identification of the interaction between booking lead time and previous no-show history as a candidate modelling relationship.

However, Week 6 does not claim that a model was improved unless downstream evidence confirms that the findings were incorporated and evaluated by Data Science.

This distinction maintains analytical and reporting integrity.

13.11 Analytical Quality Improvements

Several methodological improvements were achieved during Week 6.

1. Stronger segmentation

The analysis moved beyond individual variables to examine interactions between relevant characteristics.

2. Better KPI discipline

Primary attendance measures consistently use the eligible appointment population.

3. More careful interpretation

Observational associations are no longer presented as causal effects.

4. Better risk identification

The analysis identifies a specific high-risk segment rather than relying only on the overall no-show rate.

5. Greater modelling readiness

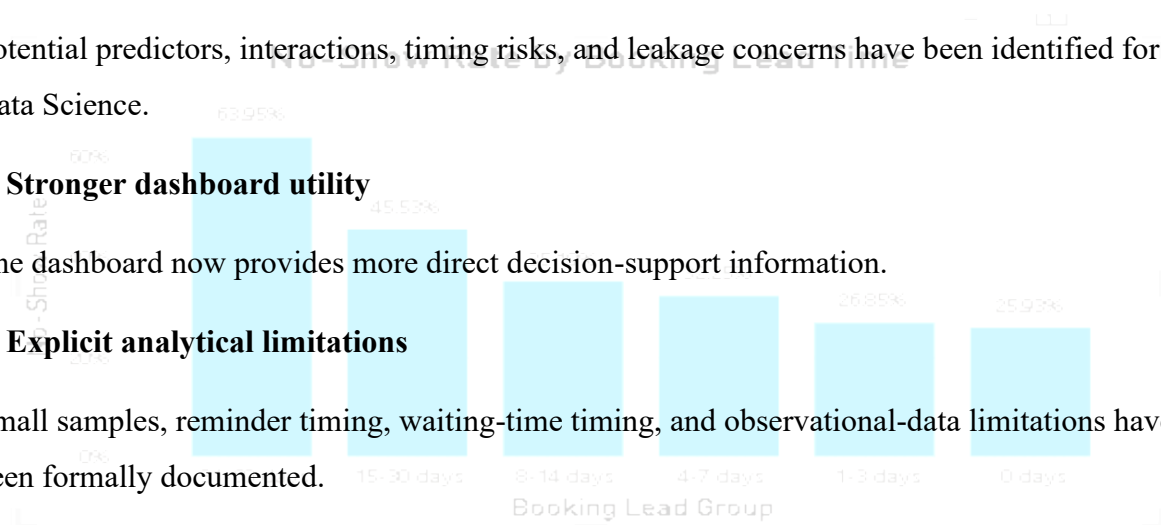
Potential predictors, interactions, timing risks, and leakage concerns have been identified for Data Science.

6. Stronger dashboard utility

The dashboard now provides more direct decision-support information.

7. Explicit analytical limitations

Small samples, reminder timing, waiting-time timing, and observational-data limitations have been formally documented.



13.12 What Week 6 Established

At the conclusion of Week 6, the Data Analytics track has established the following:

Established

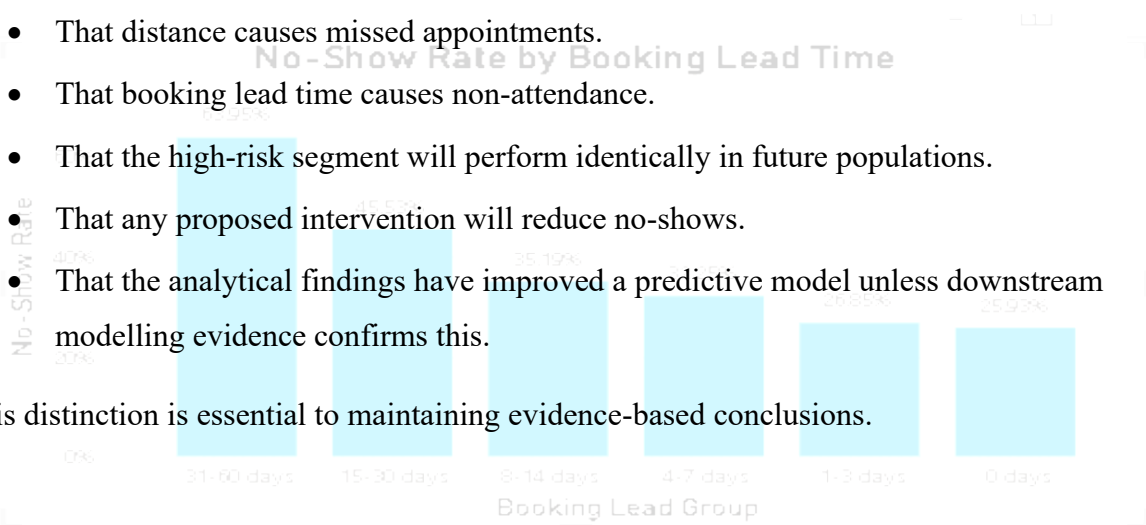
- Overall no-show rate = 51.15%.
- Attendance rate = 48.85%.
- Previous no-show history is a strong attendance-risk indicator.
- Booking lead time is a strong attendance-risk indicator.

- Their combination identifies a high-risk segment at 70.52%.
- Distance provides additional segmentation value within the high-risk segment.
- Appointment type is a weaker secondary differentiator.
- Reminder status is associated with different observed no-show rates but does not establish causal effectiveness.
- The dashboard has been upgraded for decision support.
- Candidate modelling variables and interaction effects have been identified.

Not yet established

- That reminders cause lower no-show rates.
- That distance causes missed appointments.
- That booking lead time causes non-attendance.
- That the high-risk segment will perform identically in future populations.
- That any proposed intervention will reduce no-shows.
- That the analytical findings have improved a predictive model unless downstream modelling evidence confirms this.

This distinction is essential to maintaining evidence-based conclusions.



13.13 Overall Business Contribution

The overall business contribution of the Week 6 Data Analytics work is the transformation of a broad attendance problem into a more actionable risk-segmentation problem.

The Week 5 analysis established that HealthConnect has a substantial appointment attendance challenge.

Week 6 provides greater specificity by showing that the risk is not evenly distributed.

The strongest identified segment is:

31–60-day booking lead + 1+ previous no-shows = 70.52% no-show rate.

This creates a practical basis for HealthConnect to investigate targeted attendance-support strategies and for Data Science to evaluate whether the identified characteristics can support a predictive solution.

The analysis therefore contributes to the broader project objective of using data and AI to improve patient attendance and support.

13.14 Remaining Limitations

The Week 6 contribution must be interpreted within the limitations already documented in Chapter 11.

The most important limitations are:

- Observational rather than experimental data.
- Uncertain reminder timing and content.
- Uncertain timing of waiting-time measurement.
- Small sample sizes in some detailed segments.
- Potential feature leakage in future modelling.
- Lack of evidence yet establishing intervention effectiveness.
- Need for downstream Data Science validation.
- No evidence yet that the high-risk segment will generalise beyond the current dataset.

These limitations do not invalidate the findings. Instead, they define the boundaries within which the findings can responsibly be used.

13.15 Readiness for Week 7

The Data Analytics track is positioned to enter Week 7 with a defined testing agenda.

The principal requirements are to:

- ✓ Reconcile all KPIs.

- ✓ Validate the high-risk segment.
- ✓ Test segment stability.
- ✓ Review small-sample findings.
- ✓ Validate dashboard calculations and interactions.
- ✓ Test reminder associations further.
- ✓ Confirm data quality.
- ✓ Validate predictive-feature timing.
- ✓ Review potential feature leakage.
- ✓ Confirm actual Data Science integration.
- ✓ Evaluate the operational usefulness of the high-risk segment.
- ✓ Document any changes to the Week 6 conclusions.

Week 7 should therefore focus on **testing and refinement**, not restarting the analytical process.

13.16 Week 6 Analytical Contribution to the Overall Project

The Week 6 Data Analytics contribution can be summarized as follows:

Raw Appointment Data



Week 5 Exploratory Analysis



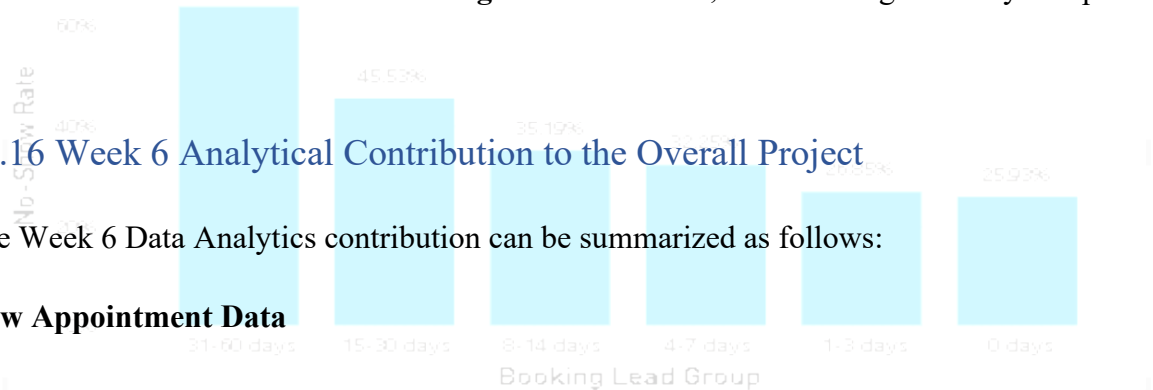
Validated KPIs and Attendance Patterns



Week 6 Advanced Segmentation



High-Risk Segment Identification





Decision-Support Dashboard



Candidate Modelling Inputs for Data Science



Week 7 Testing and End-to-End Validation

This progression demonstrates how the analytical track contributes to the wider HealthConnect solution.

The Data Analytics function is not limited to producing charts. It provides the evidence required to identify meaningful patterns, Prioritize business problems, inform modelling decisions, support operational decision-making, and establish what must be tested before implementation.

13.17 Overall Conclusion

Week 6 represents a significant progression in the HealthConnect Data Analytics work.

The analysis moved beyond the broad Week 5 findings and examined the strongest attendance-risk indicators in greater depth. Booking lead time and previous no-show history remained the most important observed differentiators, and their combination identified a high-risk segment with a **70.52% no-show rate**.

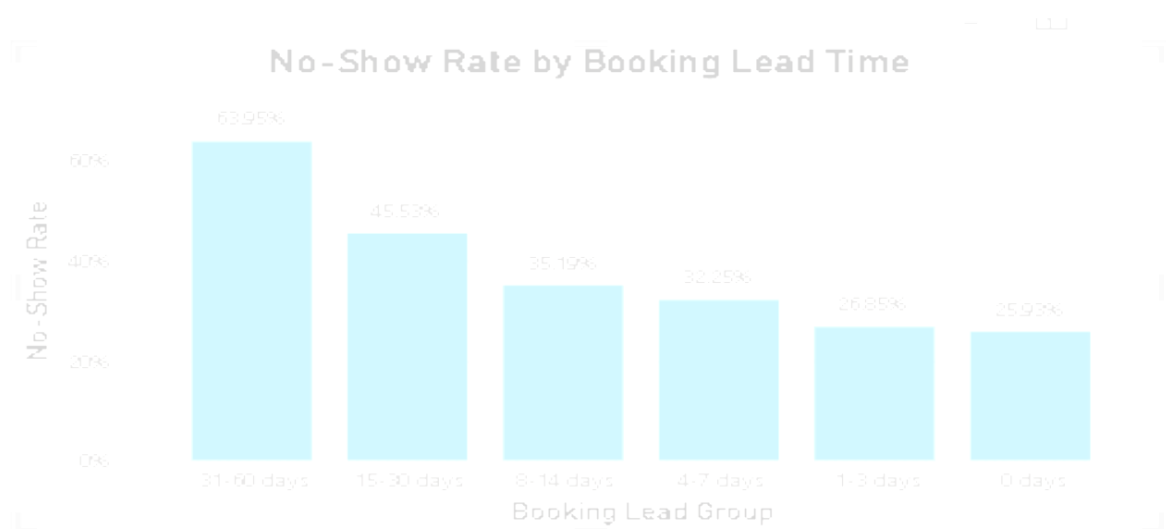
Further analysis showed that distance may provide additional segmentation value within this group, while appointment type appears to be a secondary differentiator. Reminder status showed a modest association with lower observed no-show rates overall and within the high-risk population, but the evidence remains observational and does not establish causal effectiveness.

The most important analytical contribution is therefore not simply the identification of a high no-show rate. It is the development of a more precise understanding of **where attendance risk is concentrated**.

The Week 6 findings support a transition from broad monitoring toward targeted, evidence-based attendance support and predictive modelling investigation. At the same time, the documented limitations ensure that the findings are not overstated.

The Data Analytics track is now prepared for Week 7 testing, where the emphasis should be on stability, reproducibility, dashboard validation, feature timing, cross-track validation, and practical decision-support testing.

Overall, Week 6 strengthens the HealthConnect project by transforming validated appointment data into a more focused analytical foundation for **testing, modelling, operational decision support, and eventual end-to-end solution integration.**



Appendix A — Advanced Analysis Tables

This appendix presents the detailed analytical tables supporting the advanced findings, KPI validation, segmentation analysis, and decision-support conclusions reported in Week 6.

Table A1 — Booking Lead Time × Previous No-Show History

Booking Lead Time	No Previous No-Shows	1+ Previous No-Shows
0 days	18.75%	36.36%
1–3 days	17.74%	39.13%
4–7 days	27.32%	39.52%
8–14 days	33.06%	38.79%
15–30 days	41.09%	51.27%
31–60 days	59.07%	70.52%

Interpretation: The combination of long booking lead time and previous no-show history identifies a substantially higher observed no-show segment.

Table A2 — Booking Lead Time × Reminder Status

Booking Lead Time	Reminder Sent	No Reminder	Difference
0 days	20.69%	39.13%	+18.44 pp
1–3 days	26.92%	26.67%	−0.26 pp
4–7 days	32.17%	32.47%	+0.29 pp
8–14 days	34.43%	37.33%	+2.90 pp
15–30 days	43.91%	49.72%	+5.81 pp
31–60 days	62.58%	67.63%	+5.05 pp

Interpretation: Reminder status is associated with lower observed no-show rates in several booking windows, but the magnitude is inconsistent across segments and should not be interpreted causally.

Table A3 — Previous No-Show History × Reminder Status

Previous No-Show History	Reminder Status	Eligible Appointments	No-Shows	No-Show Rate
No previous no-shows	Reminder sent	2,013	912	45.31%
No previous no-shows	No reminder	732	359	49.04%
1+ previous no-shows	Reminder sent	1,439	809	56.22%
1+ previous no-shows	No reminder	553	343	62.03%

Table A4 — Booking Lead Time × Previous No-Show History × Reminder Status

Booking Lead Time	Patient History	Reminder	No-Show Rate	n
0 days	No previous no-shows	Sent	12.12%	33
0 days	No previous no-shows	Not sent	33.33%	15
0 days	1+ previous no-shows	Sent	32.00%	25
0 days	1+ previous no-shows	Not sent	50.00%	8
1–3 days	No previous no-shows	Sent	16.30%	92
1–3 days	No previous no-shows	Not sent	21.88%	32
1–3 days	1+ previous no-shows	Sent	42.19%	64
1–3 days	1+ previous no-shows	Not sent	32.14%	28
4–7 days	No previous no-shows	Sent	29.41%	136
4–7 days	No previous no-shows	Not sent	21.28%	47
4–7 days	1+ previous no-shows	Sent	36.17%	94
4–7 days	1+ previous no-shows	Not sent	50.00%	30

Booking Lead Time	Patient History	Reminder	No-Show Rate	n
8–14 days	No previous no-shows	Sent	33.08%	263
8–14 days	No previous no-shows	Not sent	32.99%	97
8–14 days	1+ previous no-shows	Sent	36.65%	161
8–14 days	1+ previous no-shows	Not sent	45.28%	53
15–30 days	No previous no-shows	Sent	39.89%	524
15–30 days	No previous no-shows	Not sent	44.44%	189
15–30 days	1+ previous no-shows	Sent	49.35%	387
15–30 days	1+ previous no-shows	Not sent	55.76%	165
31–60 days	No previous no-shows	Sent	57.72%	965
31–60 days	No previous no-shows	Not sent	62.78%	352
31–60 days	1+ previous no-shows	Sent	69.21%	708
31–60 days	1+ previous no-shows	Not sent	73.98%	269

Key observation: The highest observed subgroup is appointments booked 31–60 days in advance for patients with previous no-shows and no recorded reminder, with a 73.98% no-show rate (n=269).

Table A5 — High-Risk Segment × Distance to Clinic

High-risk segment: 31–60-day booking lead time + 1+ previous no-shows.

Distance to Clinic	Eligible n	No-Shows	No-Show Rate
0–5 km	224	150	66.96%
>5–10 km	339	235	69.32%
>10–15 km	218	157	72.02%

Distance to Clinic	Eligible n	No-Shows	No-Show Rate
>15–20 km	90	66	73.33%
>20–30 km	68	53	77.94%
>30 km	16	12	75.00%

Note: The >30 km group contains only 16 eligible appointments and should therefore be interpreted cautiously.

Table A6 — High-Risk Segment × Appointment Type

Appointment Type	Eligible n	No-Shows	No-Show Rate
Follow-up	279	203	72.76%
Diagnostic Test	131	94	71.76%
General Consultation	409	286	69.93%
Specialist Consultation	158	106	67.09%

Interpretation: Appointment type provides secondary segmentation value within the high-risk group, but differences are smaller than those associated with booking lead time and previous no-show history.

Table A7 — Validated Week 6 KPI Framework

KPI / Indicator	Result	Status
No-Show Rate	51.15%	Retained
Attendance Rate	48.85%	Retained
Repeat No-Show Rate	57.83%	Retained
No-Show Rate by Reminder Status	49.86% vs 54.63%	Reframed as observational

KPI / Indicator	Result	Status
High-Risk Segment No-Show Rate	70.52%	Added/refined
Cancellation Rate	5.26%	Contextual metric

Denominator: Primary attendance and no-show KPIs use 4,737 eligible appointments (Attended + No-Show). Cancellations are excluded from these denominators.

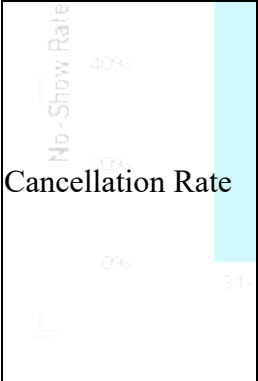
Table A8 — Week 6 Finding Prioritisation

Priority	Finding	Evidence	Business Importance
1	31–60 days + 1+ previous no-shows	70.52% no-show	Very High
2	Previous no-show history	57.83% vs 46.30%	Very High
3	Booking lead time	63.95% at 31–60 days	Very High
4	Distance within high-risk segment	66.96%–77.94%	High
5	Reminder status	4.77 pp overall difference	Moderate
6	Appointment type within high-risk segment	67.09%–72.76%	Moderate

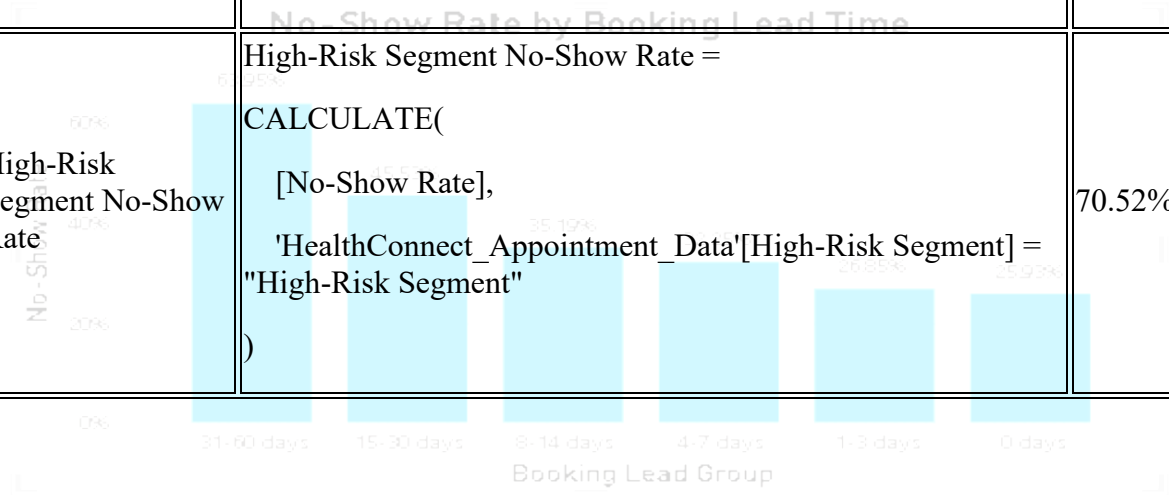
Appendix note: All percentages represent observed associations in the HealthConnect appointment dataset. They should not be interpreted as evidence of causation.

Appendix B — KPI Validation and DAX Measures

KPI	DAX Measure	Validated Result
Total Appointments	Total Appointments = COUNTROWS('HealthConnect_Appointment_Data')	5,000
Eligible Appointments	Eligible Appointments = CALCULATE(COUNTROWS('HealthConnect_Appointment_Data'), 'HealthConnect_Appointment_Data'[appointment_outcome] IN {"Attended", "No-Show"})	4,737
No-Shows	No-Shows = CALCULATE(COUNTROWS('HealthConnect_Appointment_Data'), 'HealthConnect_Appointment_Data'[appointment_outcome] = "No-Show")	2,423
Attended Appointments	Attended Appointments = CALCULATE(COUNTROWS('HealthConnect_Appointment_Data'), 'HealthConnect_Appointment_Data'[appointment_outcome] = "Attended")	2,314
No-Show Rate	No-Show Rate = DIVIDE(	51.15%

KPI	DAX Measure	Validated Result
	[No-Shows], [Eligible Appointments], 0)	
Attendance Rate	Attendance Rate = DIVIDE([Attended Appointments], [Eligible Appointments], 0)	48.85%
Cancellation Rate	 Cancellation Rate = DIVIDE([Cancelled Appointments], [Total Appointments], 0)	5.26%
Repeat No-Show Rate	Repeat No-Show Rate = CALCULATE([No-Show Rate], 'HealthConnect_Appointment_Data'[previous_no_shows] >= 1)	57.83%
No-Show Rate — Reminder Sent	No-Show Rate - Reminder Sent = CALCULATE(	49.86%

KPI	DAX Measure	Validated Result
	[No-Show Rate], 'HealthConnect_Appointment_Data'[reminder_sent] = "Yes")	
No-Show Rate — No Reminder	No-Show Rate - No Reminder = CALCULATE([No-Show Rate], 'HealthConnect_Appointment_Data'[reminder_sent] = "No")	54.63%
High-Risk Segment No-Show Rate	High-Risk Segment No-Show Rate = CALCULATE([No-Show Rate], 'HealthConnect_Appointment_Data'[High-Risk Segment] = "High-Risk Segment")	70.52%



Appendix C — Data Quality, Validation and Cross-Track Evidence

C.1 Data Quality and Analytical Validation

Validation Check	Result	Analytical Implication
Total appointment records	5,000	Confirms expected dataset size
Unique appointment IDs	5,000	No duplicate appointment identifiers
Full-row duplicates	None identified	No duplicate records requiring removal
Eligible appointments	4,737	Attended and No-Show records used for attendance analysis
Attended appointments	2,314	Valid attendance outcome count
No-Show appointments	2,423	Valid no-show outcome count
Cancelled appointments	263	Excluded from attendance/no-show denominator
Booking date after appointment date	0	No logical date-order violation identified
Booking lead-time mismatches	0	Calculated lead time was consistent with source dates
Previous no-shows greater than previous appointments	0	No logical history violation identified
Age range	18–80 years	No out-of-range ages identified
Missing reminder channel	1,366	Structural missingness associated with appointments where no reminder was sent
Missing distance to clinic	90	Retained and treated as missing rather than imputed for the reported segmentation
Missing waiting time	60	Retained as missing; interpretation restricted because timing relative to attendance was not established

C.1.1 Outcome Reconciliation

The eligible appointment population was defined as appointments with an outcome of **Attended** or **No-Show**.

$$4,737 = 2,314 + 2,423$$

Therefore:

- **Eligible Appointments = 4,737**
- **Attended = 2,314**
- **No-Show = 2,423**
- **Cancelled = 263**
- **Total Appointments = 5,000**

The resulting core rates were:

- **No-Show Rate = 51.15%**
- **Attendance Rate = 48.85%**
- **Cancellation Rate = 5.26%**

This denominator approach was retained throughout the Week 6 analysis to ensure consistency between the KPI calculations, advanced segmentation and dashboard reporting.

C.2 Week 6 KPI Validation

The Week 6 analysis retained and refined the KPI framework established during Week 5.

KPI	Validated Result	Week 6 Decision
No-Show Rate	51.15%	Retained as core KPI
Attendance Rate	48.85%	Retained as core KPI
Repeat No-Show Rate	57.83%	Retained as behavioural priority KPI
No-Show Rate — Reminder Sent	49.86%	Retained as association metric
No-Show Rate — No Reminder	54.63%	Retained as association metric

KPI	Validated Result	Week 6 Decision
Reminder-associated difference	4.77 percentage points	Retained with non-causal interpretation
High-Risk Segment No-Show Rate	70.52%	Added as decision-support indicator
Cancellation Rate	5.26%	Retained as contextual KPI

C.3 Advanced Finding Validation

The most important Week 6 finding was identified through combined segmentation of **booking lead time** and **previous no-show history**.

Appointments meeting both conditions below recorded a **70.52% no-show rate**:

1. Booking lead time of **31–60 days**; and
2. At least **one previous no-show**.

Within this high-risk segment:

- **Reminder sent:** 69.21%, $n = 708$
- **No reminder:** 73.98%, $n = 269$

The observed difference was therefore **4.77 percentage points** within this segment.

This result was interpreted as an **observed association**, not evidence that reminders cause or prevent no-shows.

Additional validation showed that previous no-show history remained an important differentiating factor:

- **No previous no-shows:** 46.30%
- **1+ previous no-shows:** 57.83%
- **Difference:** 11.53 percentage points

The analysis also identified increasing no-show rates across longer booking lead-time groups, with the **31–60-day group recording 63.95%** compared with **25.93% for same-day bookings**.

C.4 Cross-Track Integration Evidence — Data Analytics → Data Science

The Data Analytics track provided analytical findings and candidate variables to support subsequent Data Science work.

Integration Element	Week 6 Evidence
Track collaborated with	Data Science
Project dependency	Data Science modelling requires validated analytical findings, relevant candidate features and clearly defined outcome variables
Information/output received	Data Analytics used the validated appointment dataset and established KPI framework as the analytical foundation
Information/output provided	Validated risk patterns, KPI findings, segment definitions and candidate modelling features
Integration activity completed	Advanced segmentation was used to identify variables and interactions that may be relevant for predictive modelling
What changed as a result	The analytical focus shifted toward targeted risk segmentation, particularly booking lead time combined with previous no-show history
Evidence	High-risk segment analysis, validated KPI results, interaction analysis and candidate feature register documented in this report

C.4.1 Candidate Data Science Features

The following variables and derived relationships were identified as potential modelling inputs:

- previous_no_shows
- **Previous No-Show Status**
- booking_lead_days
- **Booking Lead Group**
- distance_to_clinic_km

- **Distance Group**
- appointment_type
- reminder_sent
- **Booking Lead Time × Previous No-Show History**
- **High-Risk Segment × Reminder Status**

The variable **appointment_outcome** was retained as the outcome/target variable and was not treated as a predictive input.

C.4.2 Modelling and Data Leakage Considerations

Two variables require additional validation before predictive modelling:

- **Reminder status:** The timing of the reminder relative to the intended prediction point must be established before using it as a predictive feature.
- **Waiting time:** The timing of waiting_time_minutes relative to the appointment outcome was not established. It was therefore treated primarily as a service-experience measure rather than automatically classified as a pre-appointment predictor.

The high-risk segment was provided as a **decision-support and feature-engineering input**, not as evidence of a completed predictive model.

C.5 Week 7 Validation Requirements

The Week 6 evidence establishes the analytical findings that should be tested and refined during Week 7.

Priority validation areas include:

1. Reconciliation of all core KPIs against the source dataset.
2. Stability of the **70.52% high-risk segment** across relevant subgroup analyses.
3. Validation of the relationship between booking lead time and no-show behaviour.
4. Validation of previous no-show history as a behavioural risk indicator.
5. Testing of the Booking Lead Time × Previous No-Show History interaction.
6. Further examination of reminder status and reminder-channel associations.
7. Review of small-sample segments, particularly the **>30 km** distance group.

8. Confirmation of the timing and modelling suitability of reminder-related variables.
9. Confirmation of whether waiting time can legitimately be used as a predictive feature.
10. End-to-end reconciliation between the analytical findings, Power BI dashboard and subsequent Data Science development.

